



**U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL AVIATION ADMINISTRATION**

**ORDER
8130.2M**

National Policy

07/22/2026

SUBJ: Airworthiness Certification of Aircraft

This order establishes policies and procedures for issuing airworthiness certificates, export certificates of airworthiness, and special flight authorizations for aircraft. This order applies to Federal Aviation Administration (FAA) aviation safety inspectors and certain persons designated to act as representatives of the FAA.

FAA employees and designees are required to comply with this document. For non-FAA entities, this is a guidance document and conformity with the guidance document is voluntary only. Its content is not legally binding in its own right and will not be relied upon by the Department as a separate basis for affirmative enforcement action or other administrative penalty. Nonconformity will not affect rights and obligations under existing statutes and regulations.

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Chapter 1 Introduction

1-1. Purpose of This Order. This order establishes policies and procedures for issuing:

- FAA Form 8100-2, Standard Airworthiness Certificate,
- FAA Form 8130-7, Special Airworthiness Certificate,
- Special flight authorizations (SFA),
- FAA Form 8130-4, Export Certificate of Airworthiness (export C of A), and
- Experimental certificates and special flight permits for OPA or UAS, when used in conjunction with Order 8130.34, Airworthiness Certification of Unmanned Aircraft Systems and Optionally Piloted Aircraft.

1-2. Audience. FAA manufacturing and airworthiness aviation safety inspectors (ASI) and certain persons designated to act as representatives of the FAA. Airworthiness ASIs must also refer to FAA Order 8900.1, Flight Standards Information Management System, for related policies and procedures.

1-3. Where Can I Find This Order and the Code of Federal Regulations (CFR)? This order and CFRs specified in this order are available on the FAA public website, [FAA Dynamic Regulatory System \(DRS\)](#).

1-4. Showing and Finding. In seeking a certificate or an authorization under this order, an applicant is responsible for presenting evidence necessary to show compliance with applicable requirements to the satisfaction of the FAA; that is, the burden of proof is on the applicant. The FAA is responsible for reviewing and inspecting the applicant's evidence to make findings of compliance with applicable requirements. If an applicant's evidence is insufficient to enable the FAA to make its findings, reviews and inspections may be stopped until the applicant presents new evidence to support the FAA's findings.

1-5. Aircraft Located Outside the United States. Except for issuing an airworthiness certificate under paragraph 21-2 of this order, before processing an application for a certificate under this order for an aircraft located outside the United States, the issuing office must find that the activity can be accomplished without placing an undue burden on FAA resources. Using designees to issue a certificate under this order for an aircraft located outside the United States, would not constitute an undue burden on the FAA but must be authorized in accordance with FAA Order 8000.95, Designee Management Policy.

1-6. Cross Utilization. The FAA performs a variety of functions in issuing certificates and authorization under this order. Many of these functions should be accomplished via coordination among manufacturing, airworthiness, operations ASIs, air traffic, and

other FAA stakeholders with expertise in particular subject matters. Consult with other FAA offices as necessary.

1-7. Explanation of Policy Changes. The following changes are included in this revision.

- a. This revision incorporates extensive changes throughout the document. Users should perform a complete review of this revision.
- b. This revision implements related provisions of the Modernization of Special Airworthiness Certification (MOSAIC) final rule, which amended Title 14, Code of Federal Regulations (14 CFR), Parts 1, 21, 22, 36, 43, 45, 61, 65, 91, and 119, affecting:
 - (1) The design, manufacture, certification, operation, maintenance, and alteration of light-sport category aircraft.
 - (2) Special purpose operations for restricted category aircraft.
 - (3) The duration of experimental airworthiness certificates for the purposes of showing compliance with regulations, research and development, crew training, and market survey.
 - (4) Eligible purposes for experimental aircraft with the new purpose of operating former-military aircraft under § 21.191(j).
 - (5) Operating limitations in 14 CFR Part 91 for light-sport and restricted category aircraft and for experimental aircraft, including experimental aircraft engaged in space support vehicle flights.
- c. Other significant changes to this order include:
 - (1) Added the definitions of “space support vehicle” and “space support vehicle flight.”
 - (2) The acronyms “ELSA” and “SLSA” were removed throughout the order to align with the removal of the definition of light-sport aircraft in § 1.1.
 - (3) Chapters 9 and 17 were rewritten to align with new rules for light-sport category aircraft and experimental kit-built and former light-sport category aircraft, respectively.
 - (4) Chapter 18 was added for the new experimental purpose of operating former military aircraft under § 21.191(j).
 - (5) Most operating limitations in appendix D, Table D-1, were revised.
 - (6) Removed all provisions related to FAA approval of production certificate holder’s and modifier’s procedures for operating experimental aircraft.

1-8. Cancellation. The following are canceled upon the effective date of this order:

- a. FAA Order 8130.2L, Airworthiness Certification of Aircraft, dated 01/30/2026.
- b. All clarification, deviation, and policy memorandums issued for Order 8130.2 before the effective date of this order.

1-9. FAA Forms. Table 1-1 of this chapter lists forms the FAA completes under this order.

1-10. Acronyms, Definitions, and Related Publications. Lists of acronyms and definitions are provided in appendix H and appendix I, respectively. Additionally, this order references related publications; unless otherwise stated, it always refers to the latest revision. See related publications listed in appendix J.

1-11. Conflicts. If a conflict exists between this order and an applicable bilateral agreement, follow the bilateral agreement.

1-12. Effective Date. This order is effective beginning on July 24, 2026.

Table 1-1. FAA Forms Completed by the FAA Under This Order

Form Number	Title	Source
FAA Form 8100-1	Conformity Inspection Record	FAA public website
FAA Form 8100-2	Standard Airworthiness Certificate	AWC or FAA employee website for fillable PDF format
FAA Form 8130-1	Application for Export Certificate of Airworthiness	AWC or FAA public website
FAA Form 8130-4	Export Certificate of Airworthiness	AWC or FAA employee website for fillable PDF format
FAA Form 8130-6	Application for U.S. Airworthiness Certificate	AWC or FAA public website for fillable PDF format
FAA Form 8130-7	Special Airworthiness Certificate	AWC or FAA employee website for fillable PDF format
FAA Form 8130-10	Surplus Military Aircraft Inspection Record	FAA public website
FAA Form 8130-15	Light-Sport Category Aircraft/Kit Statement of Compliance	FAA employee website for fillable PDF format

Chapter 2. Common Policies and Procedures for Issuing an Airworthiness Certificate

2-1. Introduction. This chapter provides common policies and procedures for issuing, amending, exchanging, and replacing standard and special airworthiness certificates. Additional classification-, category-, and purpose-specific procedures for issuing an airworthiness certificate are included in subsequent chapters of this order. This chapter does not apply to the issuance of a special flight permit (SFP) except as referenced by chapter 19 of this order.

Note: Where there are differences between common procedures and additional specific procedures, the specific procedures take precedence.

Note: At a minimum, document completed inspections on FAA Form 8100-1, Conformity Inspection Record, per the instructions on the back of the form, to include all applicable requirements found in Paragraph 2-3 (a through i) and in addition, for Experimental Certificates, document all applicable requirements found in Chapter 4 Section 2 of this Order.

2-2. Persons Authorized to Issue an Airworthiness Certificate.

a. ASIs. FAA ASIs are authorized to issue airworthiness certificates. Manufacturing ASIs in the Aircraft Certification Service (AIR) are primarily responsible for original certifications, while airworthiness ASIs in the Flight Standards Service (AFS) are primarily responsible for recurrent certifications.

b. Delegation. The FAA may delegate the issuance of an airworthiness certificate per 14 CFR Part 183 and the FAA Order 8000.95, Designee Management Policy.

c. Original and Recurrent Airworthiness Certifications.

Note: Appendix I of this order contains definitions of original and recurrent certification and other terms.

(1) An AWC application for an original airworthiness certificate must be assigned to an AIR CM Section. When the AIR CM Section is unable to perform the certification activity, it may request assistance from a flight standards office. If the flight standards office agrees to conduct the activity, the AIR CM Section may reassign the AWC application to that office.

(2) Certain airworthiness certification tasks must remain with the AIR CM Section and should not be delegated to a designee, including:

- (a) Issuing the first special airworthiness certificate for a new light-sport category aircraft model, including prototype flight testing under 14 CFR 21.191(a).
- (b) Issuing the first special and the first standard airworthiness certificate for an aircraft intended to support a new or amended type certificate (including supplemental type certificate) or production approval.

(c) Advising applicants about obtaining a type certificate, amended type certificate (including supplemental type certificates), production certificate, or inclusion on the “FAA Make/Model Directory for light-sport category aircraft,” which can be accessed at https://www.faa.gov/aircraft/gen_av/light_sport.

(d) Issuing a Special flight permit for production flight testing issued to a light-sport category manufacturer or production certificate manufacturer.

(3) Processing an application for an experimental airworthiness certificate used to show compliance with regulations is primarily the responsibility of the AIR CM section. In remote areas or under special circumstances, or when a project requires specialized airworthiness oversight, the AIR CM section manager should request that the Flight Standards office manager assign an airworthiness ASI. The selected inspector must possess experience in type certification programs of a type and complexity comparable to the certificate being requested.

(4) When an application is submitted through the FAA's AWC system for a recurrent airworthiness certification, it is normally routed to a flight standards office. However, certain aircraft associated with AIR-managed certification or production projects should be handled by an AIR CM Section instead. These offices support aircraft involved in projects managed by organizations such as:

(a) Holders of a type certificate.

(b) Holders of a production certificate.

(c) Organizations performing approved aircraft modifications (for example, under supplemental type certificate projects).

In those situations, the AIR CM Section should issue the recurrent airworthiness certification rather than referring the application to flight standards. This division of responsibility is part of the FAA's broader airworthiness certification framework and reflects the involvement of Aircraft Certification Service offices in managing certain certification and production-related projects.

Note: For original or recurrent certifications not previously described in this section, the managers of the responsible AIR and AFS offices should coordinate to establish a mutual agreement on which office is responsible for certificate issuance. Airworthiness and manufacturing ASIs must notify their managers of these situations so that responsibility for certificate issuance can be determined.

Note: As a courtesy, the responsible CM Section may notify the geographic AFS office of any unusual or novel technologies or operations presented by an applicant during an original airworthiness certification. FAA offices are encouraged to jointly identify and resolve risk mitigation of unusual or novel technologies or operational procedures.

2-3. Common Procedures for Issuing an Airworthiness Certificate. Common procedures for issuing an airworthiness certificate are illustrated in Figure 2-1 of this chapter and specified in this paragraph.

a. The Airworthiness Certification Application (AWC).

(1) AWC provides an electronic means to process an application for certain certificates and authorizations under this order. ASIs, designees, and organization designation authorization (ODA) unit members must establish an account, process FAA Form 8130-6, issue standard and special airworthiness certificates, FAA Form 8130-1, and denials using AWC. Unless otherwise stated in this order, all requirements concerning completion of FAA Forms 8130-6 and 8130-1 must be accomplished in AWC. Advisory Circular 21-12, *Application for a U.S. Airworthiness Certificate*, provides additional information for using AWC for FAA Form 8130-6. This order provides additional information for completing FAA Form 8130-1.

(2) The use of AWC by applicants is not mandatory but is the preferred method for application submission. If an applicant is unable or elects not to submit the application in AWC, the FAA will upload to AWC the hard copy of the application from the applicant and transcribe the information to AWC. The ASI must include a statement that the application information was entered on behalf of the applicant. When completing the application on the Documentation-Certification page of AWC, enter the following statement in the SIGNATURE block of section III, OWNER'S CERTIFICATION: "Application information entered on behalf of the applicant by (Name), (Title). Original attached." In such cases, do not sign the hard copy or the uploaded application from the applicant, but digitally sign appropriate sections of Form 8130-6 generated by AWC. Per paragraph B-4 of this order, submit both the application signed by the applicant and the AWC-generated application to the [FAA Registry](#) (hereafter referred to as Registry).

(3) Access AWC portals as follows. For ASIs, register on the FAA ASI AWC Internal portal. For ODA unit members and designees, register on the [External ODA/Designee AWC Portal](#); follow the prompts for "Login/New User Sign-Up." Contact the AWC Help Desk at 9-avs-awc-notifications@faa.gov for help with access problems.

(4) You may only issue airworthiness certificates and operating limitations outside of AWC in the event it is unavailable due to FAA network or internet outage, or in instances of remote locations where certification activity is being conducted and internet is not accessible. In such cases, revert to the paper process (or portable document format (PDF) equivalent (see table 1-1)) to document and capture data, process applications, and issue certificates. Once AWC is available and accessible, upload documents you created offline to AWC using the Upload Approval page in AWC and use the comments block to record what you uploaded and the reason for issuing the certificate outside of AWC.

b. Review Application.

(1) With the exception of an exchange or replacement airworthiness certificate as described in paragraph 2-4, a properly completed FAA Form 8130-6, Application for U.S. Airworthiness Certificate, is required to begin the process of issuing an airworthiness certificate. Review FAA Form 8130-6 to verify that the applicant completed the form properly. Refer to FAA AC 21-12 for related guidance as necessary.

(2) Letter of Authorization/Agent Letter. If an agent completed the application, verify that the aircraft's registered owner provided a notarized agent letter of authorization and that the information in this letter is consistent with the application. A copy of the notarized agent letter is acceptable to upload in AWC.

Note: Requirements and forms of notarization vary among countries.

(3) Type Certificate Data Sheet (TCDS). For an application for an aircraft in a category that requires a type certificate, review the applicable TCDS or aircraft specification to verify that the aircraft is eligible for issuance of the requested airworthiness certificate by make, model, and serial number (S/N). An aircraft listing maintained by the responsible certification branch may consist of aircraft model and S/Ns of newly manufactured aircraft that are not yet listed on the TCDS. If none of these documents clearly show the eligibility of the applicant's aircraft by make, model, and S/N, contact the responsible certification branch to verify eligibility.

Note: [DRS](#) contains TCDSs and aircraft specifications. Terms for type certificate data have changed over time, including aircraft specifications and TCDS. See the latest version of FAA Order 8110.4, Type Certification, for further information.

Note: For restricted category aircraft, refer to FAA Order 8110.56, Restricted Category Type Certification. The TCDS format for military surplus aircraft includes approved serial numbers instead of eligible serial numbers typically used for aircraft manufactured under a production approval. The applicant may include the list of approved serial numbers in the TCDS, or in a separate FAA-approved report referenced in the TCDS, stating that the serial number report contains only the approved serial numbers.

c. Review FAA Airworthiness Records on File at the Registry. This review includes reviewing the owner information that was submitted on FAA Form 8130-6 and in AWC.

(1) Current Registration. Access the Registry to verify that the aircraft is currently registered per 14 CFR Part 47.

(a) Except for paragraphs (b) through (d), the aircraft is registered when an inquiry of the Registry shows that the nationality and aircraft's registration number (registration mark), hereafter referred to as the N-number, is "assigned" and the status is "valid" or "valid-asn to mfr."

(b) For a new owner of a used aircraft, lag time in processing the application for registration may result in records of ownership within the Registry database that lag

actual transfers in ownership from one United States owner to another. In such cases, the prior owner remains the registered owner of record within the Registry's database, and registration by the current owner may only be verified as part of the aircraft records review under paragraph 2-3.e.

(c) For a manufacturer's temporary N-number, the aircraft is considered registered when an inquiry of the Registry shows that the N-number is "reserved," and the status is "manufacturer (temp)."

(d) For the assignment of a new special registration number, the aircraft is registered when an inquiry by the Registry shows that the "present registration number" is "assigned," and the status is "valid" or "pending."

Note 1: When searching the Registry for antique or replica aircraft displaying nationality and registration marks as provided for under 14 CFR 45.22(b), omit the inserted symbol allowed by § 45.22(b).

Note 2: For a recurrent certification, including previously exported aircraft returning to the United States, you may retrieve permanent aircraft records from the Registry using the aircraft make, model, and S/N. For aircraft that do not have a data plate or S/N, you may retrieve records using the N-number or prior N-number for previously exported aircraft returning to the United States.

(2) Prior Denials. Except for new aircraft, check the [Civil Aviation Registry Electronic Service](#) (CARES), or [order a copy of the aircraft record](#) to determine if the FAA previously issued a denial for this aircraft. If previously denied, verify that the reasons for the denial have been corrected, and record the action taken to clear the denial on the FAA Form 8100-1, Conformity Inspection Record.

d. Identify Potential Safety Hazards. Review available information as necessary to assist you in becoming familiar with the aircraft, aircraft engine, propeller models, and potential safety hazards. For example, review information as applicable and necessary to understand the manufacturer/builder; configuration; installed equipment; novel or unique features, characteristics, or systems; recent customer findings for new aircraft deliveries; recent quality escapes for new aircraft; histories of service difficulties, incidents, and accidents; ownership history of the aircraft; and potential for degradation from long-term storage. Designees should review the designee alert list for any pertinent information. The "Designee Alert List" is found in AWC under the "Announcements" tab.

Note: The National Transportation Safety Board (NTSB) maintains a searchable [database of accidents and incidents](#) involving civil aircraft.

Note: For rotorcraft, review guidance on the Congressional mandate for rotorcraft crash-resistant fuel systems. Refer to AIR Memo No. AIR-600-20-600-PM04 latest revision.

e. Review Aircraft Records.

(1) Registration. Review the certificate of aircraft registration to verify that the aircraft is currently registered and that the information on the registration and application matches. The manufacturer's name may be abbreviated on the application and/or registration. If the registration does not have an expiration date, the registration is not effective. Evidence of aircraft registration may be shown via any of the following [hereafter referred to as registration]:

- (a) Aeronautical Center Form 8050-3, Certificate of Aircraft Registration.
- (b) Aeronautical Center Form 8050-6, Dealer's Aircraft Registration Certificate.
- (c) Form REGAR-FAX-4, Temporary Certificate of Registration.

Note: The permanent AC Form 8050-3 and AC Form 8050-6 may be delivered to the registered owner by email or mail. The FAA plans to discontinue issuing temporary certificates (Form REGAR-FAX-4) because the 8050-3 can be emailed the same day.

- (d) Aircraft Registration Application, FAA AC Form 8050-1, under the provisions of [§ 47.31\(c\)](#).
- (e) A copy of Assignment of Special Registration Number, AC Form 8050-64, and the current, AC Form 8050-3, Certificate of Aircraft Registration.

(2) Exemptions. For exemptions that the applicant claims to apply to this aircraft, verify that each exemption is included in [DRS](#) and applies to this aircraft make, model, and S/N.

(3) Maintenance Records. Verify the following:

- (a) The applicant has all maintenance records required by 14 CFR 91.417.
- (b) All maintenance required as of the date you inspect the aircraft is complete, including compliance with applicable airworthiness directives (AD) and applicable aircraft and component life limits.
- (c) For used, type-certificated aircraft, maintenance, preventive maintenance, and alterations were accomplished per 14 CFR Part 43.
- (d) If applicable, verify that the aircraft log includes the reinspection date for the parachute and associated deployment system components.

(4) Weight and Balance. Review the aircraft weight and balance report, including center of gravity (CG) limitations, to verify that it reflects the current configuration.

(5) Manuals. Verify that the applicant has applicable manuals and documents. Applicable manuals and documents may include an aircraft flight manual (AFM), equipment list, Instructions for Continued Airworthiness (ICA), and maintenance manual(s). Such

manuals and documents may not be required for some aircraft certificated under the Civil Air Regulations (CAR).

(6) Flight Testing. Determine if the aircraft has been flight tested, if required, or as necessary to verify the condition for safe operation. As necessary, issue a special airworthiness certificate for showing compliance with the airworthiness regulations (14 CFR 21.127, 21.137(e)(1), 21.189(a)(2), 21.185(d), 22.195(d)(2), and 91.319(b)) per the applicable requirements of this order.

f. Inspect Aircraft. Physically inspect the aircraft to verify—

(1) Nationality, Registration, and Special Marks. The nationality and registration marks (N-number) meet the requirements of §§ 45.21 through 45.31.

(a) Size. When verifying an N-number is the proper size, verify the height of the marks were measured vertically. The font may be slanted, but do not use the length of the slant line to determine the height of the numbers.

Note: If a surface authorized for displaying required marks under § 45.25 is not large enough to display full-size marks, § 45.25 allows for marks as large as practicable to be displayed. Assessing “as large as practicable” requires judgment based on the aircraft make/model. For example, for some strongly contoured surfaces, smaller marks may be more legible and, therefore, more practicable than the largest marks that could fit on that surface.

(b) Legibility, Style, Borders, and Shading. The N-number must be legible and written in capital Roman letters without ornamentation. Borders or shading is allowed only if it makes the number more legible. The color of the registration marks must contrast with the background. The N-number must be legible from 500 feet away when viewing it perpendicular to the side of the aircraft during daylight hours.

Note: Section 45.21(b) prohibits a person from placing on any aircraft a design, mark, or symbol that modifies or confuses the N-number. Note that this requirement is very broad and includes any feature that would modify or confuse the N-number. For example, N-number paint type, paint color, font type(s), or background patterns must not modify or confuse the N-number.

(c) Display of Foreign Marks for Exporting New Aircraft. Per § 45.31, a person who manufactures an aircraft in the United States for export delivery may display foreign marks on the aircraft as required by the importing state of registry (SOR) of the aircraft while the aircraft is registered in the United States. The manufacturer may operate the aircraft with foreign marks within the United States for testing, demonstration, or delivery; note that these flights are performed under an SFP because the aircraft is registered in the United States.

(d) Marking of certain aircraft with a special airworthiness certificate. Except for unmanned aircraft, when marks include only the Roman capital letter “N,” and the registration number is displayed on limited, restricted, experimental, or

provisionally certificated aircraft, the operator must also display on that aircraft near each entrance to the cabin, cockpit, or pilot station, in letters not less than 2 inches nor more than 6 inches high, the words “limited,” “restricted,” “experimental,” or “provisional,” as applicable. Refer to § 45.23.

Note: Although § 45.23 requires these marks “near each entrance to the cabin, cockpit, or pilot station,” this rule does not specify whether these marks must be inside or outside the aircraft; either is acceptable.

(e) Special Marks for Antique or Replica Aircraft. § 45.22(b) provides special marks for exhibition and amateur-built aircraft.

(2) Product Marking.

(a) The aircraft is identified as prescribed in § 45.11 except for an SFP or an experimental airworthiness certificate issued for the purposes of research and development, showing compliance with regulations, crew training, exhibition, air racing, market surveys, or operating former military aircraft. Each installed aircraft engine, propeller, propeller blade, and propeller hub manufactured under a TC or production certificate is marked per § 45.11.

(b) Product identification (ID) plates include the information specified in § 45.13 and must meet § 45.11. Verify that the make, model, and serial number on the aircraft engine and aircraft propeller product ID plate(s) match the application. Verify that the aircraft make, model, and S/N on the aircraft ID plate match the registration. Note that the builder’s name may be abbreviated on the application and registration.

Note 1: For aircraft manufactured outside the United States pursuant to a § 21.29 TC, the TC and PC numbers on the ID plate may reflect the foreign design and production approval numbers. The ID plate information should provide a means to determine the applicable U.S. TC number if the TC number of the state of manufacture (SOM) is used.

Note 2: In some cases, the FAA has granted regulatory exemptions permitting alternate mounting locations of aircraft ID plates for certain aircraft of qualifying air carriers. For any aircraft whose ID plate is mounted in a location other than that required in § 45.11, verify the provisions of a current regulatory exemption for alternate ID plate location cover this.

(c) When an aircraft has been modified to conform to another make/model, verify that the supplemental ID plate contains the information specified in § 45.13. If the FAA Aircraft Registry has not yet issued the new registration with the new model designation, verify that the aircraft make, and the S/N match the application and current registration. If ownership of the aircraft has not changed, an application for aircraft registration reflecting the new model designation, need not be submitted; the FAA Aircraft Registry will change the model and issue a new registration after receipt of the certification files. Verify that the original ID plate has not been altered in any manner.

- (d) Inspect the aircraft and review aircraft records to determine whether a violation of §§ 45.13(c) or (e) is apparent. If a violation of §§ 45.13(c) or (e) is found, deny the airworthiness certificate and initiate an enforcement action.
- (3) Conformity to TC. For an application for an aircraft in a category that requires a TC, verify that the aircraft conforms to its TC. This includes conformity to applicable ADs, major repairs, and major alterations. Verify that minor repairs or alterations conform to FAA-accepted data.
- (4) Engines, Propellers, and Associated Instruments. Engine(s), propeller(s), and associated instruments have been checked per the manufacturer's instructions or other FAA-approved instructions, placards, or flight manual supplements.
- (5) Systems. Aircraft systems are functioning properly, to the extent necessary, including the flight control and pitot-static systems.
- (6) Equipment. The installed equipment, including all equipment on the aircraft equipment list, has been checked and verified as functioning properly.
- (7) Instruments and Placards. Instruments and placards are correctly located, installed, and properly marked in the English language. In addition, verify that measurements are displayed in standard U.S. units of measure for those instruments necessary for operation in the U.S. national airspace system (NAS).
- (8) Airframe Emergency Parachutes (if installed). Verify that the aircraft log matches the reinspection date for the parachute and associated deployment system components. As applicable, verify that airframe emergency parachutes that are ballistic, assisted, or deployable are properly marked, identified, and within their service dates. Verify aircraft marks clearly identify all explosive devices used in conjunction with ballistic parachutes; marks indicating the aircraft is equipped with explosive devices must be applied externally and be readable from the ground.
- (9) Foreign Objects. No foreign objects are present that could pose a safety hazard, such as unsecured laptops, tools, scrap material, or miscellaneous debris.
- (10) Airworthiness. The aircraft is airworthy.
- g. Document Reviews and Inspections.** Document all completed reviews and inspections on FAA Form 8100-1, per the instructions on the form. Verify that the applicant corrects all unsatisfactory findings and documents those corrections on FAA Form 8100-1 before issuing an airworthiness certificate. FAA Form 8100-1 must be completed even if the action isn't a full aircraft inspection or a full records review such as for an amended or replacement airworthiness certificate. Inspectors and designees should always record when an airworthiness certificate is issued, regardless of how routine the situation is. This builds a historical record of the aircraft's certification status over time.
- h. Issue Airworthiness Certificate.** If you find the aircraft airworthy and all other requirements are met for the requested certificate:
 - (1) Complete sections V and VIII of FAA Form 8130-6 per the block-by-block instructions in section 2 of appendix A to this order.

(2) Complete the standard or special airworthiness certificate, as applicable, per the block-by-block instructions in sections 3 or 4, respectively, of appendix A to this order. Review the certificate to verify that it was completed without error.

(3) Maintenance record entry. Make the following statement in the maintenance records: “I find that this aircraft meets the requirements for the certificate requested and have issued a *[standard airworthiness certificate or special airworthiness certificate for the purpose(s) of (enter purpose) and operating limitations]* dated _____. *[Signature: John Smith, Aviation Safety Inspector, AIR-883].*”

(4) Give the completed airworthiness certificate to the applicant. Advise the applicant to display it in the aircraft at the cabin or cockpit entrance so it is legible to passengers or flight crew per § 91.203(b).

Note: In certain circumstances, such as replacements and others deemed appropriate by the local office manager, transmitting the airworthiness certificate by email is permissible.

i. Issue Letter of Denial.

(1) If, after notifying the applicant of all unsatisfactory findings, the applicant does not correct those findings to your satisfaction in a timely manner, write a letter to the applicant denying issuance of the certificate and stating the reason(s) for the denial. Identify which steps may be accomplished to meet the certification requirements if feasible.

(2) Complete sections V and VIII of FAA Form 8130-6.

j. Review Records and Forward to the Registry.

(1) Review your records to verify completeness, accuracy, legibility, and compliance with applicable requirements.

(2) Review and forward certification files per appendix B to this order.

Note: AWC allows a closed certification file to be reopened for corrections within 30 calendar days. Designees must coordinate with their managing specialists, notifying them of the reason for conducting this action. Once the 30 calendar day period has elapsed, an AWC certification file may not be reopened.

2-4. Amendments, Exchanges, Replacements, Surrenders, and Transfers.

a. Amendment.

(1) Eligibility. An airworthiness certificate may be amended when there is—

(a) A change in the aircraft model specified on an airworthiness certificate.

(b) An alteration to the aircraft, such as one approved by a supplemental type certificate (STC) or amended TC, that changes the aircraft category specified in block 4 of the standard airworthiness certificate.

(c) A change in the operating limitations for an aircraft with a special airworthiness certificate.

(2) Procedures for Amending an Airworthiness Certificate. Follow the procedures in paragraph 2-3 of this order except as follows:

(a) Review Aircraft Records. A complete review of aircraft records per paragraph 2-3.e of this order is not required. Review aircraft records as necessary in consideration of the reason for the amendment.

(b) Inspect Aircraft. A complete aircraft inspection per paragraph 2-3.f of this order is not required. Inspect the aircraft as necessary in consideration of the reason for the amendment.

Note: In amending a special airworthiness certificate, all operating limitations must be issued per the current version of this order, appendix D, and Table D-1.

b. Exchange.

(1) Eligibility. Owners that hold an unexpired FAA Form 1362A, *Certificate of Airworthiness*, FAA Form 1362B, *Certificate of Airworthiness*, or unexpired Form ACA-1362, *Certificate of Airworthiness*, may apply for FAA Form 8100-2, *Standard Airworthiness Certificate*, or FAA Form 8130-7, *Special Airworthiness Certificate*, as applicable.

(2) Procedures for Exchanging an Airworthiness Certificate. Follow the procedures in paragraph 2-3 of this order except as follows:

(a) Review Application. The existing certificate serves as the application; an FAA Form 8130-6 is not required.

(b) Review Aircraft Records. A complete review of aircraft records per paragraph 2-3.e of this order is not required. Review aircraft records as necessary to verify that the applicant's request is valid.

(c) Inspect Aircraft. A complete aircraft inspection per paragraph 2-3.f of this order is not required. Inspect the aircraft per paragraph 2-3.f(1) and 2-3.f(2)(b) to verify that the exterior aircraft N-registration numbers and ID plate information match the application and registration.

c. Replacement.

(1) Eligibility. The FAA may issue a replacement airworthiness certificate when a certificate is declared lost, has been mutilated, is no longer legible, or contains inaccurate and/or erroneous information, or when the aircraft registration number has changed. If inaccurate and/or erroneous information is discovered on the certificate within 30 calendar days following the issuance of the certificate, and it has been determined that it was caused by the issuing inspector, the managing office may reopen the existing AWC record to correct the error and reissue the certificate. Retrieve the certificate with the error. If inaccurate and/or erroneous information is discovered on the certificate after 30 calendar days, the issuing inspector or managing office should initiate and reissue the certificate as a replacement certificate in AWC and follow the procedures for replacement certificates in this chapter.

All corrected documents and supporting materials required for Registry submission must be forwarded through the AWC portal.

When reopening an AWC record to correct FAA-originated errors, the managing office must upload a signed letter or memorandum that:

- Identifies the specific corrections being made.
- Confirms that any newly issued airworthiness certificate (if applicable) has been verified as placed in the aircraft and that the disposition of the original has been documented.
- Confirms that all corrected supporting documents required for Registry submission have been uploaded in AWC and submitted to the Registry.

Note: If the error (inaccurate and erroneous info) is from the owner/operator, a replacement does not require an 8130-6. A submitted declaration is acceptable.

(2) Do not issue a replacement airworthiness certificate for the scenarios provided for under paragraph 2-4.a(1) of this order for amending an airworthiness certificate.

Note: Replacing an airworthiness certificate is an administrative function and does not require a finding of airworthiness or conformity. When recording the issuance of a replacement airworthiness certificate on FAA Form 8100-1, refer to this note and record that an aircraft inspection was not required. FAA Form 8100-1 must be completed even if the action isn't a full inspection (replacement). Inspectors and designees should always record when an airworthiness certificate is issued, regardless of how routine the situation is. This builds a historical record of the aircraft's certification status over time.

(3) Procedures for Replacing an Airworthiness Certificate. Follow the procedures in paragraph 2-3 of this order except as follows:

(a) Review the Request. FAA Form 8130-6 is not required; a signed statement is acceptable. Inspect the signed statement requesting the replacement certificate to verify that it is from the registered owner, his or her agent, or an operator and includes the aircraft registration number, make, model, and S/N, and a valid reason for replacing the certificate. If an agent signs the letter, it must be notarized and a copy sent to the registry per Appendix B.

(b) Review FAA Airworthiness Records on File at the Registry. A replacement airworthiness certificate may be issued if all the information from the lost or mutilated certificate can be positively established from the maintenance records, the certificate's remains, or the Registry. If there is insufficient data on which to base issuance of the replacement certificate, including the original operating limitations, do not issue the replacement certificate; advise the applicant to apply for a new airworthiness certificate.

(c) Review Aircraft Records. A complete review of aircraft records per paragraph 2-3 e of this order is not required. Review records as necessary to verify that the applicant's request is valid and that the aircraft is eligible for the requested airworthiness certificate. When replacing an airworthiness certificate because the aircraft registration number has changed, review both the registration and, when applicable, Aeronautical Center Form 8050-64, Assignment of Special

Registration Numbers, to verify the aircraft is currently registered and to verify the current N-number.

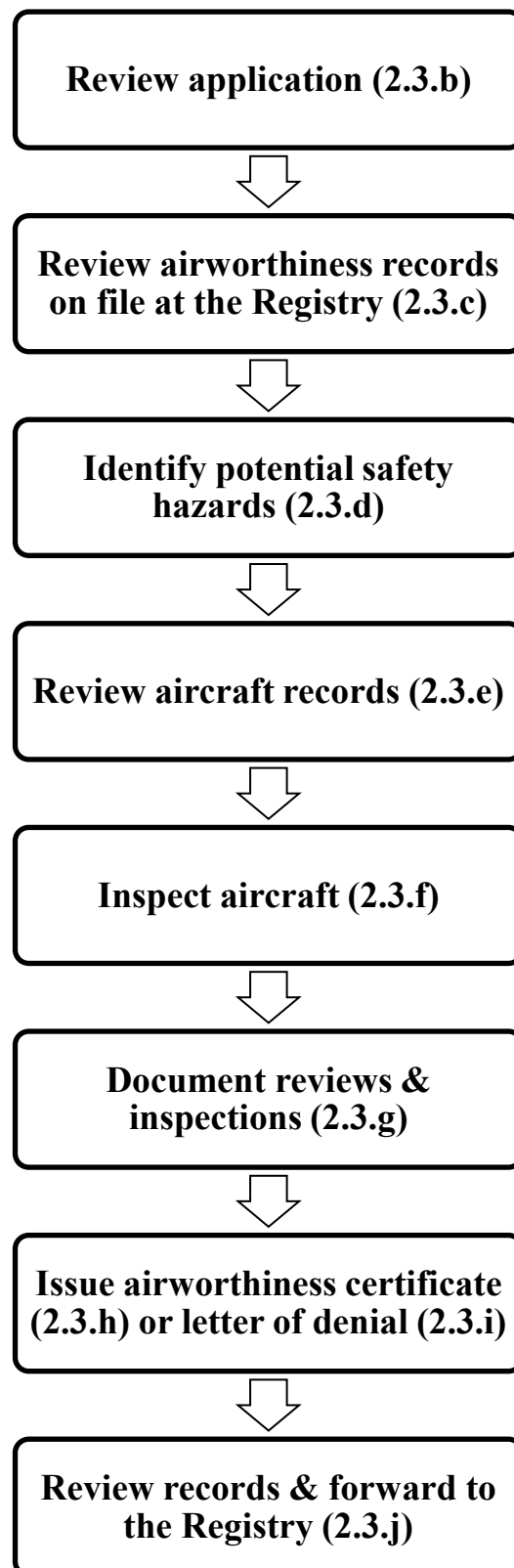
(d)Inspect Aircraft. A complete aircraft inspection per paragraph 2-3 f of this order is not required. Inspect the aircraft as necessary to verify that the applicant's request is valid and that the aircraft is eligible for the requested certificate.

(e)Issue Airworthiness Certificate. When issuing a replacement for a special airworthiness certificate, the operating limitations must match the operating limitations on the original certificate verbatim. Only modifications to the operating limitations for correcting administrative mistakes are allowed. When issuing a replacement for a standard airworthiness certificate that lists an exemption or exemptions in block 5, follow paragraph A-3 e to complete block 5.

d. Surrender. Verify that the aircraft owner's written authorization to surrender the airworthiness certificate states why the certificate is being surrendered. Except for an expired experimental airworthiness certificate, email a digital copy of the authorization and certificate to the FAA Aircraft Registry at Submit.Aircraft.Registry@faa.gov for retention in the permanent airworthiness files. The original should be destroyed after the digital copy is submitted.

e. Transfer. An airworthiness certificate is transferred with the aircraft (§ 21.179), for example, if there is a change of ownership or transfer of registration.

Figure 2-1. Common Steps for Issuing an Airworthiness Certificate



Chapter 3. Issuing Standard Airworthiness Certificates (§ 21.183)

3-1. Introduction. This chapter provides policies and procedures for issuing standard airworthiness certificates under § 21.183 for aircraft in the normal, utility, acrobatic, commuter, or transport category; manned free balloons; and special class aircraft designated by the FAA under § 21.17(b).

3-2. Common Requirements for Reviewing Aircraft Records. In addition to the common requirements for reviewing aircraft records in paragraph 2-3.e of this order, the following requirements also apply to certain airplanes per § 21.183:

Note: This list is not exhaustive. Other requirements may apply and require verification during your review of aircraft records, such as § 91.1505, *Repair assessment for pressurized fuselages*, and § 91.1507, *Fuel tank system inspection program*.

a. Noise. Per § 21.183(e), for an application to issue an original standard airworthiness certificate, review records to verify compliance with the applicable noise requirements of 14 CFR Part 36 or Part 91, subpart I, as applicable.

b. Emergency Exit. Review records for transport category airplanes manufactured after October 16, 1987, to verify compliance with the passenger emergency exit requirements of § 21.183(f).

c. Fuel Venting and Exhaust Emission. Per § 21.183(g), review records to verify compliance with the fuel venting and exhaust emission requirements of Part 34.

3-3. New Aircraft Manufactured Under a PC (§ 21.183(a)). Aircraft manufactured under a PC are eligible for issuing an airworthiness certificate without further showing in accordance with § 21.183(a). The submission of FAA Form 8130-9 is not required. The FAA may inspect each aircraft to determine conformity with the approved type design. The inspection frequency may be adjusted by the geographic CM Section or the CM branch, which has certificate management responsibility over the certificate holder. Consider the following in adjusting inspection frequencies: recent changes in the aircraft design, the quality system, personnel, and the supply base; and recent customer-identified deficiencies, quality escapes, service difficulties, incidents, and accidents. When inspecting an aircraft, follow the applicable procedures in paragraphs 2-3 and 3-2 and the following:

a. Review Aircraft Records.

(1) Review the PC and the production limitation record (PLR) to verify that the holder is authorized to manufacture the applicable aircraft make and model.

(2) Review the manufacturer's records to verify that quality system procedures have been completed satisfactorily, including production flight testing of the aircraft.

b. Reserved.

3-4. New Aircraft Manufactured Under a TC Only (§ 21.183(b)). Follow the common procedures for issuing an airworthiness certificate in paragraphs 2-3 and 3-2 of this order and the following:

a. Review Application.

- (1) Verify that the manufacturer is still authorized to manufacture under a TC since such authorizations are of limited duration.
- (2) Review FAA Form 8130-9, *Statement of Conformity*, to verify proper completion and to verify the applicant certifies the aircraft is airworthy and was flight tested. Refer to FAA Order 8110.4 for additional FAA Form 8130-9 information.

b. Review of Aircraft Records.

- (1) Review the FAA's inspection records for each aircraft to verify that any prior findings have been properly addressed.
- (2) Review the manufacturer's records to verify that any interim quality system procedures have been completed satisfactorily, including production flight testing of the aircraft.
- (3) Verify that any nonconformities involving material review actions have been resolved through the responsible certification branch before issuing an airworthiness certificate.

c. Inspect Aircraft. You must inspect each aircraft presented for issuance of a standard airworthiness certificate to find the aircraft airworthy. Adjust inspections as necessary for any significant changes in manufacturing systems, procedures, or personnel or when changes have been introduced into the aircraft. Inspect the aircraft to verify conformity to its TC, including applicable ADs. This does not imply that every article must be subjected to a conformity inspection. Conformity inspections of articles should only be conducted when, in the FAA's judgment, conformity cannot be substantiated by any other means.

3-5. New Import Aircraft, Foreign SOM (§ 21.183(c)). Follow the procedures for issuing an airworthiness certificate in paragraphs 2-3 and 3-2 and chapter 21 of this order.

3-6. Used Aircraft (§ 21.183(d)). This paragraph applies to issuing a standard airworthiness certificate for a used aircraft, including a surplus aircraft of the U.S. Armed Forces. Follow the common procedures for issuing an airworthiness certificate in paragraphs 2-3 and 3-2 of this order and the following:

a. Review Aircraft Records.

- (1) Review aircraft maintenance records to verify the aircraft was inspected and found airworthy per § 21.183(d)(2) within 60 days before the date of the application. The rule requires the aircraft to have been inspected per the *performance* rules for 100-hour inspections in § 43.15, meaning the use of a checklist that includes at least the scope and detail of appendix D to Part 43. In lieu of a 100-hour type inspection, it is acceptable for the aircraft to have been inspected with an equivalent

performance standard acceptable to the FAA, such as any inspection program prescribed under § 91.409, if inspected per an approved progressive or continuous airworthiness inspection program, this means all inspections must be current. For aircraft with no manufacturer's recommended inspection program, the inspection program must be one approved by the FAA. This inspection program must have the equivalent performance standard defined to be eligible for compliance with any aircraft certificate conformity inspection used to issue an airworthiness certificate. Verify that one of the following persons performed this inspection and found the aircraft airworthy:

- (a) The manufacturer;
- (b) The holder of a repair station certificate issued under 14 CFR Part 145;
- (c) The holder of a mechanic certificate as authorized in 14 CFR Part 65; or
- (d) The holder of a certificate issued under 14 CFR Part 119 and having a maintenance and inspection organization appropriately rated for the type of aircraft involved.

(2) Surplus Aircraft of the U.S. Armed Forces. Review:

- (a) FAA Form 8130-31, *Statement of Conformity – Military Aircraft*, or its predecessor, FAA Form 8130-2, *Conformity Certificate – Military Aircraft*. This form is issued to a military service at the time of manufacture to identify deviations from the FAA TC for a new military aircraft that was derived from an FAA, type-certificated aircraft. Verify that any deviations have been corrected. Refer to FAA Order 8110.101, *Type Certification Procedures For Military Commercial Derivative Aircraft*, concerning the use of these forms.
- (b) Military maintenance records to determine if operating limitations were exceeded; if so, contact the responsible certification branch for an airworthiness assessment of all exceedances.
- (c) Records of engines, gearbox assemblies, landing gear, instruments, or other articles establishing that they originally conformed to the TC and have been maintained per applicable FAA requirements. Military maintenance and/or repair station records may be used for this purpose.

b. Used, Imported Aircraft. For an application for a standard airworthiness certificate for a used, imported aircraft, refer to chapter 21 of this order for additional related policies and procedures.

3-7. Manned Free Balloons. Follow the applicable procedures for issuing a standard airworthiness certificate in paragraphs 3-3 through 3-6 of this order and the following:

a. Inspect Aircraft. Inspect the aircraft to verify that the applicant is presenting a complete aircraft for issuance of a standard airworthiness certificate. For a manned free balloon, a complete aircraft consists of an envelope with an eligible burner, fuel system, and basket; verify that the combination is allowed per the TCDS. You may not issue a standard airworthiness certificate for an article only.

b. Delivery of New Envelopes. Manufacturers of manned free balloons may deliver a balloon envelope when the envelope is the only article ordered. A manufactured balloon envelope assembled to a burner and basket combination authorized by the TC holder and flight tested is eligible for a standard airworthiness certificate. The envelope, along with the standard airworthiness certificate and the logbook, may be delivered without the burner and basket. The envelope may then be assembled to a different burner and basket per the TC. A person may interchange the burner and basket as a preventive maintenance task described below.

c. Changing the Burner and Basket. A new airworthiness certificate is not required when the aircraft is disassembled, and a different burner and basket combination is installed, as the TC allows. Reassembly of the envelope and bottom-end articles into a complete aircraft may be performed as preventive maintenance under 14 CFR Part 43, appendix A, paragraph (c)(27). The aircraft records must properly reflect the installation of the bottom-end articles and record the new empty weight. The bottom-end components must be in a current “annual or 100-hour” inspection status. The individual records of the bottom-end articles must be maintained.

d. Replacement Envelopes. If an envelope is provided only as a replacement article without obtaining a new aircraft S/N, registration number, or ID plate, the installation of the replacement envelope is a maintenance item under Part 43. This requires appropriate documentation of the work performed and a return to service entry in the aircraft records by a person authorized to perform the maintenance. The aircraft ID plate, S/N, and registration number are carried over from the previous aircraft envelope.

Chapter 4. Issuing Special Airworthiness Certificates

Section 1. General

4-1 Introduction.

- a. This section provides common procedures for issuing a special airworthiness certificate but does not apply to SFPs.
- b. Additional policies and procedures specific to certain aircraft categories, experimental purposes, and SFPs are provided in subsequent chapters of this order.

4-2. Common Procedures for Issuing a Special Airworthiness Certificate. Follow the common procedures in paragraph 2-3 of this order for issuing an airworthiness certificate and appendix D to this order for issuing operating limitations. As applicable, include paragraph 4-4 in record reviews and the aircraft inspection.

4-3. Flight Operations Outside the United States. A special airworthiness certificate does not authorize flight operations over a foreign country. However, the FAA may issue the certificate when the applicant intends to operate the aircraft over other countries if the applicant meets the requirements for that certificate. A U.S. special airworthiness certificate is often helpful to a foreign civil aviation authority ([CAA](#)) when issuing permission to operate the aircraft in or over its country. If the FAA knows the affected [CAA](#) will not permit the operation of the aircraft in or over its country, the FAA cannot deny issuance of the certificate if the applicant has met the requirements for that certificate. In any case, an ASI should make the [applicable CAA\(s\)](#) aware of the aircraft, its category or experimental purpose, and the operating limitations for the aircraft.

4-4. Aircraft Equipped with Explosive Devices or Jettisonable Stores. These systems are usually associated with former-military or certain restricted category aircraft.

- a. **Maintenance of Jettisonable Stores.** Verify that the applicant's inspection program includes maintenance and inspection of jettisonable stores and associated articles.
- b. **Ejection Seat Systems.** Verify that the ejection seat system has been approved for use in the aircraft model by the aircraft manufacturer or by a military service.
- c. **Procedures for Securing Aircraft.** Verify that the applicant has provisions and procedures for securing the aircraft whenever it is parked to prevent inadvertent operations of a system that uses an explosive device.
- d. **Airport Notification.** Verify that the applicant has notified the airport manager or sponsor concerning the presence of jettisonable stores, any explosive devices, and planned operations of the aircraft from that airport.
- e. **Marks for Explosive Devices.** Verify markings on the exterior of the aircraft indicating the aircraft is equipped with permanently installed explosive devices, e.g., explosive devices in ejection seats, ballistic parachutes, and jettisonable systems.

f. Operations with Jettisonable Stores. Except for certain restricted category aircraft (per the applicable restricted type certificate), only aircraft with an experimental airworthiness certificate for R&D may be operated with a jettisonable store, e.g., external fuel tanks and special weapons, and only for a test that requires jettisoning that store. Limited category aircraft and aircraft certificated for other experimental purposes may carry jettisonable stores or externally mounted equipment if mounted in such a manner that in-flight jettison is not possible. See applicable operating limitations in appendix D for additional information.

Note: MIL-STD-8591 defines two categories of stores: ejected and expendable stores. Both ejected and expendable stores are considered jettisonable for the purposes of this order. Ejected stores are released and pushed away from the aircraft laterally or downward, typically by the action of a cartridge-activated ejector, such as a flare. An expendable store is a store that is normally separated from an aircraft in flight, such as a fuel tank, missile, rocket, bomb, mine, torpedo, pyrotechnic device, sonobuoy, signal underwater sound device, cargo drop container, drone, and other similar items.

Section 2. Experimental Airworthiness Certificates

4-5 Introduction. This section provides common policies and procedures for issuing special airworthiness certificates for experimental purposes.

4-6 Procedures for Issuing an Experimental Airworthiness Certificate (§ 21.191).

a. Review Application. Follow the procedures in section 1 of this chapter and the following:

(1) Verify the program letter meets § 21.193 (see Appendix C of this order for additional guidance):

(a) Verify the program letter clearly describes how the aircraft will be used and that the operation meets a § 21.191 purpose. After demonstrating compliance with § 91.319(b)(1) and (2), the aircraft can be used to support other aircraft in the program or other experimental programs the applicant has underway; if applicable, verify that such support activities are described in the program letter.

Note: Except as provided under §§ 21.191(f) and 21.195, brokering or marketing of experimental aircraft is not a valid experimental purpose; this includes an individual who manufactures, imports, or assembles an aircraft and then applies for an experimental airworthiness certificate to help sell the aircraft.

(b) Verify the program letter provides enough information to describe the operation, equipment, or test, as applicable for the duration of the certificate. Verify that the level of detail in the program letter is sufficient to enable the FAA to prescribe all appropriate operating limitations. Details include, but are not limited to, gross takeoff weight, ejection seat status (operational or not), and a sufficient description so that a review for modifications can determine the aircraft's configuration, including aircraft previously modified for IFR.

(c) Verify the program letter contains the estimated time or number of flights required for the operation for those experimental purposes specified in § 21.191(a) through (f).

(d) Verify the program letter describes the areas over which the applicant will conduct operations (see paragraph 4-7).

(e) Verify the program letter contains enough data to identify the aircraft, such as the inclusion of registration and serial numbers that match those on the aircraft.

(f) Unless converted from a previously type-certificated aircraft without an appreciable (large or noticeable enough to be measured or observed) change in its external configuration, verify the program letter includes three-view drawings or three-view dimensioned photographs of the aircraft.

Note: If any information in the program letter is marked proprietary, confidential, or company-sensitive, or subject to the Uniform Trade Secrets Act, International Traffic in Arms Regulations (ITAR) or Export Administration Regulations (EAR); or contains any other markings indicating the document is restricted from public or foreign disclosure, return these documents to the applicant and ask them to replace them with information that would not require restrictive labels or restrict disclosure. Do not send documents with such restrictions to the Aircraft Registration Branch.(g) Chapter 17 contains additional information for aircraft certificated for the experimental purpose of operating light-sport category kit-built aircraft.

(2) Multiple Purposes. If the applicant is seeking an experimental airworthiness certificate for multiple purposes, verify the program letter clearly documents all the items listed in paragraph 4-6.a(1) of this order separately for each purpose. In addition, verify the program letter describes any required configuration changes when changing between purposes, including adding or removing equipment and enabling or disabling systems. Configuration changes for different experimental purposes are addressed in appendix D. Configuration changes should be included in the aircraft inspection program. Refer to paragraph 4-8 of this order for additional policies and procedures concerning multiple experimental purposes.

b. Inspect Aircraft. Follow the procedures in section 1 of this chapter and the following:

- (1) Inspect the aircraft to verify that the aircraft is marked “Experimental.”
- (2) Per § 21.193(g), upon inspecting the aircraft, obtain any pertinent information from the applicant as necessary to safeguard the general public.
- (3) Determine if the aircraft is airworthy by finding it in a condition for safe operation.

c. Issue Airworthiness Certificate. Follow the procedures in paragraph 2-3.h.

4-7. Areas of Operation. Per § 21.193(d), applicants must submit a description of the areas over which flights will be conducted. Prior to demonstrating compliance with § 91.319(b), no person may operate an aircraft that has an experimental certificate outside of an area assigned by the Administrator until it is shown that the aircraft is controllable throughout its normal range of speeds and throughout all the maneuvers to be executed; and the aircraft has no hazardous operating characteristics or design features. To enable the Administrator to assign an area, the applicant should specify the area that will be used to show compliance with § 91.319(b). Prior to demonstrating compliance with § 91.319(b), the area must be compliant with § 91.305. Inspectors must evaluate the applicant’s area of operation to determine whether it is suitable for the type of operations described in the program letter and for the experimental purpose. If the area is not suitable, inspectors must inform the applicant. Inspectors may ask the applicant for more information or deny the certificate.

Note: Table D-1 contains an operating limitation for Production Certificate holders which may allow takeoffs and landings outside of § 91.305 areas in certain scenarios. Refer to Table D-1 to determine eligibility.

4-8. Multiple-Purpose Experimental Airworthiness Certificates.

An experimental airworthiness certificate may be issued for multiple purposes under § 21.191.

Note: The issuance of multiple-purpose certificates for R&D and showing compliance with regulations should be limited to PC holders; however, this may be extended to modifiers (§ 21.195(c)) when justified.

Note: Multiple certifications for a restricted category aircraft under § 21.187 are not authorized in combination with an experimental airworthiness certificate, as explained in chapter 7 of this order.

a. Options. Use discretion in determining the most appropriate option for the intended use. Available options include issuing individual airworthiness certificates for each experimental purpose, issuing a multi-purpose airworthiness certificate, or limiting airworthiness certificate issuance to one experimental purpose at a time. Including experimental purposes with different durations (§ 21.181) on the same airworthiness certificate is not recommended due to the added burden on the applicant and the FAA of having to re-issue the certificate for the purpose which, otherwise, could have had a longer or unlimited duration. However, if the certificating office determines the aircraft is eligible for multiple-purposes and chooses to put multiple purposes with different durations on the same airworthiness certificate, the experimental purpose with the shortest duration per § 21.181 must be used for the expiry date on the FAA Form 8130-7 to remain in compliance with § 21.181.

b. Configuration Changes. Consider how the aircraft configuration may change when switching between purposes or certificates. This information should be included in the program letter, as it may affect the operating limitations issued and could require additional inspections when changing purposes or certificates. If more than one purpose is requested, verify the operator has adequate maintenance, inspection, and configuration change procedures to ensure the aircraft is in a condition for safe operation when changing purposes.

Example 1: An applicant holding an experimental airworthiness certificate for the exhibition purpose is seeking an additional experimental airworthiness certificate purpose to operate the aircraft as a chase plane in support of an R&D project. An experimental airworthiness certificate for the R&D purpose may be issued to the applicant and both certificates may be held simultaneously. Once the R&D project is completed, no further action is needed as the exhibition certificate remains valid due to its unlimited duration.

Example 2: An applicant for an experimental airworthiness certificate for operating an amateur-built aircraft indicates an intention to race the aircraft. In this case, a single certificate listing both purposes may be appropriate.

4-9. Inspection Programs for Certain Experimental Aircraft.

This paragraph applies to former military experimental aircraft that are turbine-powered. Advise the applicant to consider FAA AC 43-209, Recommended Inspection Procedures for Former Military Aircraft, in developing its inspection program.

a. Definitions. Advise the applicant that following definitions are terms that may be used in the development of inspection programs for these aircraft.

(1) Overhaul. Methods, techniques, or practices for disassembling, cleaning, inspecting, repairing as necessary, reassembling, and testing in accordance with approved standards and technical data acceptable to the Administrator. Overhaul should not be confused with life limit.

(2) Life Limit. The finite/retirement time assigned to a component that requires the removal of that component from service.

(3) Shelf Life. A recommended time determined by the manufacturer for removal of a component from service.

b. Inspection Program Submittal Requirements. The applicant should submit the following material for review:

(1) Proposed Inspection Program. The submitted program for an aircraft may be a current manufacturer's program, a current military program (preferably North Atlantic Treaty Organization (NATO)), an owner/operator-developed program, or based on a program previously approved for the same make/model. Prior FAA approval of an inspection program does not guarantee an automatic approval for a similar make/model because inspection programs are aircraft-specific and will be identified by the aircraft S/N. Inspection programs are subject to amendment whenever significant changes in operating environment and/or equipment occur.

(2) Operable Ejection Seats. The inspection program for operable ejection seats will be based on a current manufacturer's or current military program, and will include replacement intervals for shelf-life components such as pyrotechnic cartridges. Shelf-life intervals cannot be extended without the manufacturer's approval.

(3) Required Manufacturer/Military Manuals. If the manuals were not originally published in English, the applicant will submit an English translation of the original manuals. It is to the applicant's benefit to ensure the translation is performed by a technically competent individual familiar with aviation terms and practices.

(4) Substituting Materials or Replacement Parts. Changes involving the substitution of materials or replacement parts should be in accordance with FAA-accepted procedures or a recognized industry standard, or based on dimensions and technical data provided by the manufacturer or information provided by an appropriate engineering evaluation. Life-limited articles specified in the applicable technical publications pertaining to the aircraft and its articles are complied with in one of the following manners:

(a) Type-Certificated Products. Replacement of life-limited parts required by § 91.409(e) is only applicable to experimental aircraft when the required replacement times are specified in the U.S. aircraft specifications or TCDS.

(b) Non-Type-Certificated Products. Unless otherwise determined by the FAA, all articles installed in non-type-certificated products operated in the experimental category, in which the manufacturer has specified limits, must include in their program an equivalent level of safety for those articles. Although the FAA recommends adherence to part replacements, achieving an equivalent level of safety for non-type-certificated products is acceptable. The article must be inspected to ensure the equivalent level of safety still renders the product in a serviceable condition for safe operation.

(5) Extension of Component Life Limits. (reference operating limitation #25)

(a) The applicant may submit data, with a request to the FSDO, to extend the life limit on specific components of the aircraft beyond the manufacturer, military, and/or technical order recommended life limits. At a minimum, the data submitted by the applicant should contain:

- (1) The original strength, stress, and fatigue data for the aircraft and the pertinent parts, including other parts that may be affected by changes of the life limits and inspection intervals;
- (2) The methodology the designers used while developing the life limits and inspection intervals,
- (3) The operational history of the aircraft and parts (usage affects life limits and inspection intervals);
- (4) The service history of the aircraft and pertinent parts, including any repairs and modifications affecting the strength, stress, and fatigue characteristics of the parts and their effects on the parts' life limits and inspection intervals;
- (5) How the present operational usage differs from prior military usage;
- (6) Evidence that the applicant's inspection/testing techniques, for example, nondestructive inspection/nondestructive testing, are comparable to those used by the military;
- (7) Evidence that the methodology chosen by the owner (for example, damage tolerance with inspections versus safe-life with automatic removal) produces at least as safe a product as the military's approach.
- (8) A procedure to inspect the component to some appropriate physical standard, and non-destructive testing, where applicable.

(b) In cases where the data listed above is unavailable or cannot be substantiated, the components will not be eligible for any extension of life limits.

(6) Extension of Component Recommended Overhaul Times. The owner/operator may elect to continue in service any component that has reached its recommended overhaul time if an approved inspection is implemented that includes a procedure to inspect the component to an appropriate physical standard with a definitive time period for review. Testing to the standard may be accomplished in place where practicable. This inspection procedure will be submitted to the local FSDO to be included in the FAA-approved inspection program. (reference operating limitation #25)

c. Inspection Program Content.

(1) The owner/operator-developed inspection program presented for FAA approval should reference specific details from the appropriate military/manufacturer's manuals while encompassing the scope and detail of Part 43, appendix D, as appropriate.

(2) As an alternative, a military/manufacturer's inspection program may be adopted and presented for FAA approval. Specific irrelevant sections may be deleted for aircraft systems that have been removed or deactivated. However, all inspection programs should provide for a complete inspection of the aircraft within the preceding 12 calendar months.

(3) The following items should be a part of an approved inspection program:

(a) Title page that includes the aircraft manufacturer's name and the aircraft model, serial number, and registration number to which the inspection program applies; and the owner/operator's name and address.

(b) Table of contents.

(c) Log of revisions.

(d) Method of revision.

(e) List of effective pages.

(f) Introduction that includes the following:

1. A description of the inspection program with references to sections and supporting documents. These references may include standards of performance, procedures, methods, instructions, or other technical data. If section references are not specified by title, page, and revision, the referenced documents in their entirety become a part of the inspection program.

2. A statement that this inspection will be performed to ensure the aircraft is in a condition for safe operation and the inspection is performed in accordance with the procedures of the program.

3. Identification of the individual responsible for scheduling and performing the inspections, including their name and address.

4. A listing of the specific maintenance/inspection manuals for the make and model of the aircraft being certificated.

(g) Program-unique definitions and/or acronyms.

(h) A replacement schedule of life-limited/retirement items, if applicable.

(i) Procedures to ensure inspection records are kept and include:

1. Date of inspection,
2. Name and certificate number of the person performing the inspection,
3. Type of inspection, and
4. Total time of the component being inspected expressed in cycles, calendar time, hours, or any combination of these.

(j) Instructions and procedures for the conduct of inspections for the particular make and model of aircraft, including necessary tests and checks. The instructions and procedures must set forth in detail the parts and areas of the airframe, engines, propellers, and appliances required to be inspected, including survival and emergency equipment.

(k) A schedule for performing the required inspections under the program as expressed in terms of time-in-service, calendar time, number of system operations, or any combination of these. It should also include low-utilization inspections.

(l) Additional procedures, including—

1. Special inspections such as short-and long-term storage/out-of-service inspections, hard landing inspections, and structural inspections;
2. Compliance with service letters, service bulletins, time compliance technical orders, and ADs, as well as the method to ensure compliance;
3. Corrosion inspections; and
4. Any other inspection that may be required due to unusual environmental operations or aircraft design, such as composite construction.

d. Inspection Program Approval. Remind the owner/ operator to submit the inspection program to the local FSDO where the aircraft will be based. The approval of the inspection program is granted by a stamp of approval and the principal maintenance inspector/principal avionics inspector signature on the list of effective pages (LEP), or by some other official means of conveying approval.

e. Maintenance Requirements. Remind the owner/operator of the aircraft they will have the inspections performed as prescribed in the approved program and will, between inspections, have discrepancies repaired per the appropriate manufacturer/military manuals, instructions, and technical orders.

f. Maintenance Records. Remind the owner/operator they shall keep the records required under § 91.417 as applicable to the aircraft.

4-10 Former Military Aircraft Operated for Experimental Purposes.

a. Advising the Applicant. Advise the applicant to review the following FAA ACs.

- (1) FAA AC 21-54, *Experimental Airworthiness Certification of Certain Former Military Aircraft*, including guidance on program letters.
- (2) FAA AC 43-209, *Recommended Inspection Procedures for Former Military Aircraft*.

b. Identify Potential Safety Hazards. Given the variety of aircraft types, models, alterations, operational histories, and airworthiness ramifications of possible long-term storage, becoming familiar with the safety hazards of former military aircraft is especially important. Additional considerations include:

- (1) Identify the aircraft model and/or series and the type of engine(s), propellers, and other systems installed, as applicable. Obtain as much historical information as possible, including S/Ns, aircraft, and component life limits, overhaul dates, airframe cycles, and engine time and cycles.
- (2) Review the ownership history of the aircraft. This may provide information on how the aircraft was previously operated and maintained, which may have implications for the airworthiness inspection.
- (3) Become familiar with the scope of any restoration, repairs, and maintenance conducted by or for the applicant. It is also helpful to become familiar with the general condition of working/storage areas, the availability of spare parts, and equipment before conducting the formal records inspection.
- (4) Ensure the operator has, in English, a complete set of the applicable military flight, inspection, and maintenance manuals for the aircraft and inspection and maintenance manuals for the engine.
- (5) Ensure the operator has applicable military technical orders, in English, to address known issues related to airworthiness, maintenance, and servicing.
- (6) Identify any high-risk factors associated with the aircraft's design, manufacture, maintenance, and operation. Verify that the applicant adequately mitigates any high-risk factors. All mitigations must be added as an operating limitation using the process described in paragraph D-5.

c. Import Documentation for Former Military Aircraft.

(1) Records for imported former military aircraft that may have weapons installed should include the following documents issued by the [Department of Justice, Bureau of Alcohol, Tobacco, Firearms, and Explosives \(ATF\)](#):

- (a) ATF Form 6, *Application and Permit for Importation of Firearms, Ammunition, and Defense Articles*.
- (b) ATF Form 6A, *Release and Receipt of Imported Firearms, Ammunition, and Defense Articles*.

Note: If you have any questions regarding ATF Forms 6 or 6A, contact the ATF Firearms and Explosives Imports Branch.

(2) A product that was declared as parts, scrap, or for museum display on any documentation related to importing that product is not eligible for an airworthiness certificate. This includes documents issued by or for ATF, Customs and Border Protection, or the Department of State.

d. Demilitarization of Former-Military Aircraft.

(1) Removing or Disabling Systems. The FAA may require that some systems be removed or disabled to establish a condition for safe operation for the intended use (e.g., guns, cannons, targeting radars, electronic jammers, jettisonable stores, or explosive devices). Potential safety hazards related to these systems include accidental firing of guns, accidental release of stores, accidental operation of radars on the ground, compartment fires, and damage to the airframe. These hazards may pose risks to other aircraft and to persons and property on the ground and may not be able to be adequately mitigated via operating limitations. Although some of these systems may be required to support a valid R&D purpose, the applicant and the FAA need to understand the corresponding safety risks and work closely together to determine the feasibility of mitigating these risks via operating limitations.

(2) Additional Information on Weapons. For additional information on aircraft weapons systems, refer to aircraft maintenance manuals and the related weapons delivery manual. Some aircraft manufacturers may offer guidance on aircraft demilitarization. [U.S. Air Force Technical Order 00-80G-1, Make Safe Procedures for Public Static Display](#), can be used as a reference.

e. Ejection Seats.

(1) An inspection program for operable ejection seats will be based on a current manufacturer or military program and will include replacement intervals for life-limited or shelf-life-limited components such as pyrotechnic cartridges. These limits cannot be extended without the manufacturer's approval.

(2) The PIC and flight crew operating aircraft equipped with operational ejection seat(s), must have satisfactorily completed an ejection seat training program per FAA AC 91-87, *Ejection Seat Training Programs*, within the past 24 calendar months prior to operation for this make and model of aircraft.

(3) For questions concerning specific aircraft and ejection seat training, contact the responsible Flight Standards Office or AFS-830.

f. Drag Chutes.

As applicable, verify that the drag chute installation records reflect installation per applicable military installation requirements. Verify that the applicant's inspection program includes maintenance and inspection of the aircraft's drag chute.

g. Jettisonable Stores. Verify that the applicant's inspection program includes maintenance and inspection of jettisonable stores and associated articles.

h. Military Aircraft Structural Integrity Program Requirements: Former military aircraft from the Armed Forces of the U.S. have aircraft structural integrity program requirements, which are similar the FAA's Continued Airworthiness and Safety Improvement regulations (14 CFR Parts: 91 subpart L, 121 subpart AA, 125 subpart M). U.S. Armed Forces aircraft are required to be inspected and maintained to the military inspection requirement(s) that MIL-STD-1530D established. Foreign former military aircraft may also have similar ASIP requirements.

Former military aircraft that are turbine powered airplanes with a Maximum Takeoff Weight of greater than 100,000lbs will be required to include the military aircraft structural integrity program (ASIP) requirements (or similar requirements) in the aircraft inspection program (AIP) and comply with those requirements.

FAA ASIs will review the military technical publications that are applicable to the former military aircraft and validate the military ASIP requirements stated in those military documents are included in the aircraft inspection program (AIP).

4-11. Pioneer Era Aircraft. Replica, reproduction, restored, and similar aircraft based on aircraft from the pioneer era, before the year 1914, require special consideration. Many of these aircraft have limited maneuverability and are only capable of flying for very short distances. An application for an airworthiness certificate for these aircraft should be coordinated with the Organization & System Policy Branch (AIR-630) and the General Aviation and Commercial Division (AFS-800) by emailing the [coordination mailbox \(9-AVS-AIR-81302OPSLIMITS@faa.gov\)](mailto:9-AVS-AIR-81302OPSLIMITS@faa.gov) to ensure the operating limitations are appropriate for the intended operations.

4-12 Space Support Vehicle Flights. An aircraft intended for operation as a space support vehicle may be issued a special airworthiness certificate for one or more experimental purposes under § 21.191 and may be used for carrying persons or property for compensation and hire to simulate space flight conditions per § 91.331. There is no purpose under § 21.191 specific to space support vehicle flights, and, therefore, the space support vehicle flights can be conducted under an appropriate experimental purpose, such as research and development or crew training, as long as the aircraft conducting a space support vehicle flight meets § 91.331(a)(2). The launch license database can be accessed at [Commercial Space Data](#).

Chapter 5. Primary Category (§ 21.184)

5-1. Introduction. This chapter provides policies and procedures for issuing a special airworthiness certificate for a primary category aircraft under § 21.184.

5-2. General Requirements. Follow the procedures in chapter 4, section 1 of this order and the following:

a. Review Aircraft Records. Verify that any special inspection and preventive maintenance program for the applicable aircraft make and model was accepted by the FAA as follows:

- (1) For airplanes, acceptance by the AED (MKC-AED) with engineering input by the certification branch where the TC application was made.
- (2) For rotorcraft, acceptance by the AED (FTW-AED) with engineering input by the certification branch where the TC application was made.

b. Issue Airworthiness Certificate. Advise the applicant that only a properly qualified pilot/owner may perform preventive maintenance under the special inspection and preventive maintenance program. To be properly qualified, a pilot/owner must successfully complete an FAA-approved course given by an approved aviation maintenance technician school, the PC holder for the pilot/owner's aircraft or another entity approved by the FAA.

5-3. New Aircraft Manufactured Under a PC (§ 21.184(a)). Follow this order's procedures in paragraphs 3-3 and 5-2.

5-4. New Imported Aircraft (§ 21.184(b)). Follow the procedures in paragraph 5-2 and chapter 21 of this order.

5-5. Aircraft Having a Current Standard Airworthiness Certificate (§ 21.184(c)). As provided for under § 21.184(c), an applicant may exchange a current standard airworthiness certificate for a special airworthiness certificate in the primary category if the aircraft meets § 21.24(a)(1). Note that conversion from standard to primary categories allows an owner/pilot to perform preventive maintenance. The conversion will be made via STC. To issue a special airworthiness certificate under § 21.184(c), follow the procedures in paragraph 5-2 of this order and the following:

a. Issue Airworthiness Certificate.

- (1) Obtain the standard airworthiness certificate from the applicant.
- (2) Advise the applicant that its aircraft cannot be returned to a standard airworthiness certificate without showing that it meets all the requirements for a standard airworthiness certificate as prescribed by the regulations. Such a showing historically has been difficult when an aircraft has remained in a different classification or category for a lengthy period. To facilitate the return to a standard airworthiness certificate, the aircraft records should indicate, among other requirements, that the aircraft has been maintained according to the manufacturer's instructions and that any alterations to the aircraft were either removed or approved by the FAA.

b. Reserved.

5-6. Other Aircraft (§ 21.184(d)). For an application for a special airworthiness certificate in the primary category for an aircraft that meets the criteria of § 21.24(a)(1) and is not covered by paragraphs 5-3, 5-4, or 5-5 of this order, follow the procedures of paragraph 5-2 of this order and the following:

a. Review Aircraft Records. Review applicable records to verify:

(1) The aircraft conforms to an approved primary, normal, utility, or acrobatic type certificate, including compliance with all applicable ADs.

(2) The aircraft has been inspected and found airworthy within the past 12 calendar months per § 91.409(a)(1).

b. Reserved.

Chapter 6. Restricted Category (§ 21.185)

6-1. Introduction. This chapter provides policies and procedures for issuing a special airworthiness certificate for a restricted category aircraft under § 21.185.

6-2. Common Requirements. Follow the procedures in chapter 4, section 1 of this order and the following:

a. Review Application.

(1) Verify that the restricted category use and special purpose operation is indicated in section II, block B.3 of the application. Refer to FAA Order 8110.56, Restricted Category Type Certification, for the current listing of all special purpose operations approved under § 21.25(b).

(2) If an applicant wishes to obtain approval for a new special purpose operation, refer to FAA Order 8110.56 for guidance in proposing new special purpose operations.

(3) Verify the aircraft is type-certificated in the restricted category under § 21.25, CAR 8, or § 21.29 or previously type-certificated in another category and modified for the requested special purpose operation(s).

(a) In the case of an aircraft previously type-certificated in another category and modified for a restricted special purpose operation under § 21.25 or CAR 8, the previously approved TC and the STC or approved data can be considered as the equivalent of a restricted TC. The TC and STC, or approved design data, should define the design parameters that make up the restricted category TCDS

(b) Non-U.S.-manufactured aircraft type-certificated in any other category under § 21.29 are not eligible for certification in the restricted category unless the aircraft was issued a new or amended TC under § 21.29 or an STC under § 21.25 and subsequently was modified per the TC or STC. In this instance, § 21.185(b) is the basis for issuing the restricted airworthiness certificate.

(4) Verify that the special purpose operation(s) indicated on the application is listed in the TCDS certification basis or approved by an installed STC.

b. Review Aircraft Records. Per § 21.185(d), review records to verify compliance with the applicable noise requirements of Part 36.

c. Issue Airworthiness certificate.

(1) Operating Limitations. If necessary, immediately preceding the first operating limitation, list the applicable use(s) and special purpose operation(s) not listed in the Purpose box of the certificate as follows: “In addition to the Restricted use(s) listed in the Purpose box, this special airworthiness certificate is issued for the following Restricted category use(s) and special purpose operation(s):” “[enter Restricted use]: [enter only the approved special purpose operation(s)].”

(2) Reserved.

6-3. New Aircraft Manufactured as Restricted Category (§ 21.185(a)).

- a. Under a PC.** Follow this order's procedures in paragraphs 3-3 and 6-2.
- b. Under a TC.** Follow this order's procedures in paragraphs 3-4 and 6-2.

6-4. Aircraft Used by or Has a Service History with the U.S. Armed Forces (§ 21.185(b)). Follow the procedures in paragraph 6-2 of this order.**6-5. Aircraft Previously TC'd in Another Category (§ 21.185(b)).** For modified aircraft that were either former military aircraft of a U.S. Armed Force or previously type-certificated in another category and modified for a restricted special purpose operation under § 21.25 or CAR 8, the previously approved TC and the STC or approved data can be considered as the equivalent of a restricted TC. The TC and STC or approved design data should define the design parameters that make up the restricted category TCDS. Follow the procedures in paragraph 6-2 of this order, and verify the following:

- a. The aircraft is:** Type certificated for a special purpose operation in the restricted category.
- b. The aircraft was:**
 - (1) Manufactured in accordance with the requirements of, and accepted for use by, the U.S. Armed Forces and has a service history with the U.S. Armed Forces acceptable to the FAA; or
 - (2) Previously type certificated in another category; and
- c. The aircraft has been:** inspected by the FAA and found to be in a good state of preservation and repair and in a condition for safe operation.

6-6. Recurrent Certificates for Restricted Category Aircraft (§ 21.185(a) or (b)). Follow the procedures in paragraph 6-2 of this order.**6-7. Import Aircraft (§ 21.185(c)).** Follow the procedures in paragraph 6-2 and chapter 21 of this order.

Chapter 7. Issue of Multiple Airworthiness Certifications for Restricted Category Aircraft (§ 21.187)

7-1. Introduction. This chapter provides policies and procedures for issuing multiple airworthiness certificates for a restricted category aircraft under § 21.187.

Note: Issuing multiple certificates for a restricted category aircraft under § 21.187 is different than issuing an experimental airworthiness certificate with multiple purposes or issuing multiple experimental airworthiness certificates for multiple purposes. Issuance of multiple certificates and a multi-purpose certificate for experimental aircraft is covered by paragraph 4-8 of this order.

7-2. Procedures for Issuing Multiple Airworthiness Certificates for an Aircraft.

The procedures for issuing multiple airworthiness certificates are a combination of the procedures covering standard (chapter 3 of this order) and restricted (chapter 6 of this order) airworthiness certificates, or limited (chapter 8 of this order) and restricted airworthiness certificates, and the following:

a. Review Application.

(1) Determine if the applicant needs one or more certificates. If an aircraft already holds an eligible airworthiness certificate under § 21.187, it is unnecessary to reissue that certificate when exercising the provisions of § 21.187 for the issuance of multiple airworthiness certificates for a restricted category aircraft. Issue only the certificate needed. For example, only the restricted category certificate can be issued if the aircraft already holds a standard airworthiness certificate. AWC includes an instructional aid that provides supplemental information on the issuance of a special airworthiness certificate while allowing the applicant to retain a current standard airworthiness certificate when applying for multiple certificates for restricted category aircraft. Ensure section II.D of FAA Form 8130-6 is annotated to identify the eligible airworthiness certificate the aircraft currently holds and will retain.

(2) Verify the aircraft is eligible –

(a) For multiple certificates for the requested aircraft categories. An aircraft in the restricted category may be eligible for multiple airworthiness certificates if it can be converted to the normal, utility, acrobatic, commuter, transport, or limited category by removing or adding equipment by simple means. This provision does not extend to an aircraft in the primary or light sport category, or to an aircraft issued a certificate for an experimental purpose.

(b) By S/N for both categories according to the applicable TCDSs or STC.

b. Review of Aircraft Records. Determine if operations in the restricted category would exceed the operating limitations of the other aircraft category. If so, the aircraft may not be eligible for a multiple airworthiness certificate. Any operations outside of the other category operating limitations while operating in the restricted category, unless approved for that aircraft, may make it impossible to return the aircraft to the other category unless a complete engineering evaluation is made. Therefore, to retain eligibility for return to the other category after being operated in the restricted category, the following applies:

(1) While being operated in the restricted category, any changes made to the aircraft that are to be retained when returning to the other category, or any operations that are outside of the operating limitations of the other category, must be approved per the regulations and procedures applicable to the other category.

(2) If the maximum gross weight or operating limitations for the restricted category are higher than those for the other category, the aircraft is not eligible for operation in the other category after having been operated in the restricted category unless—

(a) The TCDS specifically states that the aircraft is eligible for operation in the other category after having been operated at the limitations applicable to the restricted category; or

(b) If the TCDS does not have such a note or any other reference, the operations outside of the other category operating limitations, including increased gross weights, must be FAA-approved.

c. Inspect Aircraft.

(1) Witness the applicant's method of compliance with §§ 21.187(a)(1) and 21.187(a)(2) and determine if the instructions for converting the aircraft from one category to the other are adequate.

(2) Reserved.

d. Issue Airworthiness Certificate(s).

(1) If the application is for restricted and limited categories, issue FAA Form 8130-7 with appropriate operating limitations for each category. If the application is for the restricted category and a category in the standard classification, issue FAA Form 8100-2 for the standard classification and FAA Form 8130-7 with appropriate conditions and operating limitations for the restricted category. If the aircraft already holds one of the certificates, only issue the certificate that is needed.

(2) If one of the multiple certificates is a standard airworthiness certificate, and the aircraft will be used for the carriage of passengers for compensation or hire in the standard configuration, evaluate the restricted special purpose operation to determine whether the airworthiness inspection prescribed in § 21.187(b) will be required each time the aircraft is converted from a restricted airworthiness certificate to a standard airworthiness certificate. Normally, if the special purpose operation involves carriage of maximum loads or is subject to contamination by pesticides, herbicides, corrosives, or other hazardous materials, the airworthiness inspection must be required, and an operating limitation to this effect should be prescribed. In such cases, specify in that operating limitation whether this airworthiness inspection must be conducted by the FAA or an appropriately certificated mechanic.

Chapter 8. Limited Category (§ 21.189)

8-1. Introduction. This chapter provides policies and procedures for issuing a special airworthiness certificate for an aircraft in the limited category under § 21.189.

8-2. Procedures for Issuing an Airworthiness Certificate. Follow the procedures in chapter 4, section 1 of this order and the following:

a. Review Aircraft Records.

(1) Review the TCDS, aircraft specifications, and/or applicable aircraft listing to verify that the aircraft is eligible for issuance of a special airworthiness certificate in the limited category. Refer to Table 8-1 of this chapter for aircraft make and models with TCs in the limited category. If you are unsure whether an aircraft is eligible according to its TCDS, contact the responsible office listed in DRS for that TCDS.

(2) Review the aircraft maintenance records to verify the applicant met the § 21.189(a)(2) requirement for flight testing the aircraft. Verify that these records include the flight test findings and were signed by the pilot who conducted the flight test.

(3) Review available documents, such as military technical orders and inspection records, to help you assess the airworthiness of the aircraft.

(4) Look for evidence of long-term storage and inactivity, which may impact the scope of your aircraft inspection.

b. Inspect Aircraft.

(1) Because surplus military aircraft may have deteriorated due to prolonged storage, prolonged inactivity, or age, the applicant must show that the aircraft was thoroughly inspected to determine its state of preservation and repair and to ensure it is in a condition for safe operation. The inspection may require removing rivets and cutting openings to check the condition of faying surfaces and closed areas. If this is the case, the applicant should be advised that the inspection would be expedited if an airworthiness inspection is performed by an appropriately rated repair station or mechanic per the requirements of Part 43.

(2) Reserved.

Table 8-1. Sample of Limited Category TCs

Original Manufacturer	TC Holder	Model	Limited Spec No.	DRS Listing
Boeing	Transcontinental and Western	B-17F and B-17G	AL-1	LTC-1
North American	S and R	B-25, B-25C, B-25G, B-25H, B-25J, and B-25N	AL-2	AL-2
Douglas	P.V. Shields	A-26B and A-26C	AL-3	TCS L-3
Douglas	Seaboard and Western	A-24 (Navy SBD-5)	AL-4	TCS L-4
Consolidated Vultee	R.M. Lewis	PB2Y-3, PB2Y-3R, PB2Y-5, PB2-5R, and PB2Y-5Z	AL-5	LTC-5
Consolidated Vultee	Consolidated Vultee	LB-30	AL-6	LTC-6
Sikorsky	Holmes	R-4B Helicopter	AL-7	LTC-7
Grumman	Air Trading Corp	TBF-1, TBF-1C, TBM-1, TBM-1C, TBM-3, and TBM-3E	AL-8	LTC-8
Douglas	Hughes	A-20B, A-20C, A-20G, A-20H, and A-20J	AL-9	TC L-9
Lockheed	Lockheed	P-38E, P-38J, P-38L, P-38M, F-5E, F-5F, and F-5G	AL-10	LTC-10
North American	Cal Pacific	P-51C, P-51D, and P-51K	AL-11	LTC-11
Beech	Cameron	AT-10, AT-10BH, AT-10GF, and AT-10GL	AL-12	LTC-12
Lockheed	Air Trading Corp	B-34, PV-1, and PV-2	AL-13	LTC AL-13
Northrup	Northrup	P-61, P-61A, and P-61B	AL-14	LTC-14
North American	Edmondson	A-36A	AL-15	LTC-15
Curtis	Holmberg	O-52	AL-16	LTC-16
Grumman	Utterback	J2F-3, J2F-4, J2F-5, and J2F-6	AL-17	LTC-17
Curtis-Wright	Prevost	P-40L and P-40N	AL-18	TCS AR-1/LTC-18
Sikorsky	Hel-i-cop Advertising	R-5A Helicopter	AL-19	LTC-19
Martin	Krogman	PBM-5	AL-20	LTC-20
Bell Aircraft	Bell Helicopter Textron	P-63C and P-63E	AL-21	LTC-21
North American	Mustang Aviation	BC-1	AL-22	LTC-22
Grumman	Grumman	F8F-1	AL-23	LTC-23
Chance-Vought	Chance-Vought	OS2N-1, OS2U-1, OS2U-2, and OS2U-3	AL-24	LTC-24
Grumman	Carlisle	FM-2	AL-25	LTC-25
Consolidated Vultee	Executive Airlines	L-1, L-1A, L-1B, L-1C, L-1D, L-1E, and L-1F	AL-26	LTC-26
North American	Wyatt	BT-9, BT-9A, BT-9B, and BT-9C	AL-27	LTC-27
Culver	Superior	PQ-14A, PQ-14B, YPQ-14B, YPQ-14A, and Navy TD2C1	AL-28	LTC-28

Original Manufacturer	TC Holder	Model	Limited Spec No.	DRS Listing
Sikorsky	Stolp-Adams	R-5A and YR-6A, HOS-1 Helicopter	AL-29	LTC-29
Consolidated	Odom	C-87A	AL-30	LTC-30
Curtis-Wright	Rehr	AT-9 and AT-9A	AL-31	LTC-31
North American	Franklin	BT-14	AL-32	LTC-32
Martin	Tennessee Gas Transmission Co.	B-26	AL-33	B02CE

Chapter 9. Light-Sport Category Aircraft (§ 21.190)

9-1. Introduction. This chapter provides policies and procedures for issuing a special airworthiness certificate for light-sport category aircraft under § 21.190.

9-2. General.

a. Consensus Standards. As used in this chapter, industry consensus standards—

(1) Are means of compliance with the design, production, and airworthiness requirements for light-sport category aircraft in 14 CFR Part 22, subpart B and § 21.190. Per § 21.190, a manufacturer's statement of compliance (SOC) must state compliance with the applicable industry consensus standards accepted by the FAA.

(2) Acceptance by the FAA is indicated via publication of a notice of availability (NOA) in the [Federal Register](#). See the [FAA webpage](#) for a current listing of consensus standards accepted by the FAA.

Note: Effective July 24, 2026, the codification of new design, production, and airworthiness requirements in Part 22, subpart B, requires all new consensus standards, which specifically function as means of compliance with Part 22. Previous consensus standards will no longer be valid for issuance of airworthiness certificates under § 21.190. Only the consensus standards identified in the most recent Notice of Availability as summarized on the FAA webpage are acceptable on a manufacturer's SOC for issuance of a special airworthiness certificate under § 21.190. A new light-sport category aircraft of a class that does not have corresponding consensus standards listed on the FAA webpage is not eligible for an airworthiness certificate under § 21.190.

b. Production Flight Testing. Policies and procedures for issuing an SFP for the purpose of production flight testing new production aircraft are included in chapter 19 of this order.

Note: Prototype and developmental aircraft are flown under an experimental airworthiness certificate for R&D. Once the aircraft manufacturer has established the model's configuration, pre-certification flight operations will be conducted with an SFP.

c. First Make or Model Light-Sport Category Aircraft. The manufacturing ASI assigned to issue the first make or model will contact the AIR-842 email box at 9-AVS-AIR842-SLSA@faa.gov before issuing the first airworthiness certificate for a new make or model light-sport category aircraft. For powered parachutes and weight-shift-control aircraft, this includes a new combination of a wing and fuselage that has not previously been issued a special airworthiness certificate under § 21.190. AIR-842 will coordinate to determine whether an audit will be conducted per FAA Order 8130.36, *Special Light-Sport Aircraft Audit Program*, prior to the issuance of the airworthiness certificate. The FAA maintains a [Make/Model Directory](#) for light-sport category aircraft that have already been issued a light-sport category airworthiness certificate under § 21.190. A designee may not issue the first airworthiness certificate for a new make or model light-sport category aircraft.

Note: Advise manufacturers of light-sport category aircraft to notify the FAA approximately 6 months before presenting a new make or model light-sport category to the FAA for issuance of an airworthiness certificate. Such advance notice enables more effective audit planning and increases the likelihood that FAA resources will be available to conduct the audit. Affirm the value of such audits to manufacturers in reducing the risks of undetected non-compliances or non-conformities.

d. Changes With the Manufacturer. Before issuing a special airworthiness certificate under § 21.190, notify AIR-842 of any actual or pending changes with the aircraft manufacturer that may impact compliance with FAA-accepted consensus standards, cessation of business, or change in ownership, name, or location.

e. Voluntary Noise Compliance. Manufacturer compliance with 14 CFR Part 36 is optional for light-sport category aircraft. A manufacturer's SOC concerning noise or Part 36 is also optional and has no bearing on records reviews, aircraft inspections, or the issuance of a special airworthiness certificate for a light-sport category aircraft.

f. Special Provisions for Light-Sport Category Aircraft Certificated Before July 24, 2026.

(1) Duration of Airworthiness Certificates. Light-sport category aircraft certificated prior to July 24, 2026, may not meet Part 22 requirements since they were designed and manufactured to meet consensus standards that predate Part 22. An airworthiness certificate for such an aircraft remains effective if the aircraft continues to meet § 21.181(a)(3).

(2) Exceeding any parameter of § 21.181(a)(3)(iv) in such an aircraft would require—

(a) An FAA grant of exemption,

(b) A new manufacturer's SOC with all applicable consensus standards for the applicable aircraft class—not just for the change related to the parameter that exceeds § 21.181(a)(3)(iv)—that were accepted by the FAA as means of compliance with Part 22, and an amended airworthiness certificate based on that SOC, or

(c) A new airworthiness certificate issuance for an appropriate experimental purpose since the aircraft no longer qualifies for the light-sport category.

(3) Aerial Work. Light-sport category aircraft certificated prior to July 24, 2026, may only be used in aerial work operations if the aircraft owner provides the FAA with an amended SOC from the manufacturer per § 21.190(e) and the other documentation required by § 21.190(e); FAA Form 8130-15 is used for this purpose. The § 21.190(e) process does not require the airworthiness certificate to be amended unless additional operating limitations are necessary for the new aerial work operation(s). To meet the requirement for submitting an amended SOC to the FAA, advise aircraft owners to submit each amended SOC via email to the FAA Registry at Submit.Aircraft.Registry@faa.gov for inclusion in the airworthiness file for the applicable aircraft.

9-3. Procedures for Issuing a Light-Sport Category Airworthiness Certificate.

Follow chapter 2 of this order, and the following:

a. Review Application. Review the following:**(1) FAA Form 8130-6.**

(a) Foreign Manufactured Aircraft. Since FAA acceptance of foreign-manufactured light-sport category aircraft are not treated as imports under bilateral agreements, verify that the import box is not checked. Per § 22.100(b)(1), regardless of the class of aircraft specified in the application, verify that the aircraft was manufactured in a country with which the United States has an airworthiness agreement concerning U.S. acceptance of airplanes. The agreement may take the form of a Bilateral Airworthiness Agreement, a Bilateral Aviation Safety Agreement with associated Implementation Procedures for Airworthiness, or an equivalent airworthiness agreement. Verify that the country of manufacture on FAA Form 8130-15 matches the country of manufacture on Aeronautical Center Form 8050-88A, *Affidavit of Ownership for Amateur-Built and Other Non-Type Certificated Aircraft*, for this aircraft. Contact the FAA Aircraft Registry to obtain Aeronautical Center Form 8050-88A. Also, per § 22.100(b)(2), verify that the aircraft is eligible for an airworthiness certificate, flight authorization, or other similar certification in its country of manufacture.

Note: Advise applicants or manufacturers that including a copy in the aircraft maintenance records of the flight authorization for production flight testing issued by the applicable foreign CAA for that aircraft is a best practice for showing compliance with § 22.100(b)(2).

(b) An aircraft originally certificated in the light-sport category and subsequently certificated for any experimental purpose is not eligible for re-certification in the light-sport category if there are any major alterations or major repairs not authorized by the aircraft manufacturer or a person acceptable to the FAA, or that conflict with the eligibility requirements of § 21.190 under the original certification.

(2) Manufacturer's SOC.

(a) As part of the application for a light-sport category airworthiness certificate, an applicant must provide the FAA with the aircraft manufacturer's SOC on FAA Form 8130-15, *Light-Sport Category Aircraft/Kit Statement of Compliance*, as described in § 21.190(d). Review the aircraft manufacturer's completed FAA Form 8130-15 per section 5 of appendix A to this order for accuracy and completeness.

Note: Light-sport category aircraft certificated on or after July 24, 2026, may conduct aerial work operations that the aircraft manufacturer has determined may be safely conducted. Per § 21.190, such aerial work operations must be listed in the pilot's operating handbook and specified on the manufacturer's SOC. **Note:** Light-sport category aircraft may be designed with simplified flight controls per the design requirements in § 22.180. Per § 21.190(d), the aircraft manufacturer must state on the manufacturer's SOC whether the aircraft complies with consensus standards for compliance with § 22.180 design requirements for simplified flight controls.

(3) Pilot's Operating Handbook. Verify the aircraft's POH contains:

- (a) Operating instructions and limitations that apply to the aircraft make, model, and serial number being inspected.
 - (b) A flight training supplement.
 - (c) A listing of any applicable aerial work operations and the instructions and limitations necessary to safely conduct those operations. If the aircraft manufacturer did not list any aerial work operations in the POH, advise the applicant that the aircraft cannot be used for aerial work operations.
 - (d) Any instructions or limitations necessary to safely conduct towing operations.
- (4) Maintenance. Verify the manufacturer's Maintenance Manual or Instructions for Continued Airworthiness applies to this make, model, and serial number. When maintenance or inspections were conducted before issuance of the special airworthiness certificate, verify the person who performed those tasks was authorized to do so via the aircraft manufacturer's quality assurance manual.

b. Review of Other Aircraft Records and Documents.

- (1) Per § 22.100(a)(6), verify the aircraft described on the application has not been issued a standard, primary, restricted, limited, or provisional airworthiness certificate or an equivalent airworthiness certificate by another CAA.
- (2) Verify the aircraft complies with the eligibility requirements of § 22.100.
- (3) Verify all documentation is in the English language.
- (4) Weight and Balance Chart. Compare the completed aircraft weight and balance chart to the data in the POH for accuracy. The weight and balance chart should include the aircraft's forward and aft CG limits, and empty and maximum weights. Verify the light-sport category aircraft manufacturer has determined the aircraft's CG location based on its current weight. If a discrepancy is found, the aircraft must be re-weighed. The weight scales used must meet the aircraft manufacturer's quality assurance system for calibration and be within the limits of the calibration interval. The aircraft presented must match the POH, the aircraft equipment listing, and the regulations, or it may not be certificated. Verify that the completed weight and balance report, including load limits for occupants, oil, fuel, and baggage, is available in the aircraft.
- (5). Aircraft Maintenance Records. Verify that aircraft maintenance records show:
 - (a) Satisfactory completion of required ground and flight tests and inspections as described in the appropriate consensus standard for each aircraft.
 - (b) Compliance with all applicable ADs. This applies to an AD for a specific light-sport category aircraft make and model, and for an AD issued against a type-certificated product or equipment installed in the aircraft.
- (6). Maintenance Work Prior to Certificate Issuance. When any work was performed on the aircraft, such as alterations, repairs, installations, assembly, or reassembly, after completion of original production acceptance ground and flight tests, verify:
 - (a) The work performed on the aircraft is covered within the manufacturer's quality system documentation. This could be in the form of an authorization or instruction.

When any of the requirements for a light-sport category aircraft cannot be substantiated by an applicant, the aircraft is not eligible for an airworthiness certificate in the light-sport category.

(b) Each major repair or major alteration was authorized by the manufacturer or a person acceptable to the FAA per § 91.327, and the work was properly documented in the aircraft records.

(c) The aircraft has a final inspection record showing acceptance by an authorized person.

(d) The weight and balance has been recalculated.

(7) Consistency Among Documents. Verify that the data contained in the documentation is consistent. For example, the POH, maintenance manual, and the aircraft's fuel tank placard must all prescribe the same fuel requirements.

(8) Additional Document Reviews. Review the following additional items:

(a) Manufacturer's Ground Test Documentation. Verify that the documentation indicates test acceptance, location, the qualification and title of the responsible person, and that they are identified in the manufacturer's quality system documentation.

(b) Manufacturer's Flight Test Documentation.

(1) Verify that the version level as called out on FAA Form 8130-15 matches the manufacturer's quality assurance system version level. When this documentation does not comply with the authorized processes, the documentation cannot be accepted to validate FAA Form 8130-15 for issuance of a special airworthiness certificate under § 21.190.

(2) Verify that the documentation indicates successful production acceptance testing, the person responsible, the qualification and title of the production flight test pilot, and the location where the production flight testing was performed. Verify the data contents of any acceptance records are within the requirements of the POH operating limitation ranges and parameters. Verify the documentation shows successful flight testing for towing and any aerial work operation, as specified in the aircraft's POH.

(3) When a light-sport category aircraft manufacturer delegates flight testing, verify the manufacturer provided written authorization for that delegation and that anyone delegated to make findings of compliance meets the training requirements of § 22.190.

(c) Continued Operational Safety Program. Verify that documentation or electronic media is available.

c. Inspect Aircraft. Start the physical inspection after the aircraft is complete in every respect.

(1) Verify the aircraft complies with eligibility requirements of § 22.100.

(2) Verify the aircraft configuration matches flight testing documentation and the POH.

(3) Verify the installed equipment is per the POH.

(4) Verify that instruments and controls are appropriately marked and required placards are located for easy reference giving particular attention to the airspeed indicator. Verify that the marks within the airspeed indicator match the POH indicated airspeed limitations (such as V_{NE} , red line; caution range, yellow arc; normal operating range, green arc; when equipped, flap operating range, white arc with lower limit of V_{SO} at maximum weight).

d. Correction of Discrepancies. Inform the applicant of any omissions, errors, or other discrepancies identified during your inspections. Only when the required corrections have been made can an airworthiness certificate be issued.

e. Issue Airworthiness Certificate. Return the original manufacturer's documentation to the applicant.

9-4. Changing From an Experimental Purpose to the Light-Sport Category.

a. Eligibility. An aircraft holding an experimental airworthiness certificate may be eligible for certification in the light-sport category under the following conditions:

(1) When the aircraft has been flown by the manufacturer for R&D, such as a prototype light-sport category aircraft.

(2) When an applicant is seeking to return a former light-sport category aircraft back to the light-sport category. Experimental kit-built aircraft certificated under §§ 21.191(i)(2) or 21.191(k) are not eligible for certification in the light-sport category because the aircraft manufacturer did not build, ground test, or flight test the aircraft. Neither the manufacturer nor the aircraft owner could state compliance with applicable consensus standards necessary for airworthiness certification in the light-sport category.

b. Procedures for Issuing an Airworthiness Certificate. Follow paragraph 9-3 and the following:

(1) Review all original documentation required by § 21.190.

(2) Review the manufacturer's SOC that was used for the original issuance of the light-sport category airworthiness certificate.

(3) Verify the following items are met. If not met, the aircraft is not eligible for return to the light-sport category:

(a) All major repairs or major alterations to products produced under an FAA-approval were performed using FAA-approved data and were approved for return to service by an authorized person.

(b) All major repairs or major alterations to products not produced under an FAA-approval, were authorized by the aircraft manufacturer, or a person acceptable to the FAA, and were performed by an authorized person;

- (c) All repairs and alterations meet the applicable consensus standards and are documented in the aircraft's records per Part 43.
 - (d) The manufacturer (or other) authorization specifies the aircraft make, model, and serial number and specifies the applicable version of FAA-accepted consensus standards in the manufacturer's SOC for the aircraft (or later version).
- (4) Verify the manufacturer (or other) authorization specifies the aircraft make, model, and serial number. Verify the authorization specifies the applicable version of FAA-accepted consensus standards in the manufacturer's SOC for the aircraft (or later version). All major repairs or major alterations to products produced under an FAA-approval must be included in the aircraft's permanent record and documented in the aircraft's records per Part 43. If this was not done, the aircraft is not eligible for return to the light-sport category.

Chapter 10. Experimental Purposes of Research and Development and Showing Compliance with Regulations (§ 21.191(a) and (b))

10-1. Introduction. This chapter provides policies and procedures for issuing special airworthiness certificates for the experimental purposes of research and development (R&D) under § 21.191(a) or showing compliance with regulations under § 21.191(b).

10-2. Procedures for Issuing an Airworthiness Certificate. Follow chapter 4 of this order and the following:

a. Review Application. Verify, through review of the applicant's program letter, that the applicant has provided information required by § 21.193 and the project is consistent with the requested experimental purpose:

- (1) R&D. Per § 21.191(a), this purpose includes flights that test new aircraft design concepts, new aircraft equipment, new aircraft installations, new aircraft operating techniques, or new uses for aircraft to determine whether an idea warrants further development.
- (2) Showing Compliance with Regulations. Per § 21.191(b), this purpose includes flights to show compliance with applicable regulations for the issuance of TCs, STCs, and major design changes, including showing compliance with applicable function and reliability requirements. This purpose would be considered valid when the applicant for a TC or an aircraft modifier has revised the TC design data or has applied for an STC or field approval. The purpose is to show compliance with the CFR after the applicant has completed testing under R&D, if applicable, and is ready for flight testing by the FAA.
- (3) Supporting Aircraft. The operation of a chase plane, an airborne icing tanker used for in-flight icing tests, or other such aircraft operating in direct support of an R&D or certification project is within the scope of these purposes.
- (4) Space Support Vehicle Flights. Per § 91.331(a)(iv)(B) and (C), the R&D purpose may be used for aircraft conducting space support vehicle flights when simulating space flight conditions in support of the testing of hardware to be used in space flight and/or research and development tasks that require unique aircraft capabilities.

b. Reserved.

10-3. Holding an Airworthiness Certificate in Suspension. This policy and procedure allows for the issuance of an experimental airworthiness certificate for R&D and/or showing compliance with regulations for aircraft already issued a standard airworthiness certificate under § 21.183, a special airworthiness certificate issued for a restricted category aircraft under § 21.185, or a light-sport category aircraft under § 21.190. This procedure enables an applicant to conduct short-term projects such as flight testing for an STC project or for a light-sport category aircraft manufacturer to flight test major repairs or alterations without having to permanently surrender its original current airworthiness certificate. To do so, the original airworthiness certificate is obtained from the applicant and held in suspension upon issuance of an airworthiness certificate for R&D purposes or upon demonstrating compliance with regulations, or both. Do not use this procedure if the applicant cannot ensure the

aircraft will remain in compliance with all the maintenance and preventive maintenance programs required under the airworthiness certificate that is intended to be held in suspension.

- a. When an applicant for a change in type design alters an aircraft per a proposed type design change, a flight test may be required to demonstrate compliance with airworthiness regulations.** A flight test also may be required if an applicant wishes to conduct R&D testing of an altered aircraft. However, the altered aircraft is no longer in compliance with its TC; therefore, the aircraft does not have an effective airworthiness certificate under which to conduct the flight test. In these cases, the FAA may issue an experimental airworthiness certificate for R&D flight testing or showing compliance with airworthiness regulations, as applicable. If necessary, the experimental airworthiness certificate may be issued for both purposes.
- b. The applicant must comply with the requirements of § 21.193 and submit a completed FAA Form 8130-6 to request a special airworthiness certificate for R&D testing or showing compliance with regulations.** Process the application in the usual manner with the following differences:
 - (1) Obtain the aircraft's original airworthiness certificate so it can be held in suspension by the responsible CM Section or the responsible Flight Standards Office. If testing will be completed in less time than required to send the airworthiness certificate to the CM Section or Flight Standards Office, the ASI or designee may hold the airworthiness certificate in suspension. The owner or applicant does not retain the suspended airworthiness certificate.
 - (2) If the applicant's need for the special airworthiness certificate for R&D testing or showing compliance with regulations exceeds 30 days, verify the need is valid, and reverify the need every 30 days thereafter until the process is complete. If the need is valid, no change is required. If the aircraft meets the maintenance and inspection requirements of the suspended airworthiness certificate, no change is required. If the need is not valid or inspection requirements are not met, advise the applicant that the suspended airworthiness certificate is now surrendered, and upon completion of testing, the applicant must apply for a new airworthiness certificate. The duration of the special airworthiness certificate for R&D or showing compliance with regulations may not exceed the requirements of § 21.181.
 - (3) Since the aircraft was airworthy prior to the alteration for R&D testing or showing compliance with the regulations, you are not required to conduct a full airworthiness assessment of the aircraft. You may limit your aircraft records review and aircraft inspection to those areas related to finding the airworthiness of the alterations and supporting the issuance of appropriate operating limitations.
 - (4) When issuing an experimental airworthiness certificate for R&D or showing compliance with regulations, include the following in the required maintenance record entry to streamline reinstatement of the airworthiness certificate.
 - (a) Explain the reason for the issuance of the experimental airworthiness certificate.

(b) Provide all information that expedites reinstatement of the standard, restricted, or light-sport category airworthiness certificate (for example, type of alteration performed).

(5) Enter “Do Not Code” in the FAA Form 8130-6 FAA Coding block. This ensures the FAA Aircraft Registry does not change the aircraft's airworthiness status in the Registry database.

c. After the flight testing has been completed and the applicant requests the return of the suspended certificate, perform the following:

(1) Review aircraft records and inspect the aircraft only as necessary to confirm what has occurred from suspension to return of the original suspended airworthiness certificate unless you identify a need for further reviews and inspections. Verify that the aircraft is airworthy with respect to either the original design or the applicable approved design for the alteration

(2) Make a signed and dated entry in the aircraft maintenance records fully explaining what has occurred, and include the following statements as applicable:

(a) R&D flight tests. “I find this aircraft meets the requirements for the reinstatement of the [*standard or restricted*] airworthiness certificate, following completion of R&D flight testing, based on an inspection confirming reconfiguration of the aircraft to the approved type design.”

(b) Light-Sport Category Aircraft R&D flight tests. “I find this aircraft meets the requirements for the reinstatement of the light-sport category airworthiness certificate, following completion of R&D flight testing, based on an inspection confirming reconfiguration of the aircraft to the manufacturer’s approved design.”

(c) Show Compliance Flight Tests. “I find this aircraft meets the requirements for the reinstatement of the [*standard or restricted*] airworthiness certificate and the inspection was performed based on all installations and alterations related to [*insert STC number, amended TC number, or other form of project description*], performed from [*insert date the alteration(s) was first installed*] to [*insert current date*].”

(d) All Flight Tests. “This certification process does not replace or change the dates or times of scheduled inspection requirements (for example, annual or 100-hour inspections).”

Chapter 11. Experimental Purpose of Crew Training (§ 21.191(c))

11-1. Introduction. This chapter provides policies and procedures for issuing a special airworthiness certificate for the experimental purpose of crew training per § 21.191(c).

11-2. Procedures for Issuing an Airworthiness Certificate. Follow chapter 4 of this order and the following:

a. Review Application. Verify that the application, including the program letter, demonstrates eligibility for the requested experimental purpose of crew training. Aircraft are only eligible for an airworthiness certificate for the experimental purpose of crew training if the purpose is limited to training flight crews employed by the applicant. This normally includes manufacturer's employees who need to be trained in experimental aircraft.

Note 1: This experimental purpose cannot be used for sales, customer demonstration flights, customer crew training, or flight training offered to the public. An aircraft manufacturer could conduct customer crew training or sales demonstration flights with an experimental airworthiness certificate for market survey or with an SFP under § 21.197(a)(5). These situations are addressed in chapters 14 and 19, respectively. Flight training offered to the public must be conducted in accordance with § 91.326 utilizing an appropriate experimental purpose as described in Advisory Circular AC 91-94.

Note 2: Although the crew training purpose generally only authorizes training the applicant's flight crews, persons conducting space support vehicle operations to simulate space flight conditions are authorized to conduct training for potential space flight participants, government astronauts, or crew in accordance with § 91.331.

b. Reserved.

Chapter 12. Experimental Purpose of Exhibition (§ 21.191(d))

12-1. Introduction. This chapter provides policies and procedures for issuing special airworthiness certificates for the experimental purpose of exhibition per § 21.191(d).

12-2. Procedures for Issuing Airworthiness Certificates. Follow chapter 4 of this order and the following:

a. Review Application. Verify that the application, including the program letter, demonstrates eligibility for the requested experimental purpose of exhibition. This purpose includes exhibiting the aircraft's flight capabilities, performance, or unusual characteristics at air shows, competitions, and other aviation events, for motion picture, television, and similar productions; and for the maintenance of exhibition flight proficiency, including (for persons exhibiting aircraft) flying to and from such events and productions. For motion pictures, television, and similar productions, the exhibition purpose allows the aircraft to be the subject of the production, not the filming platform. Exhibition aircraft may not be used to carry photographers, filming crews, or cameramen for compensation or hire. Section 21.191(d) is limited to exhibiting *the aircraft's* flight capabilities, performance, etc., for motion pictures, television, etc.; not the capabilities, performance, etc. of *another aircraft*.

Note: Flyovers, short takeoff and landing competitions, and similar events are considered to be within the scope of the exhibition purpose.

b. Reserved.

Chapter 13. Experimental Purpose of Air Racing (§ 21.191(e))

13-1. Introduction. This chapter provides policies and procedures for issuing special airworthiness certificates for the experimental purpose of air racing per § 21.191(e).

13-2. Procedures for Issuing Airworthiness Certificates. Follow chapter 4 of this order and the following:

a. Review Application. Verify that the application, including the program letter, demonstrates eligibility for the requested experimental purpose of air racing. This purpose includes participating in air races, practicing for air races, and flying to and from racing events.

b. Reserved.

Chapter 14. Experimental Purpose of Market Surveys (§ 21.191(f))

14-1. Introduction. This chapter provides policies and procedures for issuing a special airworthiness certificate for the experimental purpose of market surveys, which includes use of the aircraft for purposes of conducting market surveys, sales demonstrations, and customer crew training, per §§ 21.191(f) and 21.195.

Note: An aircraft manufacturer could also conduct sales or customer demonstration flights with an SFP under § 21.197(a)(5). This situation is addressed in chapter 19.

Note: The training of the manufacturer's employees is accomplished under the experimental purpose of crew training (§ 21.191(c)) in chapter 11.

14-2. Procedures for Issuing an Airworthiness Certificate. Follow chapter 4 of this order and the following:

a. Review Application. Verify the application, including the program letter, demonstrates eligibility for the requested market surveys, sales demonstrations, or customer crew training. An aircraft is eligible for the requested certificate if the applicant meets §§ 21.191(f) and 21.195.

b. Review of Aircraft Records. Verify the aircraft and the required maintenance entries comply with the requirements of § 21.195(d).

14-3. Reducing Flight Hour Requirements of § 21.195(d)(2). An ASI may reduce the flight hours requirement if that aircraft meets the requirements of § 21.195(d)(1) and (d)(2). Designees must coordinate with their managing specialist before reducing flight hours. In assessing the feasibility and risks of such reductions, consider the applicant information in the program letter, such as the following:

a. Type Certificated Aircraft That Have Been Altered:

- (1) Complexity of alteration.
- (2) Changes to aerodynamics or external shape.
- (3) The alteration's effect on controllability.
- (4) The alteration's effect on operations.
- (5) Adequacy of the flight test program to demonstrate that the aircraft is safe and in compliance with § 91.319(b)(1) and (2).

b. Other Factors for Consideration: Operational history of similar or derivative models.

Chapter 15. Experimental Purpose of Operating Amateur-Built Aircraft (§ 21.191(g))

15-1. Introduction. This chapter provides policies and procedures for issuing special airworthiness certificates for the experimental purpose of operating amateur-built aircraft under § 21.191(g).

15-2. General.

Amateur Builder. Under § 21.191(g), the FAA receives applications from and issues airworthiness certificates for operating amateur-built aircraft. An amateur builder (hereafter, builder) is the person who fabricates and assembles the major portion of the aircraft for their own recreation or education. A person is defined in § 1.1. A kit manufacturer or commercial assistance provider is not the builder.

a. Major Portion. “Major portion” means the fabrication and assembly of more than 50 percent of the aircraft, also known as the “51 percent rule.”

b. Builder’s Education or Recreation. An aircraft manufactured or assembled by a business does not meet the education or recreation requirements of § 21.191(g). Prototype aircraft built to prove the design for sale as an amateur-built kit are not produced by persons “solely for their own education or recreation” and, therefore, are not eligible for an experimental airworthiness certificate under § 21.191(g).

c. Designs. Amateur-built aircraft may be constructed from a builder’s original design, purchased plans, from a kit, or a combination of these.

d. Kit Evaluations.

(1) The FAA performs a kit evaluation to determine if an aircraft built from that kit per the kit manufacturer’s instructions, may allow a builder to meet the major portion requirement. The FAA maintains a list of completed kit evaluations, [*Revised Listing of Amateur-Built Aircraft Kits*](#) (referred to hereafter as the “kit list”), and completed [*Amateur-Built Fabrication and Assembly Checklists*](#) for evaluated kits. These completed checklists will enable builders to determine how much commercial assistance may be used.

Note: For information on the kit evaluation process or for kit manufacturers desiring a kit evaluation, refer to FAA Order 8130.35, *Amateur-Built Aircraft National Kit Evaluation Team (NKET)*.

(2) The FAA does not certify or approve kits or kit manufacturers. The outcome of a kit evaluation does not mean a kit is FAA “certified,” “certificated,” or “approved,” and kit manufacturers may not represent their kits as such.

(3) A kit evaluation or inclusion on the kit list is not a prerequisite for selling a kit or for an airworthiness certificate.

f. Type-Certificated Aircraft. Altering, repairing, or rebuilding type-certificated aircraft constitutes maintenance of a type-certificated aircraft under Part 43, not fabrication and

assembly of amateur-built aircraft. This policy has been in effect since 1952 under section 1.74-3 of the Civil Aeronautics Manual (CAM), which specifically states, “structural components of other aircraft may be used (for amateur-built aircraft); however, it is not intended that this provision be used to avoid obtaining approval of major alterations to aircraft previously certificated in another category.”

15-3. When Builders or Applicants Contact the FAA. Builders and applicants who contact the FAA should be:

a. Encouraged to review the following before starting their project:

- (1) Advisory Circular 20-27, *Certification and Operation of Amateur-Built Aircraft*.
- (2) The [FAA amateur-built website](#).
- (3) The [Experimental Aircraft Association \(EAA\) website](#) to learn about other resources for builders.

b. Advised to check the kit list if building an aircraft with a kit.

15-4. Procedures for Issuing an Airworthiness Certificate. Follow the procedures in paragraph 2-3 and chapter 4 of this order and the following:

a. Review Application.

- (1) Eligibility.
 - (a) Eligibility Statement. For an application for an original airworthiness certificate, review the applicant’s notarized FAA Form 8130-12, *Eligibility Statement, Amateur-Built Aircraft*, for accuracy and completeness. If submitted via AWC, verify it is a copy of the notarized original. Verify that all sources of commercial assistance are included. The FAA Form 8130-12 is required for original certification only.
 - (b) For an application for a recurrent airworthiness certificate, review information such as the original airworthiness certificate as evidence of a prior finding of compliance by the FAA with eligibility requirements for issuance of the airworthiness certificate.
- (2) Program letter. Verify the program letter complies with § 21.193. Refer to Appendix C of this Order.

b. Review Aircraft Records.

- (1) Supporting Documentation for the Eligibility Statement.
 - (a) Review the builder’s documentation to verify it shows:
 - (1) What was fabricated, assembled, and inspected.
 - (2) Who performed these actions.
 - (3) The date the activity was performed.

(4) Where the activity was performed.

(5) The methods used.

(6) Any commercial assistance or education assistance used.

(b) Examples of documentation include:

(1) The applicable Amateur-Built Aircraft Fabrication and Assembly Checklist and detailed descriptions of the steps included in each of the listed tasks.

(2) A comprehensive builder's log that includes items such as drawings, engineering specifications, plans, references, handbooks, kit manufacturer's data, photographs, video, documentation of commercial assistance used (including receipts), documentation of education assistance used, article inventory and histories, receipts, catalogs, and logbook entries.

(c) Ask the builder questions to confirm the builder's actions as described in the supporting documentation.

(2) Review any documentation of in-process and pre-cover inspections.

(3) Verify that aircraft records include a completed statement from the owner that the aircraft has been inspected per 14 CFR Part 43, appendix D, or other approved programs, and was found to be in a condition for safe operation. The inspection will help identify any errors made during the construction of the aircraft. This statement will support the owner's inspection and airworthiness statement in block III of FAA Form 8130-6.

Note: There is no requirement for a certificated mechanic to sign off on the inspection. The builder's signature on FAA Form 8130-6, block III, attests to the airworthiness of the aircraft.

(4) Review records to verify that the aircraft has been weighed per established weight and balance procedures.

(a) If the builder designed the aircraft, these limits would be determined by the builder's calculations.

(b) If the aircraft is constructed from a kit or built from purchased plans, review the kit instructions, or build plans to verify the builder followed those instructions or plans.

(c) If the builder made changes to a kit that affects the aircraft's CG, verify the kit manufacturer's weight and balance calculations were recalculated based on the change(s).

(5) It is recommended that you obtain and review a fuel flow report from the applicant to verify an adequate fuel flow rate per paragraph 2.19.5 of AC 90-89.

c. Inspect the Aircraft.

(1) Level of Assembly/Disassembly. At the time of airworthiness certification, the aircraft must be complete.

(a) In-Process Inspections. The FAA does not typically perform in-process inspections during the fabrication and assembly process. The builder must document all in-process inspections, including the dates and names of all the people involved. The FAA recommends that in-process inspections be completed by knowledgeable persons, such as EAA technical counselors or certificated mechanics.

(b) Pre-Cover Inspections. The FAA does not typically perform pre-cover inspections to determine if the aircraft meets the major-portion requirement. The builder must document all pre-cover inspections, including dates and names of all the people involved.

(c) Requesting Disassembly of the Aircraft. During the airworthiness evaluation, do not request excessive disassembly of the aircraft if the builder can provide evidence of fabrication, assembly, and in-process inspections.

(2) When Builders Use a Kit. Aircraft fabricated and assembled from a kit must meet the requirements of § 21.191(g).

(a) An aircraft assembled from a kit composed entirely of finished prefabricated articles is not eligible for a certificate under § 21.191(g).

(b) A certain quantity of prefabricated articles, such as heat-treated ribs, bulkheads, or complex articles made from sheet metal, fiberglass, composites, or polystyrene, is acceptable. Kits that provide large articles, such as complete fuselages and wing structures requiring minimal supplemental fabrication and assembly, may not allow a builder to meet the major-portion requirement.

(c) Applicable Policy and Guidance. If the kit was evaluated and placed on the kit list before September 30, 2009, and, since the evaluation, the kit manufacturer has made no changes to the kit that would affect the allocation of task credit and the applicant is not obtaining commercial assistance, the policy and guidance in effect at the time the FAA evaluated the kit apply. For a non-evaluated kit purchased from the kit manufacturer before September 30, 2009, and the applicant is not obtaining commercial assistance, the policy and guidance in effect at the time the applicant purchased the kit apply. Otherwise, current policy and guidance apply. The significance of the date corresponds to the publication of FAA AC 20-27G and implementation of the kit evaluation process.

(3) Condition for Safe Operation. Inspect the aircraft to verify it is in a condition for safe operation. The sample checklist for a condition inspection in appendix A to FAA AC 90-89, *Amateur-Built Aircraft and Ultralight Flight Testing Handbook*, may be used for determining if the aircraft is in a condition for safe operation.

Note: The JAR-22 standard for a motor glider referenced in FAA AC 21.17-2, Type Certification—Fixed Wing Gliders (Sailplanes), Including Powered Gliders, provides guidance in determining if an aircraft meets the glider requirements.

d. Verify Major Portion. For a recurrent certification, review evidence of a prior major-portion determination by the FAA, such as the original airworthiness certificate issued

under § 21.191(g) by the FAA. If such evidence is available, do not make a new major-portion determination. For an original certification, the ASI or designee must always make a major-portion determination at the time of the airworthiness evaluation. To determine if an aircraft meets the major-portion requirement, evaluate the amount of fabrication and assembly work accomplished by the builder(s) against the total amount necessary to complete the aircraft. If the checklist is completed by the applicant, the FAA must review the applicant's completion of task documentation.

Note: Fabrication is defined as layout, bending, countersinking, straightening, cutting, sewing, gluing/bonding, lay-up, forming, shaping, trimming, drilling, deburring, machining, applying protective coatings, surface preparation and priming, riveting, welding, and heat treating or otherwise transforming an article toward or into its finished state.

(1) Using the [*Amateur-Built Aircraft Fabrication and Assembly Checklist*](#). This checklist is an aid for the ASI or designee in determining if a specific aircraft meets the major portion requirement.

(a) The FAA has also developed [checklists](#) for fixed-wing, helicopter, WSC, powered parachute, and gyrocopter aircraft.

(b) NKET kit evaluations determine whether an aircraft fabricated and assembled by the builder from the kit may meet the major portion requirement. NKET developed [checklists](#) for kits it evaluated.

(c) ASIs or designees must use the checklist when:

(1) Performing NKET evaluations.

(2) A builder used commercial assistance.

(3) A builder made alterations to an aircraft kit included on the kit list that potentially affect the major portion determination.

(4) The aircraft was built from a kit that has not been evaluated or is not on the kit list.

(5) Providing guidance to a kit manufacturer to determine if a proposed kit may meet the major portion requirement of § 21.191(g).

(6) There are questions that arise as to the determination of major portion.

(7) The aircraft was built from prefabricated major components readily available from aircraft part suppliers, other than those components listed in paragraph 15-4.d(2) of this chapter.

(8) The aircraft was built using salvaged articles from other aircraft.

(d) Review the [job aid for completing the checklist](#).

(2) Commercially Produced Products and Articles. The FAA recognizes that builders cannot be expected to fabricate every product and article that makes up the aircraft and that some products and articles will be acquired from commercial sources. Items such as engines, engine accessories, propellers, rotor blades, rotor

hubs, tires, wheel and brake assemblies, instruments, hot air balloon burners, fuel tanks, and standard aircraft hardware, including pulleys, bell cranks, rod ends, bearings, bolts, and rivets, are acceptable and may be procured on the open market. The use of these items is not counted against a builder in assessing the major-portion requirement.

(3) Evaluate Commercial and Educational Assistance.

a. Commercial Assistance. Any fabrication or assembly tasks contracted to another party, performed by a commercial assistance center, or performed by a kit manufacturer, must not prevent the builder from meeting the requirements of § 21.191(g).

b. Commercial Educational Assistance. The builder may receive commercial educational assistance for the fabrication or assembly of specific articles or the completion of tasks involved in building the aircraft. In some cases, educational assistance may be provided by a kit manufacturer. The builder may receive credit for tasks completed with this assistance as long as the assistance did not exceed the demonstration of how to perform the task.

(4) When Builders Use Articles from Other Aircraft. The use of used or salvaged articles, including military surplus articles, from other aircraft is permitted if they are in a condition for safe operation; however, all fabrication, installation, and assembly tasks accomplished with used or salvaged articles will be credited to the “Mfr Kit/Part/Component” column on the *Amateur-Built Aircraft Fabrication and Assembly Checklist*. No credit will be given toward the major-portion requirement for work on these salvaged articles. This includes any “rebuilding” or “restoring” activities to return these articles to an airworthy condition. Assembly credit may be given in those cases where used or salvaged articles are assembled with portions of the aircraft fabricated and assembled by the builder.

(5) Aircraft Previously Certificated by a Foreign CAA. For aircraft that were previously certificated as amateur-built by a foreign CAA, the applicant may provide and the FAA may accept an official document from the CAA stating that the original builder met the requirements of § 21.191(g).

(6) Contact AIR-630 as needed for assistance with a major portion determination.

e. Document Reviews and Inspections. Document all completed reviews and inspections on FAA Form 8100-1, per the instructions on the form. Verify the applicant corrects all unsatisfactory findings, and document those corrections on FAA Form 8100-1 before issuing the airworthiness certificate. FAA Form 8100-1 must be completed even if the action isn’t a full inspection (amendment or replacement). Inspectors and designees should always record when an airworthiness certificate is issued, regardless of how routine the situation is. This builds a historical record of the aircraft’s certification status over time.

f. Issue Airworthiness Certificate. Advise amateur-builders of newly-built or newly-assembled aircraft that have never demonstrated compliance with § 91.319(b), if using the process in AC 90-89, that the aircraft must remain in Phase I flight for at least forty

hours or until completion of task-based flight testing conducted in accordance with the task-based flight test program described in that AC.

(1) Duration of Assignment to Flight Test Area (hourly-based). Phase I operating limitations for amateur-built aircraft should limit operations within an assigned flight test area for at least:

(a) Forty hours when a non-type-certificated engine, propeller, or engine/propeller combination is installed.

(b) Forty hours if an installed type-certificated engine, propeller, or engine/propeller combination has been altered in a way that differs from an approved type design on a TCDS.

(c) Twenty-five hours when an unaltered, type-certificated engine/propeller combination is installed.

(d) Ten hours and at least five takeoffs and landings for gliders, balloons, and airships.

(e) Five hours after any major repair or alteration as defined in § 1.1.

(2) Duration of Assignment to Flight Test Area (task-based). Phase I flight testing may be conducted in accordance with the task-based flight test program described in FAA AC 90-89, within an assigned flight test area.

g. Major Change. FAA Form 8130-6, section 1, blocks 7 and 10 are used by the applicant to provide the aircraft engine and propeller model designations. Advise the aircraft owner to:

(1) Submit a revised FAA Form 8130-6 documenting any future changes to blocks 7 and 10 and send it to the FAA Aircraft Registry at Submit.aircraft.registry@faa.gov

(2) Notify the responsible Flight Standards office of the change and receive its response in writing.

(3) Re-enter Phase I to verify compliance with § 91.319(b)(1) and (2) following any major change or alteration.

Chapter 16. Experimental Purpose of Operating Primary Kit-Built Aircraft (§ 21.191(h))

16-1. Introduction. This chapter provides policies and procedures for issuing a special airworthiness certificate for the experimental purpose of operating primary kit-built aircraft per § 21.191(h).

16-2. Procedures for Issuing an Airworthiness Certificate. Follow chapter 4 of this order and the following:

a. Inspections. Review the application and aircraft records, and inspect the aircraft as necessary to verify:

- (1) The aircraft model has previously been issued a primary category type certificate.
- (2) The kit was manufactured by the holder of the PC for that kit.
- (3) The aircraft was not assembled under the quality system of the PC holder.

b. Reserved.

Chapter 17. Experimental Purposes of Operating Light-Sport Aircraft § 21.191(i), Operating Light-Sport Category Kit-Built Aircraft § 21.191(k), and Operating Former Light-Sport Category Aircraft § 21.191(l)

17-1. Introduction. This chapter provides policies and procedures for issuing experimental airworthiness certificates for the purposes of operating kit-built and former light-sport category aircraft.

Note: The FAA no longer issues a special airworthiness certificate for the experimental purpose of operating light sport aircraft under § 21.191(i), except for replacements. A recurrent airworthiness certificate issued for an aircraft that was originally certificated under 14 CFR 21.191(i)(2) or (i)(3) is issued under 14 CFR 21.191(k) or (l), respectively.

17-2. General.

a. Operating Light-Sport Category Kit-Built Aircraft.

- (1) The design of kit-built aircraft is based on an aircraft of the same make and model that has been certificated in the light-sport category. The experimental kit-built and the light-sport category model share the same configuration.
- (2) A kit-built aircraft must not be modified or altered prior to the experimental airworthiness certificate issuance without prior written approval from the manufacturer.
- (3) Kits of eligible light-sport category models must be designed and assembled per applicable FAA-accepted consensus standards. See the [Light-Sport Category Aircraft](#) website for a current listing of consensus standards accepted by the FAA.
- (4) The kit-built aircraft is not subject to a major portion requirement; that is, any amount of commercial assistance may be used to build the aircraft.
- (5) Because the kit manufacturer does not completely assemble or flight test the kit, the kit manufacturer cannot list certain consensus standards on the manufacturer's SOC, such as for a quality assurance system in § 22.185 that ensures each aircraft conforms to its design and is in a condition for safe operation or production acceptance that covers the ground and flight testing requirements of § 22.195.
- (6) The FAA does not evaluate or approve light-sport category aircraft manufacturers' kits, and, as such, there is no listing of FAA-evaluated or FAA-approved light-sport category aircraft kits or manufacturers.
- (7) Do not perform progressive inspections during the construction or assembly of the aircraft.

b. Operating Former Light-Sport Category Aircraft.

- (1) Eligible aircraft must have previously been issued a special airworthiness certificate in the light-sport category under § 21.190. The aircraft may have subsequently held, or may currently hold, an experimental airworthiness certificate for a valid purpose.

- (2) The provisions of § 21.191(l) allow an owner or operator to continue to operate an aircraft that can no longer comply with the requirements of § 21.181(a)(3), such as the aircraft no longer conforming to its original or properly altered configuration.
- (3) Notify the owner that they are responsible for the operation, maintenance, and continued operational safety of the aircraft.
- (4) Notify the owner that they are responsible for updating aircraft documentation, such as a POH or operating manual, to account for any operating or performance changes, such as from a major alteration or repair.
- (5) These aircraft were originally flight tested during original airworthiness certification for the light-sport category and are not required to have additional flight testing unless, per issued operating limitations, they have major repairs or alterations that were not approved in writing by the light-sport category aircraft manufacturer or a person acceptable to the FAA and recorded in the aircraft records.

17-3. Procedures for Issuing an Airworthiness Certificate. Follow chapters 2 and 4 of this order and the following:

a. Review Application.

(1) FAA Form 8130-6, for aircraft assembled from kits manufactured outside the United States. Since FAA acceptance of these kits are not treated as an import under bilateral agreements, verify that the import box is not checked. Regardless of the class of aircraft specified in the application, verify that the aircraft kit was manufactured in a country with which the United States has an airworthiness agreement concerning U.S. acceptance of airplanes. The agreement may take the form of a Bilateral Airworthiness Agreement concerning airplanes or a Bilateral Aviation Safety Agreement with associated Implementation Procedures for Airworthiness concerning airplanes, or an equivalent airworthiness agreement. Verify that the country of manufacture on FAA Form 8130-15 matches the country of manufacture on Aeronautical Center Form 8050-88A, *Affidavit of Ownership for Amateur-Built and Other Non-Type Certificated Aircraft*, for this aircraft. Contact the FAA Aircraft Registry to obtain Aeronautical Center Form 8050-88A.

(2) Manufacturer's SOC.

(a) Review the aircraft manufacturer's completed FAA Form 8130-15, as required by § 21.193(h)(4) for light-sport category kit-built aircraft, per section 5 of appendix A to this order, for accuracy and completeness.

Note: Light-sport category aircraft may be designed with simplified flight controls per the design requirements in § 22.180. Per § 21.190(d), the aircraft manufacturer must state on the manufacturer's SOC whether the aircraft complies with consensus standards for compliance with § 22.180 design requirements for simplified flight controls.

b. Review of FAA Airworthiness Records on File at the Registry.

(1) For an application for the § 21.191(k) experimental purpose, verify that an aircraft of the same make and model was manufactured and assembled by the aircraft kit manufacturer and issued a light-sport category airworthiness certificate under §

21.190. See the [Light-Sport Category Aircraft](#) website for a current listing of consensus standards accepted by the FAA.

(2) For an application for the § 21.191(l) experimental purpose, verify the aircraft was previously issued an airworthiness certificate in the light-sport category.

c. Review of Aircraft Records and Documents.

(1) Verify all documentation is in the English language.

(2) For kit-built aircraft, verify the aircraft was assembled per the manufacturer's assembly instructions.

(3) For kit-built aircraft, verify the pilot's operating handbook (POH) includes a flight training supplement and operating instructions and limitations to safely accommodate all environmental conditions and normal, abnormal, and emergency procedures likely to be encountered in the aircraft's intended operations. Verify the POH applies to the make, model, and S/N of the aircraft being inspected. It is acceptable for kits with a manufacturer's SOC completed by the kit manufacturer prior to July 24, 2026, to have been issued aircraft operating instructions (AOI) or a POH that does not include a flight training supplement, and the operating instructions and limitations described above.

(4) For former light-sport category aircraft, verify that the applicant has provided either the aircraft's original or revised AOI or POH, applicable to the make, model, and S/N of the aircraft being inspected. It is the applicant's responsibility to include any changes to the operating procedures, performance, or operating envelope of the aircraft as a result of major repairs or alterations.

Note: Advise the applicant that an experimental kit-built or former light-sport category aircraft may not be used for aerial work or towing operations, regardless of whether these operations were included by the aircraft manufacturer in the original AOI or POH, or in documentation for an amended SOC per § 21.190(e).

(5) Weight and Balance Chart. Verify the completed weight and balance chart for the aircraft accurately reflects data in the AOI or POH. Verify the aircraft as presented matches the AOI or POH, and the aircraft equipment listing, or it may not be certificated. Verify that the completed weight and balance chart, including load limits for occupants, oil, fuel, and baggage, is available in the aircraft.

(6) Verify placards and marks reflect the instructions and limitations in the POH or AOI, as applicable.

(7) Maintenance and Inspection Procedures.

(a) Verify the maintenance and inspection procedures are with the aircraft and that they apply to the make, model, and serial number of the aircraft being inspected per § 21.193(h)(3).

(b) Maintenance and Inspection Program. Verify the aircraft has a maintenance and inspection program (i.e., a manual or multiple manuals), applicable to the aircraft make, model and serial number.

(c) The owner or operator is responsible for the content of the maintenance and inspection procedures. For these aircraft, Part 43 does not apply.

(8) Continued Operational Safety (COS). FAA-accepted consensus standards for light-sport category aircraft COS programs are applicable to kits. Verify that the aircraft's documentation includes a COS program. If the COS program relies on electronic media, verify the electronic media and instructions are available.

(9) Aircraft Maintenance Records. Verify that aircraft maintenance records show:

(a) Experimental aircraft are subject to the requirements of § 91.319.

(b) Compliance with all applicable ADs. This requirement applies to an AD for a specific light-sport category aircraft make and model and for an AD issued against a type-certificated product or equipment installed in the light-sport category aircraft. If an AD is issued against a type-certificated product installed in any experimental light-sport category aircraft, the manufacturer of the aircraft is required, per the FAA-accepted consensus standard, to issue a safety directive providing instructions on how to address the AD on the specific aircraft. After issuance of the experimental airworthiness certificate for operating light-sport aircraft or operating former light-sport category aircraft, § 91.327 no longer applies to the owner or operator.

Note: Advise the applicant to follow FAA AC 39-7, *Airworthiness Directives*.

(10) Consistency Among Documents. Verify data contained in the documentation (such as the maintenance manuals, POH, placards, and other manuals incorporated by reference) is consistent. For example, the POH, maintenance manual(s), and aircraft's fuel tank placard all identify the same fuel requirements (with conversion noted).

d. Inspect Aircraft. An airworthiness inspection is accomplished only after the aircraft is complete in every respect and before the issuance of the experimental airworthiness certificate. Do not perform any fabrication, construction, assembly, testing, manufacturer's quality inspections, or closing work on the aircraft. During this inspection, the FAA may only request disassembly when a safety hazard is identified that would endanger the general public.

(1) For kit-built aircraft, verify that the installed equipment matches the aircraft's POH. For a former light-sport category aircraft, verify that the installed equipment matches the original POH, or if altered, verify the installed equipment matches the revised POH.

(2) Verify that the flight control systems and associated instruments, as equipped, operate properly.

(3) Verify the cockpit instruments are appropriately marked as found in the aircraft's POH, and that placards are installed and placed for easy reference.

(4) Verify that the airspeed indicator marks match the POH limitations.

(5) Verify the system controls (for example, fuel selector(s) and electrical switches/breakers) are clearly marked, accessible to the pilot, and function per the applicable instructions and specification documentation.

e. Correction of Discrepancies. Inform the applicant of any omissions, errors, or other discrepancies identified during your inspections. For kit-built light-sport category aircraft, the applicant is responsible for following the manufacturer's instructions. Do not issue the airworthiness certificate if any discrepancies are found with the instructions supplied for a kit. The applicant may contact the kit manufacturer to resolve any discrepancies. If you find discrepancies attributable to a manufacturing issue, contact AIR-630 for further guidance, as it may affect all aircraft produced. Only when the required corrections have been made can an airworthiness certificate be issued.

f. Document Reviews and Inspections. Document all completed reviews and inspections on FAA Form 8100-1, per the instructions on the form. Verify the applicant corrects all unsatisfactory findings, and document those corrections on FAA Form 8100-1 before issuing an airworthiness certificate. FAA Form 8100-1 must be completed even if the action isn't a full inspection (amendment or replacement). Inspectors and designees should always record when an airworthiness certificate is issued, regardless of how routine the situation is. This builds a historical record of the aircraft's certification status over time.

g. Issue Airworthiness Certificate.

(1) Always issue all Phase I operating limitations. In the Phase I flight test operating limitations:

(a) Light-sport category kit-built aircraft are limited to a § 91.305 compliant flight test area for a minimum of five hours until the aircraft has shown it is controllable throughout its normal range of speeds and throughout all the maneuvers to be executed and the aircraft has no hazardous operating characteristics or design features.

(b) Following any major change, an operating limitation assigns the aircraft to a § 91.305 compliant flight test area for an appropriate time (minimum five hours) to comply with Phase I operating limitations. Flight test is completed when the aircraft has shown it is controllable throughout its normal range of speeds and throughout all the maneuvers to be executed and the aircraft has no hazardous operating characteristics or design features.

(c) Refer to paragraph 17-2b.(5) for flight testing requirements for former light-sport category aircraft. The aircraft may not be required to complete a flight test program under Phase I if the applicant shows no unauthorized alterations or changes were made after issuance of the original light-sport category airworthiness certificate.

(2) For any unmitigated very high-risk factors such as exceeding applicable § 21.181(a)(3) weight limits, noncompliance with ADs, or major repairs or alterations

that were not authorized by the aircraft manufacturer, consider whether issuing the operating limitations of paragraph D-4.f is necessary to mitigate these risks.

Chapter 18. Experimental Purpose of Operating Former Military Aircraft (§ 21.191(j))

18-1. Introduction. This chapter provides policies and procedures for issuing special airworthiness certificates for the experimental purpose of operating former military aircraft under § 21.191(j).

18-2. Procedures for Issuing Airworthiness Certificates. Follow the procedures in chapter 4 of this order and the following:

a. Review Application. Verify the application, including the program letter, demonstrates eligibility for the requested experimental purpose. An aircraft is eligible for an airworthiness certificate for the experimental purpose of operating former military aircraft when it:

- (1) Is not an unmanned aircraft.
- (2) Was manufactured by, purchased by, modified by, or on the registry of the U.S. Armed Forces or a foreign military.
- (3) Is operated for the purposes of--
 - (a) Flying the aircraft to a base where repairs, alterations, or maintenance are to be performed and for check flights following those repairs, alterations, or maintenance,
 - (b) Flying the aircraft to a point of storage, or
 - (c) Repositioning the aircraft for operation as a public aircraft. Such repositioning is for the convenience of and at the expense of the operator and not a mission directed by a government agency

b. Review Aircraft Records. Review historical military records as necessary to verify the aircraft is a former-military aircraft.

Chapter 19. Special Flight Permits (§ 21.197)

Section 1. Common Policies and Procedures.

19-1. Introduction.

a. A special flight permit (SFP) is a special airworthiness certificate issued for a U.S.-registered aircraft that may not currently meet applicable airworthiness requirements but is capable of safe flight. An SFP does not authorize:

- (1) Flight over a foreign country. See paragraph 4-3 for additional guidance for issuing a special airworthiness certificate outside the United States.
- (2) A deviation from a requirement of 14 CFR.

b. Section 1 of this chapter provides common policies and procedures for issuing an SFP. Section 2 of this chapter provides additional policies and procedures for issuing SFPs for specific purposes.

c. Section 21.197(c) applies to air carriers and fractional ownership operators. Procedures for issuing an SFP under § 21.197(c) are contained in FAA Order 8900.1, volume 4, chapter 13, section 1.

d. Use AWC per paragraph 2-3.a for issuing an SFP.

19-2. Who May Issue an SFP?

a. The FAA may delegate the issuance of an SFP. See FAA Order 8000.95 concerning related designee management and oversight policies and procedures.

b. Except as provided in paragraphs 19-2.c and 19-2.d, an SFP for purposes other than production flight testing and customer demonstration flights will be issued by the Flight Standards Office, AIR CM Section, or International Field Office (IFO) that has accepted responsibility for the issuance of the SFP. If the applicant's aircraft is outside the geographic area office receiving the request, the applicant should be referred to the appropriate office. AIR and Flight Standards are responsible for defining respective geographic areas of responsibilities for their respective offices.

c. For a certificated operator:

- (1) With a continuous authorization requesting an SFP for a purpose outside the scope of its authorization, the responsible Flight Standards Office will issue the SFP.
- (2) Without a continuous authorization, the responsible Flight Standards Office for that operator will issue the SFP. However, with the concurrence of the responsible Flight Standards Office, the SFP may be issued by the office having geographic responsibility.

d. Under special conditions, SFPs may be issued to owners or operators at Part 145 repair facilities for delivering aircraft from international locations to the United States. In this instance, the SFP will be issued by the responsible Flight Standards Office having jurisdiction over the repair facility under the following conditions:

- (1) It is a U.S.-registered aircraft that currently does not meet the conditions of its standard airworthiness certificate, due to the installation of non-standard auxiliary fuel

systems. Auxiliary fuel system installations must be accomplished by an FAA-certificated repair facility which is specifically airframe rated for the desired installation.

(2) Procedures relating to the application and issuance of SFPs, the installation of auxiliary fuel systems, and any conditions and limitations for flight must be incorporated into the repair facility's operations specifications.

Note: The FAA office issuing the SFP under these special procedures must meet all other applicable requirements of this chapter. The responsible Flight Standards Office may request the IFO with geographic responsibility for the area in which the flight is to originate to inspect the aircraft before issuance of the SFP and/or before flight.

19-3. Purposes. Section 21.197 prescribes the general purposes for which an SFP may be issued. These specific flight operations are considered within the scope of § 21.197 if the aircraft is capable of safe flight, for the following purposes:

- a. Flying the aircraft to a base where repairs, alterations, or maintenance are to be performed, or to a point of storage. This may include incidental check flights as necessary to verify that the aircraft is safe to fly to a base where repairs are to be performed. Such check flights could be included under the SFP to verify the proper function of auxiliary fuel tanks, confirm there are no hydraulic or fuel leaks for aircraft coming out of storage, etc. This provision cannot be used to circumvent § 91.407.
- b. Delivering or exporting the aircraft.
- c. Production flight testing new production aircraft. This purpose is intended for aircraft that are produced per § 21.123 and § 22.195.
- d. Evacuating aircraft from areas of impending danger.
- e. Conducting customer demonstration flights in new production aircraft that have satisfactorily completed production flight tests.
- f. Flying an aircraft at a weight in excess of its maximum certificated takeoff weight for flight beyond the normal range over water, or over land areas where adequate landing facilities or appropriate fuel is not available.
- g. A special flight permit with a continuing authorization may be issued to certificate holders under Part 119 or management specification holders under Part 91, subpart K, for those aircraft they operate under § 91.141

19-4. Procedures.

- a. Review and Complete Application. Review the application per paragraph 2-3.b. of this order.
- b. Review FAA Airworthiness Records on File at the Registry. Access the Registry to verify the aircraft is currently registered per paragraph 2-3.c.
- c. Review Aircraft Records and Inspect Aircraft.
 - (1) Review records and conduct inspections or tests as necessary to ensure the aircraft is capable of safe flight for the intended purpose. Reviews and inspections should

include those areas where the aircraft does not meet applicable airworthiness requirements as identified in section VII of the application. The FAA may require a certificated mechanic or repair station to conduct the necessary review of records and aircraft inspection(s) to support issuance of an SFP. However, the FAA should physically inspect damaged aircraft or an aircraft for which the safety of flight is questionable in any respect. If the intended operation would exceed FAA approved limits such as certification maintenance requirements or airworthiness limitations, ensure the application includes supporting data for safe flight from the applicable certification branch.

Note: All designees must (1) physically review the records and physically inspect the aircraft to ensure an aircraft is capable of safe flight for the intended purpose, (2) document all completed reviews and inspections on FAA Form 8100-1, per the instructions on the form, (3) verify the applicant corrects all unsatisfactory findings, and (4) document the inspection in the aircraft maintenance record per paragraph 2-3h.

(2) If a special inspection is recommended by the manufacturer, review records to verify compliance. When the FAA requires the applicant to make the inspection, advise the applicant that such inspections must be:

(a) Accomplished by an appropriately certificated mechanic or authorized repair station familiar with the procedures and requirements of this chapter.

(b) Documented in the aircraft maintenance record by the authorized person who conducted the inspection.

(3) ADs.

(a) For Aircraft. If the AD does not prohibit issuance of an SFP to fly the aircraft to a repair facility to do the work required by the AD, you may issue an SFP per 14 CFR 39.23.

(b) Not for Aircraft. For the purpose of this paragraph, product means aircraft engine, propeller, or article. If the AD does not allow operation of the product during a flight, then it may not be operated in flight under an SFP. If the aircraft on which the product is installed can be operated safely with that product inoperative, an SFP could be issued per § 21.197(a) with a limitation that the product be rendered inoperative for flight.

d. Document Reviews and Inspections. Document all completed reviews and inspections on FAA Form 8100-1, per the instructions on the form. Verify the applicant corrects all unsatisfactory findings, and document those corrections on FAA Form 8100-1 before issuing an airworthiness certificate.

e. Issue Airworthiness Certificate.

(1) Operating Limitations. Establish operating limitations as necessary for safe operation. Because individual circumstances may vary greatly, a list of limitations applicable in every case cannot be provided. If necessary, solicit technical assistance from other FAA offices or specialists. Limitations should be clear and concise so they can be easily

understood. Consider Table D-1 as an aid for identifying and issuing appropriate operating limitations. Consider the following in establishing operating limitations:

- (a) The specific reasons identified in section VII of the application why the aircraft does not meet applicable airworthiness requirements, any related hazard to persons and property on the ground, and corresponding operating limitations concerning carriage of passengers, and operations over densely populated areas.
- (b) Conformity to the aircraft's technical data.
- (c) Operational equipment necessary for safe operation of the aircraft, including the need for IFR equipage.
- (d) Aircraft weight limits.
- (e) Fuel and fuel distribution limits.
- (f) CG limits.
- (g) Maneuvers to which the aircraft is limited.
- (h) Limits on use of flight equipment, such as autopilots.
- (i) Meteorological conditions to be avoided and the inspections required if inadvertently encountered.
- (j) Airspeed limits.
- (k) Flight operations in an overweight condition, non-compliant with airworthiness directives, conducting engine-out operations, or any aircraft meeting the criteria of paragraph D-4.f must include an operating limitation to avoid cities, towns, villages, and congested areas, or any other areas where the flights might create hazardous exposure to persons or property. The operating limitation for aircraft meeting the paragraph D-4.f criteria must include the restrictions of paragraph D-4.g. Aircraft with overdue maintenance or inspections, but no other known hazards, may conduct a takeoff and landing over populated areas if necessary. Inspectors must consider the reason for the SFP to determine if allowing takeoffs and landings over populated areas is appropriate.
- (l) Runway selections, if considered necessary for safety, such as aircraft performance or avoidance of densely populated areas.
- (m) Communications required with airport tower personnel to inform them before takeoff or landing of the non-standard condition of the aircraft.
- (n) Intended operations over a foreign country. An ASI should inform the CAA if the applicant intends to conduct flight operations over another country; the designee or ASI should inform the applicant of its responsibility to obtain permission from that country prior to such operations. When required to fly over another ICAO [member state](#), include, when appropriate, the following statement: "This aircraft does not comply with the international standards of Annex 8 to the Convention on International Civil Aviation as follows: *[describe the item(s) which do not comply with the airworthiness requirements for standard aircraft]*"

- (o) Any inspection requirements.
- (p) Any other limitation that should be prescribed for the particular flight.
- (q) Any limitation required by an Airworthiness Directive.
- (r) An appropriate flight area, as applicable.

(2) Issue the SFP.

(a) When you find the application for an SFP in compliance with all requirements and the aircraft is capable of safe flight, issue FAA Form 8130-7 per paragraph A-4 of appendix A to this order with operating limitations deemed necessary for safe flight operation.

(b) Give the completed airworthiness certificate to the applicant. Except for an SFP for overweight operations, an ASI may transmit the completed and signed SFP to the applicant electronically. A designee must physically provide the SFP to the applicant.

(3) Advise the Applicant.

(a) Review the operating limitations with the applicant to ensure a clear understanding of the limitations.

(b) Advise the applicant to display the certificate in the aircraft at the cabin or cockpit entrance so the certificate is displayed per § 91.203(b). The copy of the permit that is displayed in the aircraft at the point of departure will be considered the original permit.

(4) Denials. If, after notifying the applicant of all unsatisfactory findings, the applicant does not correct those findings to your satisfaction in a timely manner, write a letter to the applicant denying issuance of the certificate and stating the reason(s) for the denial. If feasible, identify which steps may be accomplished to meet the certification requirements. That is, if you are unable to find the aircraft capable of safe flight, do not issue the SFP.

f. Review and Forward Records. Review your records to verify completeness, accuracy, legibility, and compliance with applicable requirements. Forward and/or maintain records as required by your office.

Note: Reviewing and forwarding certification files to the FAA Aircraft Registry for an SFP is only required for overweight operations.

Section 2. Additional Policies and Procedures for Particular Purposes.

19-5. Production Flight Testing of Aircraft Produced under a TC/PC. An SFP issued for production flight testing may be used by a manufacturer to meet the requirements of § 91.203 when operating new production aircraft for production flight testing, as provided in § 21.197. This permit may be used with any valid registration and is transferable from one aircraft to another. Normally, the permit is valid only for production flight testing. However, when deemed appropriate, the AIR CM Section or Integrated Certificate Management Division (ICMD) may allow both production flight testing and customer demonstration to be entered in block A of FAA Form 8130-7 as explained in paragraph 19-7 of this order. This paragraph provides policies and procedures concerning the issuance of aircraft produced under a TC/PC. For production flight testing of a light-sport category aircraft, refer to paragraph 19-6 of this order. For production flight testing of an unmanned aircraft system (UAS), refer to FAA Order 8130.34.

a. Eligibility:

- (1) A manufacturer producing aircraft under Part 21, subpart F or G, is eligible to obtain an SFP for production flight testing.
- (2) A manufacturer producing aircraft before issuance of the TC is also eligible for an SFP for production flight testing, provided the following conditions are met:
 - (a) The FAA official flight test program is in progress.
 - (b) A prototype aircraft of that model has been flown by the manufacturer under an experimental airworthiness certificate to ensure there are no adverse flight characteristics and that production test pilots are fully familiar with the aircraft.
 - (c) An FAA-accepted production flight test procedure and checklist for the aircraft involved will be used to ensure all requirements for production flight tests are fulfilled.
 - (d) The aircraft is not being flown by the manufacturer for purposes other than production flight tests, except as identified in paragraph 19-7 of this order.
 - (e) Limitations have been established to define the production flight test area.
- (3) Military Aircraft.
 - (a) Aircraft built under a military contract with military aircraft identification marks do not require registration or the issuance of an airworthiness certificate for flight testing or demonstration before acceptance by the military. Further information regarding type certification of military commercial derivative aircraft can be found in FAA Order 8110.101.
 - (b) There may be cases where a TC/PC holder is building a military aircraft not pursuant to a contract and that are not produced under its PC. The aircraft manufacturer may be eligible for an SFP for production flight testing under § 21.197(a)(3). The aircraft should have completed R&D testing prior to production flight testing. The manufacturer maintains ownership of the aircraft during production flight testing.

b. Application and Issue.

(1) An eligible manufacturer should apply for as many SFPs for production flight testing as deemed necessary for satisfactory coverage of the aircraft involved. The number of SFPs for production flight testing issued to the manufacturer is limited to actual need, not to exceed the number of temporary registration numbers (§ 47.16), as applicable.

(2) The CM Section that has issued SFPs for production flight testing should maintain suitable records that show expiration dates and the number of permits issued to each manufacturer. For tracking purposes, it is recommended that each permit issued be numbered serially in the upper right corner of the airworthiness certificate by the issuing office; for example, AIR-882 # 1. A special flight permit is effective for the period of time specified in the permit.

19-6. Production Flight Testing of Light-Sport Category Aircraft. As provided for in § 21.197(a)(3), an SFP may be issued to a light-sport category aircraft to meet the requirements of §§ 21.190(c)(7) and 91.203 to operate new production aircraft for production flight testing.

a. Coordination. The CM Section will coordinate the production flight testing activities with the geographic or assigned Flight Standards Office.

b. Eligibility for Production Flight Testing in the United States. A manufacturer producing light-sport category aircraft under § 21.190 is eligible to obtain a single SFP or transferable SFPs for production flight testing if the manufacturer shows the FAA that:

(a) A prototype aircraft of that light-sport category aircraft model and configuration has been flown by the manufacturer under an experimental R&D certificate to comply with the requirements of § 91.319(b).

(b) The purpose of the operation is production flight testing only.

(c) It will follow its quality assurance system, production assurance flight test procedures, and checklists that meet applicable FAA-accepted consensus standards and record satisfactory completion of testing in the aircraft maintenance records.

c. Issue SFP.

(1) An SFP may be issued for a single aircraft or one that is transferable from one aircraft to another.

(2) For transferable SFPs, issue an appropriate number of SFPs for production flight testing based on actual need, not to exceed the number of temporary registration numbers (§ 47.16), as applicable. See § 47.16, *Temporary Registration Certificates*, and § 47.61, *Dealer's Aircraft Registration Certificates*, for additional information. Maintain suitable records that show expiration dates and the number of permits issued to each manufacturer. For tracking purposes, it is recommended that each permit issued be numbered serially in the upper right corner of the airworthiness certificate by the issuing office; for example, AIR-882 # 1.

(3) **Flight Test Duration.** In addition to the applicable operating limitations previously identified under paragraph 19-4e.(1), include the following:

- (a) This aircraft may be operated only for production flight testing. These operating limitations are a part of an SFP and must be carried in the aircraft and available to the PIC of the aircraft.
- (b) All flights must be conducted within the geographical area described as follows. [*Use the applicant's proposed operating area, if reasonable*] Include either a written description of the geographic area, a pictorial description, or an aeronautical chart. The chart must be clear, legible, and sized appropriately to provide necessary detail. Advise the applicant that it is their responsibility to comply with all airspace procedures, requirements, and prohibitions, including avoiding overflight of densely populated areas, if specified in the operating limitations. Coordination with an operations ASI is recommended.
- (c) All flight tests must be conducted and recorded per an FAA-accepted consensus standard for production acceptance.
- (d) This aircraft is to be operated under visual flight rules (VFR), day only.
- (e) The test pilot is to be the sole occupant for production flight testing.

19-7. Conducting Customer Demonstration Flights. An SFP may be issued to an aircraft to meet the requirements of § 91.203 when operating a new production aircraft for conducting customer demonstration flights per § 21.197(a)(5). However, as stated in paragraph 19-5 of this order, customer demonstration may be listed in block A of FAA Form 8130-7 along with production flight testing but will not be issued in conjunction with any other SFP purpose. When both flight purposes are listed in block A of FAA Form 8130-7, the aircraft's operating limitations must clearly state that no customer demonstration flights are allowed until the aircraft has satisfactorily completed its production flight tests. The format for listing both flight purposes is "Production Flight Testing or Customer Demonstration."

a. Eligibility. An SFP for conducting customer demonstration flights may be issued when the following conditions are met:

- (1) The new production aircraft was produced under a PC or TC.
- (2) The PC/TC holder has satisfactorily completed production flight tests. Completion of production flight tests indicates acceptance by the production flight test pilot and no further flight tests are required or planned.

b. Application and Issue.

- (1) A letter from the manufacturer must accompany the application describing the customer demonstration flights to be made if sufficient information cannot be included on the application.
- (2) Upon receipt of a properly executed application, inspect the aircraft, and prescribe the operating limitations per paragraph 19-4.e(1) of this order and as deemed necessary for safe operation. The demonstration flight area(s) also must be listed on the operating

limitations. SFPs may be issued only for the period needed to complete demonstration flights, usually not to exceed 90 days.

(3) Permits that are transferable from one aircraft to another may be issued. It is still necessary to prescribe operating limitations per paragraph 19-4.e(1) of this order and as deemed necessary for safe operation. A reasonable expiration date should be used on the FAA Form 8130-7 and the associated limitations. Issue an appropriate number of SFPs for conducting customer demonstration flights based on actual need, not to exceed the number of temporary registration numbers (§ 47.16), as applicable.

(4) The CM Section issuing SFPs for customer demonstration flights will maintain a copy of the complete file per record retention requirements.

19-8. Operation of Overweight Aircraft.

a. General. An SFP may be issued for the operation of overweight aircraft for flight beyond the normal range over water, or over land areas where adequate landing facilities or appropriate fuel is not available. Per § 21.197(b), the excess weight that may be authorized must be limited to the additional fuel, fuel-carrying facilities, and navigation equipment necessary for the flight.

(1) The FAA has two primary concerns when issuing SFPs for the temporary operation of overweight aircraft:

(a) That the public will be guarded in the event of an accident; and

(b) That when the aircraft is returned to a standard configuration, it has not been rendered unairworthy due to the overweight operations.

(2) With safety being the primary concern, it is essential that the processing office use the technical assistance of other FAA offices or specialties as deemed necessary to ensure the highest degree of safety possible. All installations, for example, a long-range fuel system or navigational equipment, must be installed per FAA-approved data or service bulletins.

(3) All applications for rotorcraft must be coordinated with a certification branch for an engineering evaluation of the structural integrity, the flight integrity, and for any other provisions deemed necessary.

(4) Applications for which the proposed maximum weight does not exceed 110 percent of the maximum certificated weight, and for which the certificated CG limits are not exceeded, may be processed by CM Sections and the responsible Flight Standards Office without obtaining an engineering evaluation.

(5) Applications for which the proposed maximum weight exceeds 110 percent of the maximum certificated weight, or the CG limits exceed the certificated limits, must be coordinated with a certification branch for an engineering evaluation of the structural integrity and for any other provisions deemed necessary.

(6) The processing of an application must encompass a review of the airworthiness status of the basic aircraft, an evaluation of the added installations that constitute the excess weight, required flight crew member qualifications, and proposed operating limitations.

b. Added Installations.**(1) Technical Data.**

- (a) Verify that any drawings and reports submitted with the application substantiate structural integrity sufficiently and are detailed to show that the added installations are structurally and functionally safe and allow for a conformity inspection of the added installations.
- (b) The structural report should reference the drawings used for the installation(s).

(2) Record of Installation(s).

- (a) The installation(s) added to the aircraft for the intended overweight flight must be recorded per the requirements of § 43.9.
- (b) The following statement must be entered in section 3 of FAA Form 337: “No person may operate this aircraft, as altered herein, unless it has within it an appropriate and current special flight permit issued under Part 21.”

(3) Auxiliary Fuel System Installations. In the evaluation of the auxiliary fuel system installations, the following items will be considered:

- (a) The aircraft and auxiliary fuel system should meet all applicable original airworthiness requirements, except for those the aircraft cannot meet because of its overweight condition. The aircraft and auxiliary fuel system must be found safe for the intended flight.
- (b) Fuel tank(s) installed in a pressurized area are vented outside the aircraft and should be tested to show the empty tank(s) can withstand the maximum pressure differential at the maximum operating altitude.
- (c) The ability of the fuel tank(s) and surrounding area to provide adequate ventilation to prevent the accumulation of fumes that would be detrimental to the flight crew or present a fire or explosion hazard.
- (d) The ability to determine the quantity of fuel in the auxiliary tank(s) before takeoff. In addition, an ability to indicate the quantity of fuel in tanks that have a vapor/excess fuel return line, both before takeoff and during flight.
- (e) The location of the fuel tank(s) in the aircraft is a major factor in determining that the aircraft is safe for flight because the added fuel and fuel facilities have the greatest effect on the aircraft's CG. In addition, the fuel system installation must not prevent the means of egress, using normal openings, exits, or emergency exits, as provided by the aircraft's applicable airworthiness requirements. If required for landing, the aircraft should have an adequate fuel jettison system installed.
- (f) Auxiliary fuel systems that are not complete, that is, not connected to the basic aircraft fuel system, may not be considered for issuance of an SFP.

(4) Engine Oil Quantity. The ability of the oil supply provided for each engine to ensure satisfactory cooling and system circulation for the duration of the flight. If applicable, the ability of an oil transfer system to replenish the engine oil while the aircraft is in flight.**(5) Maximum Weight and Center of Gravity Limits.**

(a) Section 21.197(b) limits any excess weight over the certificated maximum weight to additional fuel, fuel carrying facilities, and navigational equipment added for the intended flight. It must be determined that this part of the maximum weight complies with this requirement.

(b) When numerous alterations are performed, it may be necessary to weigh the aircraft to establish the aircraft weight and the CG limits. The computations should be evaluated for accuracy. It also may be necessary to require flight testing at the new maximum weight and CG limits to determine that the aircraft is safe for operation. Computed weight and balance information should be reflected on FAA Form 337, section 8.

(c) Operation of rotorcraft over the certificated maximum weight presents some unique conditions over and above those encountered with fixed wing aircraft. Special attention should be given to this type of aircraft. A careful evaluation should be made to determine what effect the overweight operation may have on the retirement times of critical articles.

(6) Operating limitations must include—

(a) Operation in the overweight condition must be conducted to avoid cities, towns, villages, and congested areas, or any other areas where the flights might create hazards to persons or property.

(b) Runway [*specify*] must be used for overweight takeoff (and landing when appropriate). If an in-route stop is scheduled, the following must be added to this limitation: Contact FAA office, [*city, routing symbol, and telephone number*] for runway to be used for overweight takeoff and landing at [*city*].

(c) A copy of FAA Form 337 covering the additional fuel carrying facilities and equipment must be in the aircraft.

(d) Special entries to note required inspection of the aircraft for possible damage due to overweight operation upon completion of overweight flight(s).

c. Review Records and Forward to the Registry. Review and forward records per appendix B of this order.

19-9. Aircraft to Which § 91.409(e) Is Applicable.

a. Eligibility. An SFP may be issued for certain large aircraft for which 14 CFR Part 125, *Certification and Operations: Airplanes Having a Seating Capacity of 20 or More Passengers or a Maximum Payload Capacity of 6,000 Pounds or More*, is not applicable.

b. Application and Issue.

(1) Verify the application includes a clear and complete status of the aircraft. Verify that section VII of the application describes why the aircraft does not meet the applicable FAA airworthiness requirements, including an assessment of AD compliance, major repair, major alteration status, life-limited parts status, aircraft equipment, accessory configuration, and Certification Branch or manufacturer recommendations concerning operations that would exceed a certification maintenance requirement.

(2) Verify the application includes the minimum requirements for the aircraft to be considered safe for operation per established industry practices (e.g., MMEL, configuration deviation list, and AFM).

(3) Before you issue the SFP, the applicant must select, identify in the aircraft maintenance records, and use one of the inspection programs specified in § 91.409(f). All such programs must specify the parts and areas of the airframe, engines, propellers, rotors, and appliances, including survival and emergency equipment, that must be inspected, as specified in §§ 91.409(g)(1), 121.135(b)(18), 135.427 and 135.419(d). The FAA may specify additional inspections and/or tests required to ensure safe operation. The scope and detail of the inspections and/or tests required to ensure safe operation may vary considerably depending on why the permit is issued and/or the conditions or circumstances surrounding the subject aircraft. Aircraft that have been routinely maintained and/or inspected under an approved inspection program may not need additional inspections.

Note: Only an airworthiness ASI can approve an inspection program under § 91.409(f)(4).

(4) Issue the SFP. Restrict the operation of the aircraft to specific airports and to a specific flight path to discourage unnecessary flight operations. The destination airport is the final location for the purpose of the flight. Specify an appropriate duration.

Note: At your discretion, consult an operations ASI, type rated in the same or similar aircraft, concerning the adequacy and appropriateness of the conditions and limitations of the SFP.

(5) Examples. The following examples illustrate how the above procedures may be applied:

Example 1: Continuous Airworthiness Inspection Program: ABC Airlines, operating a Boeing 777 aircraft in air carrier service, wishes to lease another Boeing 777 from XYZ Leasing. The aircraft to be leased has been in storage for one year. ABC Airlines wishes to operate the aircraft from the point of storage to a maintenance facility before placing the aircraft in service with the airline. ABC Airlines must select an inspection program under § 91.409(f), identify it in the maintenance records, and use it. ABC Airlines selects the inspection program that is part of ABC Airlines' Continuous Airworthiness Maintenance Program (CAMP) for its Boeing 777, as permitted under § 91.409(f)(1). If the CAMP contains provisions for inspection before operation under a special flight permit, those provisions may be used to ensure safe operation of the aircraft. If the CAMP does not contain such provisions, the CAMP may still be selected; however, the FAA may require ABC Airlines to make additional inspections or tests necessary to ensure safe operation as part of the SFP process.

Example 2: Inspection Program Recommended by the Manufacturer: XYZ Leasing wishes to operate its Airbus 320 from one storage location to another. When applying for the SFP, XYZ submits a description of the inspections and tests it considers necessary to ensure safe operation of the aircraft. The inspection program selected and identified in the maintenance records is the manufacturer's

recommended program that meets § 91.409(f)(3). Upon review and evaluation of the application, the FAA issues the SFP with any specific operational conditions and limitations under which XYZ may operate its aircraft following the satisfactory completion of the inspections and tests described in the selected inspection program. XYZ must comply with all the applicable inspections and tests described in the selected inspection program prior to operating the aircraft.

Example 3: Other Inspection Program: XYZ Leasing wishes to operate its Airbus 320 from one storage location to another. When applying for the SFP, XYZ submits a description of the inspections and tests it considers necessary to ensure safe operation of the aircraft. Upon review of the submitted description, the airworthiness ASI approves the description of the inspections and tests as a one-time use inspection program under § 91.409(f)(4). The FAA issues the SFP with any specific operational conditions and limitations under which XYZ may operate its aircraft following the satisfactory completion of the inspections and tests described. XYZ must identify the inspection and tests approved by the FAA in the maintenance records as the inspection program under which the aircraft is to be operated for the purpose of this flight only.

c. Special Cases.

(1) Aircraft Coming Out of Storage. For an aircraft that has been out of service for an extended period, its storage conditions should be evaluated. In many cases, aircraft in storage have been routinely maintained, inspected, and have had preventive maintenance performed at regular intervals. These aircraft normally would require less attention before any anticipated flight. However, an aircraft that has been in storage for an extended period should undergo an extensive visual inspection by a properly certificated mechanic, an inspection of the fuel storage and delivery systems for contamination, and operational checks of all systems and equipment that may be used on the intended flight. Aircraft that have been stored for an extended period should have had the preservation procedures recommended by the engine manufacturer in place during the storage period. If not, the engine manufacturer should be contacted for a return to service work-scope to be completed on the installed engines.

(2) Aircraft Going into Storage. In some cases, the applicant may not intend to place the aircraft in service following the flight authorized by the SFP. In this case the applicant may wish to use an inspection program as specified in the § 91.409(f)(4) example above. Unless provisions for additional flights are provided for in the FAA-approved program, no additional flights are permitted.

(3) Damaged Aircraft. An aircraft that has been damaged may require engineering evaluations or special tests to determine if the aircraft is capable of safe flight.

(4) Carriage of Passengers. When the flight characteristics of the aircraft have not been appreciably altered, persons other than flightcrew and/or persons essential to the operation of the aircraft may be carried aboard during non-revenue flight operations authorized by an SFP. In those cases, the passenger carrying requirements of Part 91 will apply.

(5) The operation of noise restricted aircraft that do not comply with the requirements of § 91.801 requires a special flight authorization issued per §§ 91.858 or 91.859 and, except for emergency relief authorizations under § 91.883(a)(7), must be obtained by applying 30 days in advance to the Office of Environment and Energy (AEE). An SFP is not required in these instances and will not be issued unless the aircraft does not meet applicable airworthiness standards as provided in § 21.197. All other inspection program requirements apply.

Chapter 20. Provisional Airworthiness Certificates (14 CFR Part 21, Subpart I)

20-1. Introduction. This chapter provides policies and procedures for issuing a provisional airworthiness certificate under **14 CFR, Part 21, Subpart I**.

20-2. Procedures for Issuing an Airworthiness Certificate. Follow the procedures in chapter 4, section 1 of this order, and the following:

a. Review Application.

- (1) Applicant. Verify the applicant meets § 21.213.
- (2) Purpose. Verify the applicant's intended purpose meets § 91.317.
- (3) Statement of Conformity. Review FAA Form 8130-9, *Statement of Conformity*, to verify proper completion and to verify it contains the information required by §§ 21.221(c), 21.223(c), or 21.225(c), as applicable.
- (4) Current TC. Verify the corresponding provisional TC or provisional amendment to the TC is in effect.

b. Review of Aircraft Records. Per §§ 21.221(d) and 21.223(e), review aircraft maintenance records as necessary to verify that the aircraft was flown at least five hours by the manufacturer.

c. Inspect the Aircraft.

- (1) Condition for Safe Operation. Because the aircraft normally is one that is being used in the type certification process, the FAA should be familiar with its progress and conformity status. Therefore, per §§ 21.221(a)(2), 21.223(a)(2), or 21.225(a)(2), as applicable, inspect the aircraft as necessary to find that no feature, characteristic, or condition of the aircraft would make the aircraft unsafe when operated per the applicable operating limitations.
- (2) Flight Manual and Placards.
 - (a) For § 21.81 class I or provisional amendment to a TC, verify the aircraft was supplied with a provisional AFM or other document and appropriate placards containing the applicable operating limitations.
 - (b) For § 21.83 class II or provisional amendment to a TC, verify the aircraft was supplied with a provisional AFM containing the applicable operating limitations.

d. Issue Airworthiness Certificate.

- (1) Operating Limitations. Operating limitations established for the issuance of a provisional TC or provisional amendment to a TC are a part of the provisional airworthiness certificate issued to an aircraft. Operating limitations that are not included in placards or the provisional flight manual must be issued per appendix D to this order. Based on your inspections and assessment of potential safety hazards, prescribe additional operating limitations you consider necessary in the interest of safety.
- (2) Reserved.

Chapter 21. Import Aircraft

21-1. Introduction. This chapter provides policies and procedures for issuing U.S. airworthiness certificates for imported, type-certificated aircraft. Part 21 provides for the import of aircraft as follows:

- a. New Standard Classification Aircraft Under § 21.183(c).** For an application for an airworthiness certificate for a new, imported aircraft in the standard classification, follow the applicable procedures in paragraph 3-5 of this order and this chapter.
- b. Used Standard Classification Aircraft Under § 21.183(d).** The FAA issues airworthiness certificates under this section for used, standard classification aircraft imported into the United States. For an application for an airworthiness certificate for a used, import aircraft in the standard classification, follow the applicable procedures in paragraph 3-6 of this order and this chapter.
- c. New Primary Category Aircraft Under § 21.184(b).** For an application for an airworthiness certificate for a new, imported aircraft in the primary category, follow the procedures in paragraph 5-4 of this order and this chapter.
- d. Used Primary Category Aircraft Under § 21.184(d).** For an application for an airworthiness certificate for a used, imported aircraft in the primary category, follow the procedures in paragraph 5-6 of this order and this chapter.
- e. Restricted Category Aircraft Under § 21.185.** For an application for an airworthiness certificate for an import aircraft in the restricted category, follow the procedures in chapter 6 of this order and this chapter.

21-2. CAA Assistance with U.S. Airworthiness Certificates for New Aircraft Manufactured Outside the United States. Refer to appendix E of this order for the related procedures.

21-3. Export Certificate of Airworthiness (Export C of A).

a. Purpose. An export C of A issued by an exporting CAA assists the FAA in determining if an imported, type-certificated aircraft is airworthy. An export C of A also helps identify any exceptions that must be addressed before an airworthiness certificate may be issued.

Note: An export C of A is not an airworthiness certificate and does not authorize flight operations of an aircraft.

b. Required for New Aircraft. An export C of A is mandatory for the import of new standard classification aircraft, new primary category aircraft, and new restricted category aircraft per §§ 21.183(c), 21.184(b), 21.185(c), respectively.

c. Desirable but Not Mandatory for Used Aircraft.

(1) Although countries with which the United States has concluded bilateral agreements concerning airworthiness have agreed to provide export certificates of airworthiness with used aircraft exported under those agreements, an export C of A is not mandatory. Similarly, an export C of A is not mandatory for used aircraft exported from non-

bilateral countries. That is, the FAA may not deny issuance of an airworthiness certificate for a used imported aircraft solely because an export C of A has not been issued or provided. However, without an export C of A for a used aircraft, the applicant's showing and the FAA's finding of airworthiness may be burdensome, impractical, or not feasible due to the need for conducting a complete conformity inspection and a lack of available design data for such inspections. Advise applicants accordingly. For these reasons and the purposes described in paragraph 21-3.a of this chapter, obtaining an export C of A is always desirable.

(2) An aircraft manufacturer typically retains data enabling it to develop a conformity statement for any aircraft it manufactured, including any design changes that did not conform to U.S. type design when the aircraft was manufactured. Such manufacturer statements are acceptable for establishing the conformity status of the aircraft when manufactured. Beginning with that baseline conformity status, an applicant could show that subsequent maintenance, repair, alteration, compliance with Airworthiness Directives, etc., preserved, established, or restored conformity to the U.S. type design. Such a showing by an applicant is necessary to enable an exporting authority to issue an export certificate of airworthiness and the FAA to issue a standard airworthiness certificate. This policy affirms that the use of manufacturer statements of conformity as described above are acceptable even in the absence of an endorsement of that statement by the CAA of the State of Design, State of Manufacture, or their technical agent. This policy is not intended to supersede any agreement for an exporting authority to provide an export certificate of airworthiness, or equivalent statement, per the procedures in those agreements. This policy takes precedence over any statement in a U.S. Type Certificate Data Sheet that requires a statement of conformity from the State of Design/Manufacture.

d. Flight Operations After Issuance of Export C of A. Verify that any repairs, alterations, equipment installations, and maintenance activities completed after the issuance of the export C of A are validated. Per §§ 21.183(c), 21.184(b), and 21.185(c), the applicant must show that the aircraft has remained in or has been returned to conformity to its FAA TC and is in a condition for safe operation. This may involve extensive inspections accomplished by the FAA, the CAA of the SOM, the CAA of the SOR, the aircraft manufacturer, repair stations, etc., before a U.S. airworthiness certificate can be issued.

e. Notification of Exceptions. As specified in our bilateral agreements, a foreign CAA will notify the FAA before issuing an export C of A for an aircraft that is not airworthy. Some bilateral agreements specify that this notification be sent to the geographically responsible IFO; others specify that the notification be sent to the geographically responsible Flight Standards Office. For a new aircraft, forward the notification to the responsible AIR certification branch for disposition. For a used aircraft, forward the notification to the responsible Flight Standards Office for disposition. The responsible FAA office may accept all exceptions without further coordination if its letter of acceptance includes a statement that the applicant for a U.S. airworthiness certificate for that aircraft is responsible for addressing all exceptions to make the aircraft airworthy. In addition, include in the letter of acceptance a request that the exporting CAA attach the

letter of acceptance to the export C of A. Provide this letter to the exporting CAA to enable issuance of the export C of A.

21-4. Procedures for Issuing an Airworthiness Certificate. Follow the applicable procedures in paragraph 21-1 of this order and the following:

a. Review Application.

(1) FAA Form 8130-6. Verify the aircraft category on the application is eligible for import into the U.S. under the applicable bilateral agreement as described in paragraph 21-4.a.(3) below.

(2) Export C of A.

(a) New Aircraft. As required by §§ 21.183(c), 21.184(b), and 21.185(c) for a new standard classification aircraft, new primary category aircraft, and new restricted category aircraft, respectively, verify that the export C of A or other certifying statement from the exporting CAA states conformity to the applicable U.S. TC and condition for safe operation.

(b) Used Aircraft. For a used aircraft, verify the export C of A or other certifying statement issued by the exporting CAA states conformity to a U.S. TC and condition for safe operation.

(1) Returning U.S. SOM Aircraft. If available, for U.S. SOM aircraft returning to the United States, review the U.S. export C of A issued when the aircraft was originally exported from the United States; this provides a baseline for the airworthiness determination and identifies any exceptions listed on that export C of A that may require corrective action.

(2) Third Country Aircraft. If available, for used, third country aircraft, review the export C of A or other certifying statement issued by the CAA of the SOM when the aircraft was originally exported from the SOM to identify any exceptions listed on that export C of A that may require corrective action. Note that any exceptions listed may be exceptions to conformity to the TC of another country, not the FAA's TC.

(c) Verify that any statement(s) required by the applicable bilateral agreement and the aircraft TCDS is included on the export C of A.

(d) Verify the FAA accepted in writing any exceptions listed on the export C of A per the procedures in the applicable bilateral agreement and that this acceptance is included with the export C of A.

(3) Bilateral Agreements.

(a) For imported new, standard classification and new restricted category aircraft, verify the United States has an agreement for the acceptance of that aircraft as required by § 21.183(c) and § 21.185(c), respectively.

Note: Part 21 does not require a bilateral agreement with the exporting CAA for acceptance of any used aircraft or a new, primary category aircraft.

(b) For all imported aircraft, verify that the aircraft category selected on the application is eligible for import into the United States under the applicable bilateral agreement. This information may be found in an agreement with the *Implementation Procedures for Airworthiness* (IPA). Note that the scope of acceptance of some IPAs may specify acceptable aircraft makes and models.

(c) Review the applicable bilateral agreement to determine what findings the other CAA may make on the FAA's behalf, such as for acceptance of design data for repairs or alterations.

(d) For third-country aircraft, verify the applicable bilateral agreement includes provisions for acceptance of third-country aircraft. If so, consult the agreement to determine what documentation must be provided by the applicant and the extent of the findings the exporting CAA may make on behalf of the FAA. An export C of A issued by an exporting country for which the United States does not have a third-country provision may be acceptable if it was endorsed or validated by the CAA of the SOM, and the product is within the scope of a bilateral agreement between the SOM and the United States.

(e) Service documents such as service bulletins and structural repair manuals approved by the bilateral CAA where an affected product is manufactured are considered FAA--approved data unless otherwise noted, provided the United States has a bilateral agreement with that country. However, service bulletins or other similar instructions classified as "mandatory" by the CAA are not mandatory in the U.S. regulatory system unless required by an AD. Therefore, owners or operators of affected U.S.-registered aircraft are not required to comply with service documents or directives issued by the CAAs of other countries unless an FAA AD is issued under Part 39, *Airworthiness Directives*. However, for type-certificated products not currently registered in the United States, alternate procedures have been established involving the processing of foreign Mandatory Continuing Airworthiness Information (MCAI) that may affect the way the airworthiness certification requirements are met. The MCAI process is described in paragraph 21-5 of this order.

(4) TCDS. Verify compliance with any import requirements on the U.S. TCDS.

b. Review of Aircraft Records.

(1) Evidence of Conformity. Review the applicant's evidence (for example, export C of A, maintenance records, and historical records) used to show the aircraft is entitled to the airworthiness certificate requested.

(a) When an export C of A or equivalent statement identifies exceptions to the FAA TC (for example, alterations and major repairs that have had no prior FAA approval), verify the applicant resolved the exceptions by either having the exceptions approved by the FAA or removing the nonconforming items to establish full conformity to the FAA TC.

(b) Verify that any major alteration and major repair performed while the aircraft was under a foreign SOR was accomplished in compliance with FAA-approved data and that the aircraft conforms to its TC. Under certain bilateral agreements, the FAA

has determined that the other CAA may approve design data associated with major alterations or repairs that do not rise to the level of an amended TC or STC on certain categories of aircraft for which either country is the State of Design (SOD). When these data are approved directly by the CAA, or by a delegated individual or organization, they would then be subsequently recognized as FAA-approved data per the provisions of the bilateral agreement; in such cases, do not require the applicant to seek additional FAA approval(s) unless there is clear evidence that the data are erroneous or otherwise unreliable. In all other situations, use of an FAA designated engineering representative (DER) to expedite the design approval process should be encouraged for any major alteration or repair that may have been incorporated without FAA approval. Persons authorized under § 43.7 must record in the maintenance records that the major alterations or repairs conform to FAA-approved data.

(c) Verify that any maintenance performed while the aircraft was under a foreign SOR was performed per methods acceptable to the FAA and that the aircraft conforms to its approved TC or properly altered condition.

(d) Verify that any aircraft article repaired while the aircraft was operating under a foreign SOR was accomplished per methods acceptable to the FAA and that the article conforms to its TC. When this cannot be shown, the article must be removed.

(2) A 100-Hour Inspection for Used Aircraft. In addition to the persons listed in paragraph 3-6.a(1) of this order, an inspection performed by a foreign maintenance organization meets § 21.183(d)(2)(v) if all the following are met:

(a) The United States has a bilateral agreement with implementation procedures (IP) with the exporting country that includes acceptance of this aircraft category by the United States for import.

(b) The inspection was performed by a maintenance organization appropriately certificated by the exporting CAA.

(c) The inspection was completed while the aircraft was operated on the registry of the exporting CAA and within 60 days of submitting the application for a U.S. airworthiness certificate.

(d) The aircraft inspection records demonstrate that the scope of the inspection meets § 43.15.

c. Inspect Aircraft. In all cases, you are required by 49 USC 44704(d) to inspect the aircraft and make a finding that the aircraft conforms to an FAA-approved TC and is in a condition for safe operation before issuing an airworthiness certificate for that aircraft, including the following:

(1) Verify that no changes or alterations have been made, and that the condition of the aircraft has not deteriorated after issuance of the export C of A or other certifying statement by the CAA. If changes or alterations were made during this period, ensure the operator followed the requirements of the SOR

(2) If the aircraft was disassembled for shipment and reassembled, flight testing under § 21.191(b) may be required before issuance of the requested, category-specific airworthiness certificate.

21-5. Aircraft with Mandatory Continuing Airworthiness Information (MCAI).

a. When an unsafe condition is found to exist in a U.S. type-certificated product that is not currently on the U.S. registry, the certification branch may use an alternate procedure concerning the issuance of ADs. Under this alternate procedure each MCAI received will be reviewed to determine whether it meets established criteria for required corrective action. No further action will be taken for an MCAI that does not meet these criteria. An AD will be issued for an MCAI that meets these criteria if there is one or more aircraft of the affected design currently registered in the United States. If no aircraft of the affected design are currently registered in the United States, the FAA may elect to defer publishing any ADs on the MCAIs that meet those criteria until an aircraft of that design becomes U.S.-registered. A list of each MCAI that is deferred will be maintained by the geographically responsible certification branch. A statement like the following will be found in the *Eligible Serial Numbers* section of the TCDS for an aircraft design on which ADs have not been issued for some or all the required MCAIs:

“For issuance of an airworthiness certificate the [*airworthiness authority of the SOD*] must certify that the aircraft conforms to the U.S. type certificate and is in a condition for safe operation. In that regard, [*airworthiness authority of the SOD*] will certify that the aircraft complies with all applicable MCAIs it has issued. Also, the FAA must find that the product conforms to its U.S. type certificate and is in a condition for safe operation. To make that finding, all actions required by deferred MCAIs must be accomplished per each MCAI required compliance time, except as noted on the TCDS. Deferred MCAIs must be included in the operator’s airplane maintenance or inspection program to ensure they will be accomplished within the required compliance time. All inspections or alterations required by MCAI that have surpassed the initial compliance time must be accomplished before issuance of the airworthiness certificate.”

b. In some instances, the TCDS also indicates that certain ADs have been issued for the affected model. Compliance with any applicable AD is required, in addition to compliance with the MCAI.

c. After the first aircraft is U.S. certificated, the geographically responsible certification branch will amend the TCDS to list the required MCAIs. Compliance must be shown before subsequent aircraft can be found to be in a condition for safe operation and issued an airworthiness certificate. The certification branch will issue ADs for any subsequent MCAIs that meet the required criteria for corrective action and will not issue any retroactive ADs for any MCAIs listed as effective before the first aircraft being certificated.

d. This alternative procedure also may be used when an aircraft of the affected model previously was registered in the United States but currently is not.

Note: This procedure is not considered appropriate at this time for other products, for example, engines and propellers, because there is no reliable means to ensure none of these products have been imported and installed in U.S. registered aircraft.

Chapter 22. Issuing an Export Certificate of Airworthiness (§ 21.329)

22-1. Introduction.

- a. This chapter provides policies and procedures for issuing FAA Form 8130-4, Export Certificate of Airworthiness, (export C of A) for certain type-certificated aircraft as provided for under § 21.329. The FAA issues an export C of A to facilitate the acceptance of complete aircraft by other importing CAAs.
- b. An export C of A only attests to the airworthiness of an aircraft. An export C of A does not:
 - (1) Authorize the operation of an aircraft.
 - (2) Make any attestation concerning any agreement between the exporter and importer.
 - (3) Make any attestation concerning compliance with applicable regulations of other Federal agencies such as export restrictions. In issuing an export C of A, you are not responsible for making findings to the regulations of other Federal agencies.
- c. **FAA ASIs are authorized to issue export C of As.** The FAA may delegate the issuance of an export C of A per 14 CFR Part 183 and the applicable designee management order.

22-2. Procedures for Issuing an Export C of A.

a. Review Application (§ 21.327).

- (1) FAA Form 8130-1. Verify FAA Form 8130-1, *Application for Export Certificate of Airworthiness*, has been properly completed. Verify the application was completed by the registered owner of the aircraft or an agent of the owner. If an agent completed the application, verify the registered owner of the aircraft provided a notarized agent letter of authorization and that the information in this letter is consistent with the application. A copy of the notarized agent letter is acceptable.
- (2) Eligibility.
 - (a) Per § 21.327, verify the aircraft is U.S. registered per paragraph 2-3.c.
 - (b) Verify the aircraft is in the normal, utility, acrobatic, commuter, transport, primary, or restricted category; a manned free balloon; or special class of aircraft. Per § 21.329, other aircraft categories are not eligible for an export C of A.
 - (c) Review the applicable bilateral agreement, equivalent airworthiness agreement, or working procedure to determine if the importing country accepts the category of aircraft being exported. Some bilateral agreements require a special arrangement between the FAA and the importing CAA to export primary or restricted category aircraft; if applicable, contact AIR-40 for assistance in developing a special arrangement and information about current special arrangements.

b. Aircraft Manufactured Under Part 21 Subpart F or G. Per § 21.329(a)(1), verify the aircraft meets the applicable requirements for an airworthiness certificate:

- (1) For a standard classification aircraft without a U.S. standard airworthiness certificate, follow the applicable procedures of chapter 3 of this order to verify the aircraft meets the requirements for that certificate.
- (2) For a primary category aircraft without a U.S. special airworthiness certificate in the primary category, follow the applicable procedures of chapter 5 of this order to verify the aircraft meets the requirements for that certificate.
- (3) For a restricted category aircraft without a U.S. special airworthiness certificate in the restricted category, follow the applicable requirements of chapter 6 of this order to verify the aircraft meets the requirements for that certificate.
- (4) For a standard, primary, or restricted category aircraft manufactured under Part 21 subpart F or G and that has an airworthiness certificate, verify the certificate is valid per the procedures of paragraphs 22-2.c(1) through (3) of this chapter.

c. Aircraft Not Manufactured Under Part 21 Subpart F or G. Per § 21.329(a)(2), an exporter may obtain an export C of A for an aircraft that was not manufactured under Part 21 subpart F or G if the aircraft has a valid standard airworthiness certificate or special airworthiness certificate in the primary or restricted category. For such an aircraft, verify the airworthiness certificate is valid.

- (1) Maintenance. Inspect maintenance records to verify that all maintenance required as of the date you inspect the aircraft is complete, including compliance with applicable ADs and component life limits.
- (2) ID Plates/Marks. Inspect the aircraft and engine ID plates and propeller marks to verify the information matches the application.
- (3) Other. Conduct any other inspections as necessary to verify the airworthiness certificate is valid.

d. Comply with the Bilateral Agreement or Working Procedure. Review the applicable bilateral agreement or working procedure to identify and comply with applicable requirements of the importing CAA. Requirements may include:

- (1) Type Certificate. Many importing CAAs require a finding of conformity to the TC of the importing country and a corresponding statement on the export C of A. This may include approval by the importing authority of repairs and alterations and verification of compliance with foreign ADs.
- (2) Special Import Requirements (SIR). Many importing CAAs require the FAA to verify that the exporter has complied with their SIR. These requirements must be satisfied as a condition of shipment at the time of export. SIR may include, for example, the requirement for FAA Form 8130-4 and copies of maintenance records, flight manuals, and other documents. These requirements may be found in FAA AC 21-2, *Complying with the Requirements of Importing Countries or Jurisdictions When Exporting U.S. Products, Articles, or Parts*; appendix 2, [*Special Requirements of Importing Countries*](#). Identify any SIR applicable to this export.

Note: If inconsistencies are identified between a bilateral agreement or working procedure and an SIR while preparing an export C of A, contact AIR-40 for guidance. FAA ASIs and designees should not contact the importing CAA directly to resolve these inconsistencies. AIR-40 will contact the importing CAA and determine whether a particular export transaction is consistent with the importing CAA's expectations. If necessary, AIR-40 may also determine whether amendments to bilateral agreements or working procedure and SIRs are necessary and pursue that action with the affected CAA.

(3) Required Statements on the Export C of A. Identify any applicable statements that need to be included on the export C of A as required by the bilateral agreement or work procedure, the U.S. TCDS, the foreign TCDS, and appendix 2 to FAA AC 21-2.

e. No Bilateral Agreement or Working Procedure. Although unnecessary, the FAA may issue an export C of A to export an aircraft to a country with which the United States does not have a bilateral agreement or working procedure; in such cases, the export C of A may only certify conformity to the U.S. TC.

f. Coordinate Exceptions. If the aircraft does not conform to the applicable TC, the aircraft is not in a condition for safe operation, or the exporter will not meet the SIR of the importing country, obtain from the exporter a written description of all such nonconformities and noncompliances (hereafter, "exceptions"). If the CAA has not previously provided acceptance of all exceptions, an FAA ASI must prepare an accompanying cover letter on FAA letterhead to the importing CAA, requesting the CAA's acceptance of the exceptions and a return reply to the FAA. The responsible FAA office transmits the letter to the importing CAA; electronic transmissions may be used to expedite this process if the FAA confirms that the reply is from an authorized person within the importing CAA. Do not issue the export C of A without written acceptance of all exceptions from the importing CAA.

Note: For countries with which the United States has a bilateral agreement with separate implementation procedures (IP), instructions for transmittal of requests for acceptance of exceptions are contained in the IP paragraph concerning coordination of exceptions for an export C of A. For countries with which the United States has a Bilateral Airworthiness Agreement (BAA), the requests for acceptance of exceptions should be directed to the importing CAA's appropriate contact identified on the AIR-40 [global contacts list](#). For countries with which the United States has a working procedure with, requests for acceptance of exceptions should be directed to the importing CAA's appropriate contact identified in the working procedure. For all non-bilateral countries, if an appropriate recipient and address is unknown, contact AIR-40 for assistance.

(1) If the importing CAA does not provide written acceptance of all exceptions, write a letter to the applicant denying issuance of the certificate and stating the reason(s) for the denial. If feasible, identify which steps may be accomplished to meet the certification requirements.

(2) Complete Part III of FAA Form 8130-1.

g. Document Reviews and Inspections. Document all completed reviews and inspections on FAA Form 8100-1, per the instructions on the form. Verify the applicant corrects all unsatisfactory findings, and document those corrections on FAA Form 8100-1 before issuing

an export C of A. FAA Form 8100-1 must be completed. Inspectors and designees should always record when an Export Certificate of Airworthiness is issued, regardless of how routine the situation is. This builds a historical record of the aircraft's certification status over time.

h. Issue the Export C of A. Upon finding the applicant meets applicable requirements, issue the export C of A as follows:

(1) Export C of A.

(a) Complete FAA Form 8130-4, *Export Certificate of Airworthiness*, per the instructions in section 7 of appendix A to this order.

(b) If applicable, attach a copy of the written statement of acceptance of exceptions from the importing CAA of any nonconformity to the TC or noncompliance with SIR of the importing country. Note the original statement of acceptance will be included in the files forwarded to the FAA Aircraft Registry. Written CAA acceptance of exceptions transmitted electronically are acceptable for attaching to the export C of A.

(2) Application. Complete block 21 of the application.

(3) Make a maintenance record entry. Although not required, you should make a maintenance record entry to facilitate a finding of airworthiness should this aircraft return to the U.S registry: "This aircraft was issued an export certificate of airworthiness on [enter date] to facilitate export from the United States. [Signature: Jane Doe, Aviation Safety Inspector, AIR-883]."

(4) Review with Applicant. Give the applicant the completed export C of A and, if applicable, a copy of the importing CAA's written acceptance of exceptions. Remind the exporter to comply with § 21.335. When exporting an unassembled aircraft, advise the exporter to forward the manufacturer's assembly instructions and, if applicable, an FAA-approved flight test checkoff form to the importer.

i. Review Records and Forward to the Registry.

(1) Review your records to verify completeness, accuracy, legibility, and compliance with applicable requirements. Following export of an aircraft, strikeovers are permitted to correct errors on records.

(2) Review and forward certification files per appendix B to this order.

22-3. Unassembled Aircraft. Verify the importing country has no prohibition on importing unassembled aircraft. In exporting a new, unassembled aircraft manufactured under a PC, the exporter should provide the importer the manufacturer's assembly instructions and the FAA-approved flight test procedures. This information should also be provided for other unassembled aircraft.

22-4. Approval of Alterations. In many instances, an aircraft that conforms to the TC may be altered before export per the purchaser's requirements. The responsibility for

approval and recording of such alterations primarily depends on the state of registry status of the aircraft at the time of alteration.

22-5. Multiple Categories. To retain eligibility for issuance of an export C of A as a standard aircraft after having **been operated in the restricted category, the following items apply:**

- a.** While being operated in the restricted category, any changes made to the aircraft that are to be retained when in normal category operation, or any operations that are outside of the standard classification operating limitations, must be approved per the regulations and procedures applicable to an aircraft having a standard airworthiness certificate.
- b.** If the TCDS for an aircraft includes both standard and restricted category, and the maximum gross weight and/or other operating limitations for the restricted category are higher than that for standard classification, the aircraft is not eligible for issuance of an export C of A as a standard aircraft, after having been operated in the restricted category, unless—
 - (1) The TCDS specifically states that the aircraft is eligible for operation under a standard airworthiness certificate after having been operated at the limitations applicable to the restricted category; or
 - (2) If the TCDS does not have such a note or other reference, the operations outside of the standard classification operating limitations, including increased gross weight, had been approved as appropriate for an aircraft having a standard airworthiness certificate.

22-6. Replacement and Correction of FAA Form 8130-4.

- a. Replacements.** When FAA Form 8130-4 has been declared lost or mutilated, advise the owner to contact the FAA Aircraft Registry for replacement.

Note: Replacing an export C of A is an administrative function and does not require a finding of airworthiness.

- b. Correction of Administrative Errors.** When FAA Form 8130-4 has been issued with erroneous information, typos, and other such mistakes not impacting the aircraft records review or conformity inspection of the exported aircraft, follow the procedures of paragraphs 22-2.h and 22-2.i for issuance of a replacement certificate, reviewing and forwarding records, and as follows.

- (1) Recover the originally issued FAA Form 8130-4.
- (2) An application is not required; a hand or digitally signed statement from the original applicant or importer is acceptable. Review the signed statement requesting the replacement certificate to verify it includes the aircraft registration number, when the original export C of A was issued, make, model, S/N, and a valid reason for replacing the certificate.
- (3) Verify the error is administrative in nature and does not require a new review of aircraft records or a new aircraft inspection.

- (4) Except for the correction of administrative errors, complete FAA Form 8130-4 identically to the original. Enter the date of the original certificate, inserting "R" before the date to signify that this is a replacement certificate.
- (5) Transfer any attachments from the original to the replacement certificate.

Chapter 23. Special Flight Authorizations for Foreign Civil Aircraft (§ 91.715)

23-1. Introduction. This chapter provides policies and procedures for issuing a special flight authorization (SFA) for foreign civil aircraft, per the requirements of § 91.715.

- a. FAA ASIs are authorized to issue SFAs.
- b. The navigation of foreign registered civil aircraft in the United States is permitted under 49 U.S.C. 41703(a) as implemented by 14 CFR Part 375, Navigation of Foreign Civil Aircraft within the United States. Part 375 specifies that a foreign civil aircraft being operated in the United States must carry a current and effective registration and airworthiness certificate issued or rendered valid by the SOR. Part 375 also allows the operation of foreign aircraft that do not carry current airworthiness certificates, but that have been issued an SFA by the FAA.
- c. Foreign civil aircraft that do not have a current airworthiness certificate issued by the SOR meeting Annex 8 to the ICAO Convention require an SFA issued by the FAA per § 91.715(a).
- d. An aircraft registered in a country that is not an [ICAO member state](#) or a territory of a member state always requires an [authorization from the Department of Transportation \(DOT\)](#) and an SFA issued by the FAA before operating in the United States.
- e. FAA Order 8900.1 requires Flight Standards to use the FAA Web-based Operations Safety System (WebOPSS) to administer SFAs with continuing authorization.

23-2. Eligibility.

Foreign civil aircraft are eligible for an SFA as follows:

- a. Section 91.715 is applicable to foreign aircraft that do not have a current airworthiness certificate, or an equivalent to a U.S. standard airworthiness certificate, that indicates that the aircraft complies with a detailed and comprehensive airworthiness code as provided by Annex 8 to the ICAO Convention.
- b. An SFA is required for an aircraft carrying an airworthiness certificate, flight permit, or similar document issued by the SOR that is equivalent to a U.S. special airworthiness certificate.
- c. **An SFA is required for either of the following scenarios:**
 - (1) The aircraft is registered in an ICAO member state but does not have an airworthiness certificate attesting that the aircraft complies with Annex 8 to the ICAO Convention, or it has an invalid airworthiness certificate.
 - (2) The aircraft is registered in a non-ICAO member state regardless of the type of airworthiness certificate issued or its planned operation.

d. An SFA must not be issued for the following:

- (1) Foreign Military Aircraft. Refer an applicant requesting an SFA for a foreign military aircraft to the U.S. [Department of State](#). Such aircraft may enter the United States only with a diplomatic clearance that would be issued solely on a government-to-government, noncommercial basis.
- (2) Special Interest Flight. The aircraft is registered in a country that has special overflight approval requirements under the U.S. Department of State Special Interest Flight (SIF) program. For requests involving aircraft identified under the SIF program, the foreign owner/operator needs overflight clearance from the U.S. Department of State. For further information, contact FAA Air Traffic System Operation, AJR-2, or refer to Part 3 of the latest [Notice to Air Missions \(NOTAM\)](#).
- (3) A Canadian Owner–Maintenance category aircraft.

23-3 Procedures.

a. Application and Document Review.

- (1) General. The application for an SFA may be in the form of a letter, email, or fax from the owner or operator. If the aircraft is in the United States, the local FAA office is responsible for processing the SFA. If the aircraft is not in the United States, the office having jurisdiction over FAA matters in that country is the office responsible for processing the application. If an agent completed the application, verify the registered owner of the aircraft provided a notarized agent letter of authorization and that the information in this letter is consistent with the application. A copy of the notarized agent letter is acceptable.
- (2) Aviation Events. The application may be made to the AFS division manager or AIR division manager of the FAA region in which the event is located. Aviation events encompass many different activities. Refer to FAA Order 8900.1 for additional information.
- (3) SFAs for Operation of Canadian-Registered, Recreational Aircraft in the United States. Operation in the United States of Canadian-registered, amateur-built aircraft or basic/advanced ultralight airplanes is permitted by the issuance of an SFA under § 91.715. The SFA must be obtained before operation in the United States is permitted. An SFA may be obtained from the FAA for operation of these aircraft in NAS by following the [instructions](#) on the FAA website.
- (4) Applications for Individual Aircraft Authorizations. An application for an SFA must contain the following information, as applicable, and any other information deemed appropriate by the issuing FAA field office:
 - (a) The name and address of the applicant.
 - (b) The name and address of the aircraft owner.
 - (c) The purpose for which the SFA is requested, including-

- (1) Whether the aircraft will be used as a test aircraft in the development of a U.S. STC and will require flight testing for showing compliance with regulations; and
 - (2) If flying the aircraft to a base where repairs or maintenance are to be performed, a description of the needed repairs and the operating limitations, if any, assigned by the SOR.
- (d) The type of airworthiness document, if any, issued for the aircraft by the SOR.
- (e) Information such as total aircraft time, maintenance status, date of last inspection, type of inspection, and the name and title of the person performing the inspection. This information is necessary to establish that the requested flight(s) will not adversely affect safety.
- (f) The make, model, and S/N of the aircraft.
- (g) The assigned nationality and registration marks and a valid copy of the registration document issued by the SOR.
- (h) Itinerary.
- (1) The country of origin.
 - (2) The base of operations for the proposed flights and the areas where the flights will be conducted.
 - (3) The proposed U.S. point of entry and the itinerary while operating in the United States.
 - (4) The proposed U.S. point of departure and the ultimate destination.
- (i) The duration for which the SFA is requested.
- (5) Applications for Blanket Authorizations. A blanket SFA may be requested for an operation that will be conducted many times during a given period or for multiple aircraft engaged in the same operation, for example, an export delivery flight. A blanket SFA may be issued when deemed appropriate by the issuing office manager. An application for a blanket SFA should contain the following information:
- (a) The name and address of the promotion sponsor; or the name and address of the manufacturer when the purpose is for export.
 - (b) The purpose(s) for which the blanket SFA is requested, and the number of signed copies required to meet operating needs.
 - (c) Enough information to establish that the flights will not adversely affect safety.
 - (d) For aviation events, the name and address of the owner or operator, make, model, S/N, registration number, type of airworthiness certificate carried, reason why the aircraft does not comply with standard airworthiness requirements, and aircraft maintenance provisions. The listing of owners, pilots, and aircraft participating may be provided separately.

(e) Any other information deemed appropriate by the ASI.

b. Aircraft and Record Inspection. The aircraft may need to be inspected before issuance of the SFA to ensure it is capable of safe flight for the intended purpose(s). The ASI may make, or require the applicant to make, appropriate inspections or tests considered necessary for safety.

c. Document Reviews and Inspections. Document all completed reviews and inspections on FAA Form 8100-1, per the instructions on the form. Verify the applicant corrects all unsatisfactory findings, and document those corrections on FAA Form 8100-1 before issuing the SFA.

d. Issuance of an SFA and Operating Limitations.

(1) Numbering. Each SFA issued must include the application number assigned by AWC, or one assigned by the local office generating the SFA. When using a number not assigned by AWC, each SFA issued must be assigned (beginning with “01”) that is prefixed by the appropriate location identifier code of the FAA office, for example, MKE-GL-13 01 or OKC-AIR-881 01. If an SFA is extended, a new SFA must be issued using the number assigned to the original followed by the letter “A,” for example, MKE-GL-13 01A. In some cases, an SFA may require extension more than once. The second extension would still use the original number followed by the letter “B.”

(2) Duration. The SFA is usually issued with the duration requested by the applicant. However, the issuing office may issue the SFA with a different duration. For example, if the purpose is one for which delays may be expected, such as in STC projects, the office may establish a longer duration than was requested to preclude the need for extensions.

(3) Blanket SFA: When issuing multiple copies of a blanket SFA to an applicant, advise the applicant that the applicant is accountable for each signed copy. Refer to figures F-4 and F-6 of this order for examples of blanket authorizations.

(4) Operating Limitations. Because an SFA is issued to cover the operation of an aircraft that may not meet the airworthiness standards established by ICAO, pursuant to § 91.715(b), appropriate operating limitations must be prescribed to minimize hazards to persons or property. Appendix F of this order includes examples of operating limitations for SFAs for specific flight operations; these examples are not intended to be prescriptive or exhaustive. You may prescribe additional limitations deemed necessary in the interest of safety. In certain circumstances, such as a flight for export delivery, additional limitations may not be necessary. The limitations should be similar to what you would issue to a comparable U.S.-registered aircraft. If deemed necessary, you should coordinate draft operating limitations with a local operations inspector. Minimum operating limitations include the following:

(a) General. The following are applicable to all SFAs issued unless otherwise noted:

(1) A copy of this authorization must be available to the pilot in command when operating under the terms of this SFA.

- (2) You must comply with all limitations imposed by the SOR and this authorization.
- (3) Persons or property must not be carried for compensation or hire.
- (4) This authorization is valid in the United States only.
- (5) Upon request, this authorization must be made available to an FAA inspector.
- (6) This authorization is valid until [insert expiration date] unless superseded or rescinded.

(b) Damaged Aircraft.

(1) Aircraft Located in the United States. The determination that the aircraft has been damaged to the extent that the airworthiness certificate is ineffective is the responsibility of the SOR. Contact the IFO for any information related to the SOR. Under Annex 8 to the ICAO Convention, the SOR may either prohibit further flights of the aircraft until it is restored to an airworthy condition or may prescribe limitations under which the aircraft would be safe to fly to a base where repairs can be made. In the event the SOR requests the FAA to inspect the aircraft on its behalf, the appropriate geographical office should arrange for inspection of the aircraft by personnel from the responsible Flight Standards Office or CM Section. Any limitations considered necessary because of the inspection must be prescribed as special limitations in addition to the minimum limitations.

(2) Aircraft Located Outside the United States. An applicant with a foreign registered aircraft needing repair, who wants the repair to be accomplished at a manufacturer or repair facility in the United States, may do so regardless of the country in which the damage was sustained. The SOR remains responsible for inspection of the aircraft and for establishing any necessary special operating conditions and limitations.

(c) Change in Nationality. This paragraph applies when an airworthiness certificate has been invalidated by the new SOR. If the aircraft complies with U.S. or ICAO airworthiness requirements, except for the invalid airworthiness certificate, it may not be necessary to prescribe additional operating limitations. Note that the aircraft need not have a registration issued by the country of the foreign buyer but must bear the ID marks issued by the SOR or intended registry.

(d) Flight Testing. The ASI must evaluate the reasons the flight test must be conducted in the United States, the qualifications of the individual or company in the United States who will be primarily responsible for the flight test operations, and the nature of the flight tests. The conclusions reached from that evaluation are an important factor in determining the special operating limitations that must be prescribed in addition to the minimum limitations. The following operating limitations would generally be applicable, but may be altered or added to as deemed appropriate:

- (1) All flight tests must be conducted in compliance with § 91.305.
 - (2) All maintenance and inspection of the aircraft must be conducted under the direct supervision of qualified personnel holding appropriate licenses issued or rendered valid by the [insert SOR] CAA and according to [insert SOR] aircraft maintenance requirements.
 - (3) Except for flight tests conducted according to the terms of this authorization, additional flights within the United States must be limited to those necessary to proceed from [*specify origin*] to [*insert the name of the airport or other area from which the flight test will be conducted*] and return to [*specify destination*] by the most practical direct route, considering safety risks.
- (e) Flight training of customers, employees, or designees. In most cases, an SFA issued for this purpose would be a blanket authorization issued to an aircraft manufacturer. The following operating limitations, in addition to the required standard limitations, are worded to indicate that more than one aircraft is involved. If an SFA under this paragraph is issued for a single aircraft, an appropriate change must be made.
- (1) Each aircraft operated for customer crew training flights must carry this SFA attached to a statement that includes the name and address of the aircraft owner, the aircraft's assigned nationality and registration marks, and the dates on which the customer crew training flights are scheduled to begin and end. This limitation applies only if a blanket authorization has been issued.
 - (2) All customer crew training and aircraft maintenance must be conducted under the direct supervision of [*insert name of manufacturer*] personnel.
- (f) Ferrying an Aircraft for Export Delivery.
- (1) Individual Aircraft Authorizations. The limitations below may be omitted if the aircraft has a valid FAA export C of A with no major exceptions listed and is not carrying extra fuel or navigational equipment. Approval of major exceptions must be obtained from the CAA of the SOR. If temporary fuel system(s)/equipment are installed or the aircraft is to be operated in excess of its maximum certificated takeoff weight, the limitations below must be included as applicable:
 - (a) Operation in the overweight condition must avoid cities, towns, villages, and congested areas, or any other areas where the flights might create hazards to persons or property.
 - (b) The aircraft must not be operated with temporary fuel system(s) or temporary navigation equipment installed, or at a weight in excess of its maximum certificated takeoff weight, unless approved in writing by the CAA of the SOR.
 - (2) Blanket Authorization. The limitations applicable to an individual aircraft authorization obtained in accordance with chapter 23-3, paragraph a.

(5), of this order generally apply to a blanket authorization. When an aircraft manufacturer uses a blanket authorization for export delivery, the manufacturer must submit a request through the SFA-Update Blanket Authorization function in AWC to add aircraft. The blanket authorization must be worded so any possible situation will be covered by each copy issued. When issuing multiple copies for a blanket SFA, add a limitation requiring the applicant to attach a statement to the SFA including the name and address of the aircraft owner; nationality and registration marks displayed on the aircraft; make, model, and S/N of the aircraft; date the copy is issued for the aircraft; and signature of authorized representative. Refer to figures F-4 and F-6 in Appendix F of this order for examples of blanket authorizations.

(g) Demonstration or Test. The issuing office should find that the applicant for an SFA for demonstration has satisfied, as applicable, the items listed in Part 91. Persons having an interest in the demonstration, for example, customers, may be carried in an aircraft issued an SFA for demonstration.

(1) Examples. Examples of SFAs for various flight operations are shown in appendix F to this order.

e. Review and Maintain Records.

(1) Review your records to verify completeness, accuracy, legibility, and compliance with applicable requirements.

(2) The SFA issuing office maintains records in AWC.

Chapter 24. Other Policies and Procedures

24-1. Introduction. This chapter contains other miscellaneous policies and procedures related to issuing an airworthiness certificate.

24-2. Public Aircraft Operations (PAO).

- a. Public aircraft are defined in 49 USC 40102(a)(41).
- b. If an applicant intends to perform both public and civil aircraft operations, advise them that when changing from a public status to a civil status, the aircraft must meet all civil requirements. The operator must have written instructions on how it transitions from public to civil status.
- c. Note that a civil airworthiness certificate issued under 14 CFR is not in effect during PAO.
- d. For additional information, refer to FAA Order 8900.1, volume 3, chapter 14; and FAA AC 00-1.1, *Public Aircraft Operations--Manned and Unmanned*; see [DRS](#).

24-3. Production Flight Testing of Aircraft Built Under Military Contract. Aircraft built under a military contract with military aircraft identification marks do not require FAA registration or the issuance of an airworthiness certificate for flight testing or demonstration before acceptance by the military. However, aircraft of military design built independently by manufacturers and not having military identification are required to obtain FAA registration and an airworthiness certificate because these aircraft are considered civil aircraft. These aircraft may be issued a SFP for customer demonstration flights if they have successfully completed a production flight test.

24-4. Operation of Civil Aircraft With a Door Open or Removed for Parachuting, Skydiving, or Other Special Operations

- a. The responsible Flight Standards Office having jurisdiction over the area in which the operations are to be conducted should advise the owner or operator to provide the following information:
 - (1) Name and address of the registered aircraft owner;
 - (2) Make, model, serial, and registration number of the aircraft;
 - (3) Location where the aircraft normally is based; and
 - (4) Reason for the aircraft to be operated with a door removed.
- b. Aircraft may be approved to operate with a door open or removed by TC, amended TC, STC, authorization in the aircraft's flight manual, or field approval.
 - (1) Aircraft approved by TC, amended TC, STC, or approved flight manual authorization do not require any additional limitations or authorization from the responsible Flight Standards Office.

(2) Aircraft approved to operate by issuance of FAA Form 337, Major Repair and Alteration, must have operating limitations issued. Examples of appropriate operating limitations are in appendix G of this order. Field approval guidance is in FAA Order 8900.1. The ASI must note on the operating limitations the aircraft make, model, registration and serial number, type of operation authorized, date of issuance, ASI's name, and district office number. On an aircraft that requires removal or opening of a particular door, the ASI must specify in the limitations which door may be removed or opened.

Note: A copy of the limitations must be forwarded to the coordination mailbox.

- c.** Removal or installation of a cabin door for the specified aircraft is considered maintenance and as such must be accomplished by persons authorized under § 43.3.
- d.** If operations of rotorcraft with the doors opened or removed obstructs the nationality and registration marks from view, the operator must notify the appropriate managing office in writing detailing the nature of the proposed operation and the proposed dates of operation with doors removed. The managing office will then instruct the operator to affix temporary nationality and registration marks on an authorized surface required by § 45.27(a). The size of the marks must comply with § 45.29(b) unless no authorized surface is large enough for display of marks meeting the size requirements of this section. The rotorcraft would then be marked on the largest authorized surface with marks as large as practicable, as allowed by § 45.29(f). Any remnants of the permanent nationality and registration marks must be obliterated so as not to confuse identification of the rotorcraft with temporary markings. The temporary markings must be able to endure flight operations in various weather conditions. Flight operations must be authorized in writing by the managing office for a specified time and purpose. The managing office will verify the temporary markings comply with Part 45 and that the rotorcraft is returned to its permanent marking scheme.
- e.** Under appendix A to Part 43, paragraph (c)(15), a pilot may be authorized to remove or reinstall passenger seats. Removal or installation of control sticks and wheels must be performed in accordance with the applicable sections of Part 43.

24-5. Replacement ID Plates.

- a.** When a replacement ID plate is required, the owner or the owner's authorized representative will contact their local FAA office. The FAA determines whether the request is valid and provides a letter to the applicant with the FAA's finding.
- b.** Per § 45.13(c), no person may remove or install any identification plate required by § 45.11 without the approval of the FAA, except when necessary during maintenance operations. Prior to granting approval, the old ID plate, when available, should be voluntarily surrendered by the owner. The FAA office must make a digital copy of the original ID plate and then either physically destroy it or, if requested by the manufacturer, return it to the manufacturer via certified mail. The FAA office must then send a letter to the FAA Aircraft Registry stating that the surrendered plate has been destroyed. The FAA Aircraft Registry will include the letter in the permanent aircraft records file.

24-6. New Aircraft Manufactured Under § 21.6(b). If you receive an application for an airworthiness certificate for a new aircraft manufactured under § 21.6(b), contact AIR 630 for assistance.

Section 3. Military Surplus Flight Safety Critical Aircraft Parts, Engines, & Propellers

24-7. Evaluation.

a. This paragraph provides guidance for use in evaluating and determining the eligibility of U.S. military surplus Flight Safety Critical Aircraft Parts (FSCAP), engines, and propellers for installation on FAA type-certificated products. Many military surplus FSCAPs have the potential to be approved for installation on aircraft that hold special or standard airworthiness certificates.

b. Military engines, propellers, and articles are categorized as new or used and fall into one of the following categories:

- (1) Dual use FSCAPs;
- (2) Military unique FSCAPs;
- (3) Dual use military surplus engines, propellers, and articles; and
- (4) Military unique surplus engines, propellers, and articles.

c. Before these military engines, propellers, and articles are installed on type – certificated products, the installer must determine that they are—

- (1) Eligible for installation, and
- (2) Airworthy

d. There are certain unique design considerations and FAA certification requirements for engines and propellers. Therefore, the eligibility and evaluation processes for military surplus engines, propellers, and articles are described separately in paragraph 24-10 of this order.

Note: For eligibility and evaluation of non-flight safety critical articles, safety critical aircraft articles, engines/propellers, and their articles, use FAA *AC 20-62, Eligibility, Quality, and Identification of Aeronautical Replacement Parts*.

24-8. Dual Use FSCAP (New or Used).

a. Eligibility Screening. New or used dual use FSCAPs may be eligible for installation on FAA type-certificated products with standard or special airworthiness certificates. The eligibility determination is made based on a review of the following pertinent Department of Defense (DOD) historical records:

- (1) FSCAP identification: part number, DOD National Stock Number, and S/N.
- (2) Manufacturer, DOD Commercial and Government Entity (CAGE) code, and date of manufacture.
- (3) Total time in service.
- (4) Current status of life limited parts and FSCAP.
- (5) Time since the last overhaul of each FSCAP that is required to be overhauled on a specified time basis.
- (6) Identification of current inspection status, including time since last required inspection or maintenance performed.
- (7) Current status of applicable AD and DOD directives (for example, engineering changes, technical orders, or maintenance work orders) including the date and method of compliance. If the AD involves recurring action, the current status includes the time and date when the next action is required.
- (8) A list of current major alterations or repairs for each FSCAP.
- (9) Date any work was accomplished.
- (10) Work authentication.

b. Airworthiness Determination. After determining the FSCAP is eligible to be installed on a type-certificated product, the FSCAP must be evaluated to determine whether it is airworthy.

- (1) New Dual Use FSCAP.
 - (a) For an FSCAP to be installed on products with standard airworthiness certificates, the FSCAP must be airworthy.
 - (b) For an FSCAP to be installed on products with special airworthiness certificates, the FSCAP must be cited in the FAA-accepted maintenance manual and illustrated parts catalog (IPC) specified on the applicable TCDS and must be in a condition for safe operation.

(2) Used Dual Use FSCAP

(a) For an FSCAP to be installed on products with standard or special airworthiness certificates, the FSCAP must be evaluated using the procedures for new dual use FSCAPs above, as appropriate, to determine the FSCAP's airworthiness per § 43.13.

(b) The FSCAP also must be evaluated by persons authorized under §§ 43.7(a), (c), (d), or (e) by using the following applicable methods, means, or data sources:

- (1) Differences between military and civil versions (for example, possible DOD alterations or repairs performed);
- (2) Current manufacturer or DOD technical data and procedures to perform tests and inspections, including current life limited articles list;
- (3) Comparison of military time and/or cycle count for accumulated operational time versus civil (for example, "Did the military use a different method than civil operators to account for accumulated operational time?");
- (4) Nondestructive tests, as required;
- (5) Bench test or functional test, as required;
- (6) Results of tests and inspections recorded;
- (7) Complete alteration or repair records;
- (8) Manufacturer's ID plate;
- (9) Flight, maintenance, and/or structural manual(s), and IPC; and
- (10) Instructions for Continued Airworthiness (ICA).

c. Approval for Installation. Persons authorized under § 43.7 may approve dual use FSCAPs for installation on type-certificated products if the FSCAP successfully completed the eligibility screening and airworthiness evaluation. The installer must be able to determine that the installation of the FSCAP will leave the product in compliance with all regulations and in a condition for safe operation. The authorized individual completing the eligibility screening and/or airworthiness evaluation must make a maintenance record entry that clearly documents the results of the evaluation. Each maintenance record entry per § 43.9 should include a description of the work performed, the completion date of the work performed, and the name of the person performing the work or authorized to sign.

24-9. Military Unique FSCAP.

a. Eligibility Screening. New or used military unique FSCAPs may be eligible for installation on civil aircraft with special airworthiness certificates under § 21.305(c) in conjunction with type certification procedures for a product or per a TCDS. Military unique FSCAPs are not eligible for installation on a civil aircraft with a standard airworthiness certificate. The eligibility determination is made based on a review of the following pertinent DOD historical records:

- (1) FSCAP identification: part number, DOD National Stock Number, and S/N.
- (2) Manufacturer, DOD CAGE code, and date of manufacture.
- (3) Total time in service.
- (4) Current status of life limited FSCAPs.
- (5) Time since the last overhaul of each FSCAP that is required to be overhauled on a specified time basis.
- (6) Identification of current inspection status, including time since last required inspection or maintenance performed.
- (7) Current status of applicable ADs and DOD directives, (for example, engineering changes, technical orders, or maintenance work orders) including the date and method of compliance. If the AD involves recurring action, the current status includes the time and date when the next action is required.
- (8) A list of current major alterations or repairs for each FSCAP.
- (9) Date any work was accomplished.
- (10) Work authentication.

b. Airworthiness Determination. After determining the FSCAP is eligible to be installed on a type-certificated product with a special airworthiness certificate, the FSCAP must be evaluated to determine whether it is airworthy.

- (1) New Military Unique FSCAP. The FSCAP must be cited in the FAA-accepted, military-approved maintenance manual and IPC specified on the applicable aircraft TCDS and must be in a condition for safe operation.
- (2) Used Military Unique FSCAP.
 - (a) The FSCAP must be cited in the FAA-accepted, military-approved maintenance manual and IPC specified on the applicable aircraft TCDS and must be in a condition for safe operation.
 - (b) The FSCAP also must be evaluated to determine airworthiness per § 43.13, by using the following applicable methods, means, or data sources:
 - (1) Special equipment or test apparatus, as required;
 - (2) Current manufacturer or DOD technical data and procedures to perform tests and inspections;

- (3) Comparison of military time and/or cycle count for accumulated operational time versus civil time (for example, “Did the military use a different method than civil operators to account for accumulated operational time?”);
- (4) Nondestructive tests, as required;
- (5) Bench test or functional test, as required;
- (6) Results of tests and inspections recorded;
- (7) Complete alteration or repair records;
- (8) Manufacturer’s ID plate;
- (9) Flight, maintenance, and/or structural manual(s), and IPC; and
- (10) ICA.

c. Approval for Installation. Persons authorized under § 43.7 may approve military unique FSCAPs for installation on type-certificated products if the FSCAP successfully completed the eligibility screening and the airworthiness evaluation. The installer must be able to determine that the installation of the FSCAP will leave the product in compliance with the TCDS and in a condition for safe operation. The authorized individual completing the eligibility screening and/or airworthiness evaluation must make a maintenance record entry that clearly documents the results of the evaluation. Each maintenance record entry per § 43.9 should include a description of the work performed, the completion date of the work performed, and the name of the person performing the work or authorized to sign.

24-10. Dual Use & Unique Military Surplus Engines, Propellers, & their Articles.

a. New, used, or parted out military surplus engines, propellers, and articles should not be presumed to be eligible for installation on FAA type-certificated aircraft. Military surplus engines, propellers, and articles are either dual use or military unique.

b. The pertinent accompanying historical records documentation is essential for—

- (1) The Defense Reutilization and Marketing Office’s public sale of engines, propellers, and articles;
- (2) Categorizing the engines, propellers, and articles as dual use or military unique, and
- (3) Establishing the eligibility and airworthiness of the engine, propeller, and articles.

c. Military surplus engines and propellers may be type-certificated under § 21.17, which requires issuance of a new TC and compliance with the applicable requirements, such as Part 33, Airworthiness Standards: Aircraft Engines, for engines and Part 35, Airworthiness Standards: Propellers, for propellers. For a military aircraft issued a TC under §§ 21.25 or 21.27, the applicable engine or propeller is not required to be issued a separate TC. However, note that the engine and propeller cannot be certificated separately under these two sections. Any eligible military surplus engines or propellers will be referenced on the aircraft’s TCDS. However, military unique

surplus engines, propellers, and articles may be eligible for installation only on civil military surplus aircraft with special airworthiness certificates.

d. Engines, propellers, and articles are deemed flight safety critical if their failure, malfunction, or absence could cause a catastrophic failure resulting in loss or serious damage to the aircraft or an uncommanded engine shutdown resulting in an unsafe condition. Such conditions include, but are not limited to, release of engine or propeller debris, propeller separation, and, in rotorcraft, a transient or continuous power loss, or loss of power response. Examples of flight safety critical engine and propeller articles are life limited articles, rotating articles, and, for rotorcraft, actuating articles.

e. Dual Use Military Surplus Engines, Propellers, and Articles. Dual use military surplus engines and propellers that hold a TC, and their articles, may be eligible for installation on civil products per the applicable regulations. The authorized individual completing the eligibility screening and/or the airworthiness evaluation should make a record entry to document the result(s).

(1) Eligibility Screening. New or used dual use engines, propellers, and articles may be eligible for installation on FAA type-certificated civil or surplus military aircraft with standard or special airworthiness certification. A U.S. TC must have been issued for a corresponding civil model engine or propeller under § 21.21 at the time of manufacture, or a U.S. aircraft TC must have been issued and the engines or propellers referenced in the aircraft TCDS under §§ 21.27 or 21.25. The eligibility determination is made based on a review of the following pertinent historical records:

- (a) Engine, propeller, and article ID (article part number and S/N and manufacturer).
- (b) Contract or purchase order number under which the engine, propeller, or article was manufactured.
- (c) Evidence of engine, propeller, and article status, for example, serviceable or unserviceable, per DOD Form (DD Form) 1574-1, *Serviceable Label Materiel*, or Department of the Army (DA) Form 2410, *Component Removal/Repair/Install/Gain/Loss Record*.
- (d) Complete historical records maintained by the military, the manufacturer, and any other prior owner(s), pertaining to inspection, repair, alteration, maintenance, and operation of the engine from the time of acceptance by the military, including, but not limited to, DA Form 2408-5, *Equipment Modification Record*, and DA Form 2408-6, *Aircraft Component Historical Record*. The maintenance records should also include the date on which the work was accomplished and work authentication.
- (e) Current status of applicable ADs and DOD directives (for example, engineering changes, technical orders, or maintenance work orders) including the

date and method of compliance; and, if the AD involves recurring action, the time and date when the next action is required.

(2) Airworthiness Determination. After determining the article is eligible to be installed on a type-certificated product, the article must be evaluated to determine whether it is airworthy.

(a) New Dual Use Engines, Propellers, and Articles.

(1) For engines, propellers, and articles to be installed on aircraft with standard airworthiness certificates, each engine, propeller, and article must conform to the approved TC, must have been manufactured under an FAA-approved production system, and must be in a condition for safe operation.

(2) For engines, propellers, and articles to be installed for aircraft with special airworthiness certificates, each engine, propeller, and article must be listed in the FAA-accepted, military-approved maintenance manual or FAA-accepted civil maintenance manual and IPC specified on the TCDS, and must be in a condition for safe operation.

(b) Used Dual Use Engines, Propellers, and Articles.

(1) For engines, propellers, and articles to be installed on aircraft with standard airworthiness certificates, an evaluation should be performed by an FAA engineer or an appropriately authorized DER. When a DER is used, the DER's recommendations or decisions must be substantiated in writing using FAA Form 8110-3, *Statement of Compliance With Airworthiness Standards*, and include supporting documents. Each engine, propeller, and article must conform to the approved TC, have been manufactured under an FAA-approved production system, and be in a condition for safe operation. In addition, the following should be evaluated:

(a) Operational differences between military and civil versions (for example, possible DOD alteration or repair performed) in performance standards as listed in the TCDS (for example, thrust, shaft horsepower, revolutions per minute (RPM), and ratings), and in specifications, as listed in the TCDS and the maintenance manuals (for example, fuel type, oil, weight).

(b) Complete historical operational records. This includes extreme operational conditions such as accidents, fires, or exceeding engine operating limits.

(c) Complete historical maintenance records; for example, alterations and repairs, and complete documentation of work performed by an FAA-approved facility that was properly rated for the work performed and that conformed to the FAA-approved data.

(d) ICA.

(e) Emission requirements as stated in the TCDS (engine only).

(f) Comparison of military time and/or cycle count for accumulated operational time and cycle versus civil (for example, “Did the military use a different method than civil operators to account for accumulated operational time and what are the expended equivalent civil cycles of the articles, taking into account their past operational history and mission profile?”).

(g) Current manufacturer’s technical data to perform tests or inspections.

(h) Written results of inspections performed (for example, maintenance record entry; FAA Form 8130-3, *Authorized Release Certificate*; or FAA Form 337 for approval for return to service) and a completed FAA Form 8130-9, *Statement of Conformity*.

(i) The application of the identifying marking requirements per §§ 45.11 and 45.13, as applicable.

(j) Engine, propeller, or article overhaul records, including overhaul per civil engine/propeller manuals.

(k) Verification that the engine, propeller, or article was produced by an FAA production approval holder (PAH).

(2) For engines, propellers, and articles to be installed on aircraft with special airworthiness certificates, an evaluation should be performed by an FAA engineer or an appropriately authorized DER. When a DER is used, the DER’s recommendations or decisions must be substantiated in writing using FAA Form 8110-3, *Statement of Compliance With Airworthiness Standards*, and include supporting documents. Each engine, propeller, and article must be listed in the FAA-accepted, military-approved maintenance manual or FAA-accepted maintenance manual and IPC specified on the TCDS, and be in a condition for safe operation. In addition, the following should be evaluated:

(a) Complete historical operational records. This includes extreme operational conditions such as accidents, fires, or engine exceeding operating limits.

(b) Complete historical maintenance records; for example, alterations and repairs, and complete documentation of the work performed.

(c) ICA.

(d) Emission requirements as stated in the TCDS (engine only).

(e) Comparison of military versus civil time and/or cycle count for accumulated operational time and cycle (for example, “Did the military use a different method than civil operators to account for accumulated operational time and what are the expended equivalent civil cycles of the articles, taking into account their past operational history and mission profile?”).

(f) Current manufacturer’s technical data to perform tests or inspections.

(g) Written results of inspections performed (for example, maintenance record entry, FAA Form 8130-3, or FAA Form 337, for approval, for return to service) and a completed FAA Form 8130-9, signed by an authorized person.

(h) The application of the identifying marking requirements per §§ 45.11 and 45.13, as applicable.

(i) Engine, propeller, or article overhaul records, including overhaul per civil engine/propeller manuals.

(j) Verification that the engine, propeller, or article was produced by an FAA PAH.

(3) Approval for Installation. Persons authorized under § 43.7 may determine dual use engines, propellers, or articles for installation if the engine, propeller, or article has successfully completed the eligibility screening and airworthiness evaluation. The installer must be able to determine that the use of the engine or propeller, and/or the installation of the article, will leave the aircraft in compliance with pertinent regulations and in a condition for safe operation. The authorized individual completing the eligibility screening and/or airworthiness evaluation must make a maintenance record entry that clearly documents the results of the evaluation. Each maintenance record entry per § 43.9 should include a description of the work performed, the completion date of the work performed, and the name of the person performing the work or authorized to sign.

f. Military Unique Engines, Propellers, and Their Military Unique Articles. Military unique engines, propellers, and articles are FSCAPs that were specifically and uniquely designed and manufactured for the U.S. military for which there originally was no corresponding FAA-approved PAH engine, propeller, or article for civil application.

(1) Eligibility Screening. New or used military unique engines, propellers, and articles may be eligible for installation on surplus U.S. military aircraft type-certificated under § 21.25(a)(2). The eligibility determination is made based on a review of the following pertinent DOD historical records:

(a) Engine, propeller, article ID (article part number and S/N and manufacturer).

(b) Contract or purchase order number under which the engine, propeller, or article was manufactured.

(c) Evidence of engine, propeller, and article status; for example, serviceable or unserviceable, per DD Form 1574-1 or DA Form 2410.

(d) Complete historical records maintained by the military, the manufacturer, and any other prior owner(s), pertaining to inspection, repair, alteration, maintenance, and operation of the engine from the time of acceptance by the military, including, but not limited to, DA Form 2408-5 and DA Form 2408-16. The maintenance records also should include the date that the work was accomplished and work authentication.

(e) Current status of applicable ADs and DOD directives (for example, engineering change, technical order, maintenance work order), including the date and method of compliance; and, if the AD involves recurring action, the time and date when the next action is required.

(2) Airworthiness Determination. After determining that the engine, propeller, or article is eligible to be installed on a surplus military aircraft with special airworthiness certificates, each engine, propeller, or article must be evaluated to determine whether it is airworthy.

(a) New Military Unique Engines, Propellers, and Articles. For new military unique engines, propellers, and their associated articles to be installed on surplus military aircraft with special airworthiness certificates, each engine, propeller, and article must be listed in the FAA-accepted, military-approved maintenance manual or FAA-accepted civil maintenance manual and IPC specified on the TCDS, and must be in a condition for safe operation.

(b) Used Military Unique Engines, Propellers, and Articles. For used military unique engines, propellers, and articles to be installed on surplus military aircraft with special airworthiness certificates, each engine, propeller, and article must be evaluated by an FAA engineer or an appropriately authorized DER. When a DER is used, the DER's recommendations or decisions must be substantiated in writing using FAA Form 8110-3 and include supporting documents. Each engine, propeller, accessory, and associated article must be listed in the FAA-accepted, military-approved maintenance manual or FAA-accepted civil maintenance manual and the IPC specified on the TCDS, and must be in a condition for safe operation.

(3) Approval for Installation. Persons authorized under § 43.7 may approve military unique engines, propellers, or articles for installation on surplus military aircraft with special airworthiness certificates if they have successfully completed the eligibility screening and airworthiness evaluation. The installer must be able to determine that the use of the engine or propeller, and/or the installation of the article, will leave the product in compliance with the TCDS and in a condition for safe operation. The authorized individual completing the eligibility screening and/or airworthiness evaluation must make a maintenance record entry that clearly documents the results of the evaluation. Each maintenance record entry per § 43.9 should include a description of the work performed, the completion date of the work performed, and the name of the person performing the work or authorized to sign.

Chapter 25. Administrative Information

25-1. Distribution. This Order is distributed to all divisions of the Aircraft Certification Service, Organization Designation Authorization Office, and to the Flight Standards Service.

25-2. Authority to Change This Order. The issuance, revision, or cancellation of the material in this order is the responsibility of the AIR Policy and Standards Division (AIR-600).

25-3. Deviations. Adherence to the procedures in this order is necessary for uniform administration of this directive material. Any deviations from this guidance material must be approved by AIR-600. If a deviation is necessary, the requesting FAA employee should ensure the deviations are substantiated, documented, and concurred with by the appropriate, local supervisor. The deviation must be submitted to AIR-600 for review and approval. The limits of Federal protection for FAA employees are defined in Title 28 of the United States Code (28 USC) § 2679.

25-4. Suggestions for Improvements. Please forward all comments on deficiencies, clarifications, or improvements regarding the contents of this order to the AIR Directives Management Officer (DMO) at 9-AWA-AVS-AIR-DMO@faa.gov. Please find FAA Form 1320-19, Directive Feedback Information, located in appendix K of this order for such feedback. We welcome your suggestions. If you require an immediate interpretation, please contact AIR-600 at (202) 267-1575; and complete FAA Form 1320-19 as a follow-up to the conversation.

25-5. Records Management. Refer to current versions of FAA Order 0000.1, FAA Standard Subject Classification System; FAA Order 1350.14, Records Management; or your office Records Management Officer/Directives Management Officer for guidance regarding retention or disposition of records.

Appendix A. Completing and Reviewing FAA Forms

Section 1. General Procedures.

A-1. General.

- a.** This appendix provides instructions for completing and reviewing certain forms under this order.
- b.** Information entered on these documents by the FAA under this order should be accomplished in AWC, electronically on a PDF form provided by the FAA, or, when using paper, typewritten when possible. The use of pencil, erasures, etc., on airworthiness forms is not permitted. The use of correction fluid or correction tape is prohibited. Application forms may be corrected by the applicant or the FAA, provided the person making the changes initials beside the correction. Following the export of an aircraft, strikeovers are permitted to correct errors on records.
- c.** When the reverse side of the certificate is used, the statement “See Reverse Side” must be typed on the face of the certificate and the word “-END-” must be typed in the center of the page below the last line of information of the reverse side. This is not applicable to certificates generated with AWC.
- d.** The signature of the FAA on any original FAA certificate or FAA form prescribed for use by this order, must be made digitally or in permanent blue or black ink. Digital signatures must be generated from an FAA administered information technology application such as AWC, or electronic form (PDF) provided by the FAA. Use of an electronic signature by non-personal identification verification card (PIV) holders is authorized as prescribed by the FAA Chief Information Officer in FAA Order 1370.121, *FAA Information Security and Privacy Program & Policy*. AWC provides for electronic signatures for non-PIV card holders, i.e., an applicant or designee.
- e.** Dates on certificates should be in DD MMM YYYY or DD Mmm YYYY format; for example, “25 DEC 2016” or “25 Dec 2016.”
- f.** When copies are required under this order and obtaining a copy is impractical, you may issue a certificate or form in duplicate instead; in such cases, annotate the signed, duplicate certificate with the word, “Duplicate.”
- g.** ASIs, designees and ODAs must use AWC to complete all applicable electronic forms or upload paper forms in the application and encourage applicants to do the same. Applicants are not required to use AWC. Follow procedures in paragraph 2-3.a(4) for AWC application or system outages. When using AWC, the term “Attach when required” means scan, copy and/or upload into AWC.

Section 2. Application for U.S. Airworthiness Certificate.

A-2. Completion of FAA Form 8130-6, Application for U.S. Airworthiness Certificate. FAA Form 8130-6 is required for the issuance of an original or recurrent airworthiness certificate. This form is also used to amend or modify, per § 21.177, an airworthiness certificate using the instructions in paragraph 2-4 of this order. It is not used for replacement certificates. Advisory Circular 21-12 provides instructions to applicants for completion of FAA Form 8130-6 using AWC or the paper form. When the airworthiness certificate, or denial, of the requested certificate is issued, the FAA completes sections V and VIII of FAA Form 8130-6 as follows:

a. Section V. FAA Representative Certification. This section must be completed by the ASI, designee, or ODA office member who inspects the aircraft and issues the certificate or denial. Check all applicable blocks in items A and B.

(1) **FAA Office.** An ASI must enter the appropriate CM Section or responsible Flight Standards Office designation (that is, the current CM Section/responsible Flight Standards Office or branch identifier). Designees must enter the designation of the CM Section or responsible Flight Standards Office geographically responsible for overseeing their activities.

(2) **FAA Inspector's Signature or Designee Signature and Number.** The FAA inspector, designee, or unit member of a manufacturer's organization designation authorization office who issued the certificate must sign here. For an ODA, enter "ODA" followed by the ODA number. The designated manufacturing inspection representative (DMIR), designated airworthiness representative (DAR), or ODA office member signature must be signed with the typed or printed name. The ASI's name must be typed or printed with the signature; the typed names and signatures must be legible and must not obstruct preprinted information on FAA Form 8130-6.

(3) **ASI's Certification File Review Signature.** Review the certification file per appendix B to this order. A completed certification file review is indicated by the signature of the reviewing ASI above or adjacent to the typed or printed name in this block. This review is required and may not be performed by the ASI that issued the certificate. For special flight permits, the file review required by this paragraph may be performed by the inspector who issued the certificate.

b. Section VIII. Airworthiness Documentation. This section must be completed by the ASI or designee who inspects the aircraft and issues the airworthiness certificate or denial. However, this section is not applicable when a special flight permit is being issued.

(1) **Item A. Operating Limitations and Markings in Compliance With § 91.9,** as applicable. This block applies to all aircraft covered by this section. The FAA should check this block when an FAA-approved AFM, listing of operation limitation, placards, etc., as applicable to the category of certificate requested, are in the aircraft per § 91.9.

(2) **Item B. Current Operating Limitations Attached.** Check this block when operating limitations have been issued and a copy is attached for retention in the permanent record. This only applies to issuance of a special airworthiness certificate.

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- (3) Items C and D. Self-explanatory. Check all blocks that are applicable.
- (4) Item E. Check this box and attach FAA Form 337 if the approved alteration changes the aircraft category, model, or compliance with Annex 8 of the ICAO Convention. Do not include referenced data forming the basis for approval of the repair or alteration.
- (5) Item F. This Inspection Recorded in Aircraft Records. Check this block after making the maintenance record entry required under paragraph 2-3.h(3) of this order.
- (6) Item G. *Statement of Conformity*, FAA Form 8130-9 (Attach when required). Check this block to indicate FAA Form 8130-9 is attached when required.
- (7) Item H. Foreign Airworthiness Certification for Import Aircraft (Attach when required). Check this block to indicate that certification of another country is required for the certification action and that a copy is attached for retention in the aircraft's permanent record.
- (8) Item I. Conformity Inspection Record FAA Form 8100-1. Check this block to indicate that a completed FAA Form 8100-1 is attached for retention in the aircraft's permanent record.
- (9) Item J. Superseded, terminated, or canceled airworthiness certificate Issued in Accordance With 14 CFR Section _____ CAR _____ (Original attached). Check this box and enter the applicable CFR or CAR of the superseded, terminated, or canceled airworthiness certificate, if applicable.
- (10) Item K. Current Airworthiness Certificate Issuance via this Application in Accordance With 14 CFR. Check this box and enter the applicable section from Part 21, subpart H. For an amendment, enter § 21.177.
- (11) Item L. *Light-Sport Category Aircraft/Kit Statement of Compliance*, FAA Form 8130-15 (Attach when required). Check this block to indicate that a completed applicable copy of the manufacturer's SOC, FAA Form 8130--15, is attached for retention in the aircraft's permanent record.
- (12) Item M. Reserved.
- (13) Item N. Denial Letter. Check this box that a copy of an issued denial letter is attached for retention in the aircraft's permanent record.
- (14) Item O. Certification file forwarded to the registry DOES NOT contain information that is proprietary, confidential, company-sensitive, subject to International Traffic in Arms Regulations (ITAR) (22 CFR, subchapter M), subject to Export Administration Regulations (EAR) (15 CFR Part 734), or otherwise restricted from public or foreign disclosure. Check this box affirming the aforementioned.

Section 3. Standard Airworthiness Certificate.

A-3. Completion of FAA Form 8100-2, Standard Airworthiness Certificate. Figure A-1 of this appendix is an example of a completed standard airworthiness certificate. Complete FAA Form 8100-2 as follows:

a. Nationality and Registration Marks. Enter the capital letter “N” followed by the registration number assigned to the aircraft. For antique or replica aircraft displaying marks as provided for under § 45.22(b), do not include the additional symbols allowed for marking the aircraft in this block.

b. Manufacturer and Model. This information must exactly match the information on the aircraft ID plate. For surplus military aircraft, enter the military model in parentheses after the civil model designation.

Note: Section 45.13 uses the term “builder.”

c. Aircraft Serial Number. Self-explanatory. For surplus military aircraft, enter the military S/N in parentheses after the civil S/N.

d. Category. Enter the appropriate category or categories, as applicable, from the application. If there is no category, as in the case of aircraft certificated before adoption of the regulations that established categories, enter the aircraft specification, or TCDS, as applicable. For example, “CAR 4a” for a Bellanca 14-13; “ATC 614” for an Aeronca LC; and “Aero Bulletin 7A” for a Douglas DC3. For very light aircraft (VLA) type-certificated under § 21.17(b), enter, “VLA-Special Class.” For an import VLA, enter, “VLA-Import.”

e. Authority and Basis for Issuance. Block 5 of a standard airworthiness certificate includes pre-printed statements of airworthiness and of compliance with ICAO Annex 8. It also includes introductory text for recording exceptions to the statement of compliance with ICAO Annex 8. Although unusual, such an exception occurs when an applicable FAA exemption does not comply with ICAO Annex 8. Review the applicable aircraft TCDS to identify applicable exemptions by make, model, and S/N. Review each exemption to determine if it contains language indicating a noncompliance with Annex 8 of the ICAO Convention or limiting the aircraft to domestic operations only. You may find FAA exemptions granted to date in [DRS](#). If an exemption indicates noncompliance with Annex 8 or limits the aircraft to domestic operations only, add the following statement to block 5 of the standard airworthiness certificate: “This aircraft is limited to domestic operations only because exemption [number/numbers] [insert exemption number(s)] [does/do] not meet Annex 8 of the ICAO Convention.” Otherwise, enter “None.”

f. Date of Issuance.

(1) Prefix. When a certificate is being amended, exchanged, or replaced, insert the letter “A” or “E” or “R,” respectively, before the date.

(2) Date. For a replacement or exchange certificate, enter the date of the original certificate. Replacing an airworthiness certificate for an N-number change is an administrative function and does not require a finding of airworthiness; therefore, the original certificate date is entered. For amended certificates, enter the date the amended certificate is issued.

g. FAA Representative. The typed name and signature of the FAA representative issuing the certificate must be legible. The PDF version of the form includes two fields in the FAA representative block, one for a digital signature and the other for entering the name of the person signing the form; use only one of those fields.

h. Designation Number. Depending on who issues the certificate, enter the following applicable information:

- (1) ASI. The office identifier can be the current CM Section or responsible Flight Standards Office or branch identifier.
- (2) DMIR or DAR. The unique designee identification number.
- (3) ODA. The letters "ODA" followed by the ODA number.

i. Printing a Certificate. The preferred method of generating a certificate is through AWC. When printing a certificate using the PDF version of the form, set the page scaling to none, or actual size. The printed form should be approximately 5½" by 3-3/8". Use white 8½" by 11" paper (if the applicant wishes, they may trim the document to size) or appropriate size card stock. You should use a heavy weight paper or cardstock. The forms will print in grayscale; using a color printer is not required. Laser print may be subject to displacement. Encourage applicants to protect the document by lamination or other document protection processes.

Figure A-1. Example FAA Form 8100-2, *Standard Airworthiness Certificate*

UNITED STATES OF AMERICA DEPARTMENT OF TRANSPORTATION-FEDERAL AVIATION ADMINISTRATION STANDARD AIRWORTHINESS CERTIFICATE			
1 NATIONALITY AND REGISTRATION MARKS N54321	2 MANUFACTURER AND MODEL Cirrus Design Corporation SR20	3 AIRCRAFT SERIAL NUMBER 9676	4 <u>CATEGORY</u> Normal
5 AUTHORITY AND BASIS FOR ISSUANCE This airworthiness certificate is issued pursuant to 49 U.S.C. § 44704 and certifies that, as of the date of issuance, this aircraft has been inspected and found to conform to its type certificate and be in a condition for safe operation. This aircraft meets the requirements of the applicable airworthiness standards in Annex 8 to the Convention on International Civil Aviation, except as follows: None.			
6 TERMS AND CONDITIONS Unless sooner surrendered, suspended, revoked, or a termination date is otherwise established by the FAA, this airworthiness certificate is effective as long as maintenance, preventative maintenance, and alterations are performed per the applicable Federal Aviation Regulations and the aircraft is registered in the United States.			
DATE OF ISSUANCE 19 Aug 2023	FAA REPRESENTATIVE <u>James Edward Newberger</u>		DESIGNATION NUMBER AIR-6XX
Any alteration, misuse, or reproduction of this certificate for a fraudulent purpose may be punishable by certificate revocation, fine, and / or imprisonment. THIS CERTIFICATE MUST BE DISPLAYED IN THE AIRCRAFT PER THE APPLICABLE FEDERAL AVIATION REGULATIONS.			
FAA Form 8100-2 (11-2016) Previous Edition Dated 04-11 May be Used Until Depleted			

Section 4. Special Airworthiness Certificate

A-4. Completion of FAA Form 8130-7, Special Airworthiness Certificate. Figures A-2 through A-6 are examples of completed special airworthiness certificates. Complete FAA Form 8130-7 as follows:

a. Category/Designation. Enter the appropriate category or designation from the application: restricted, limited, primary, light-sport, experimental, provisional, or special flight permit. When issuing an experimental airworthiness certificate for a kit-built or former light-sport category aircraft, enter “Experimental,” not “Light-sport.”

b. Purpose. Enter the operating purpose as it is described in the applicable CFR section that authorizes the certificate. You may abbreviate the wording if the meaning remains clear.

(1) Limited or Primary. For a limited or primary category airworthiness certificate, enter “N/A.”

(2) Experimental. You may abbreviate experimental purposes as follows: “R&D” for research and development and “Show Compliance” for showing compliance with regulations.

(3) Restricted. For a restricted category airworthiness certificate, enter the use(s) and the special purpose operation(s). For example, Agricultural: spraying, dusting, insect control. Due to limited space in the Purpose box, two or more restricted category uses may be abbreviated as necessary. If all approved special purpose operations do not fit in the Purpose box, enter the approved uses only. In this case, the applicable special purpose operation(s) must be identified by listing them immediately preceding the first operating limitation. This action does not necessitate the use of the coordination mailbox in appendix D.

Note: See figure A-2. If necessary, immediately preceding the first operating limitation, list the applicable use(s) and special purpose operation(s) not listed in the Purpose box of the certificate as follows: “In addition to the Restricted use(s) listed in the Purpose box, this special airworthiness certificate is issued for the following Restricted category use(s) and special purpose operation(s):” “[*enter Restricted use*]: [*enter only the approved special purpose operation(s)*]”

c. Manufacturer. Enter the name and address of the manufacturer only if the application is for an SFP for the purpose(s) of production flight testing and/or customer demonstration. In all other cases, enter “N/A” in both spaces under this section. For surplus military aircraft, enter the manufacturers' name and model designation; if not available, enter the military model designation.

d. Flight From/To.

(1) For an SFP for purposes other than production flight testing, the flight “From” and flight “To” spaces must be the same as those shown on FAA Form 8130-6, section VII, item B.

(2) For an SFP production flight testing or multiple purpose of production flight testing and customer demonstration, enter “N/A” in both spaces.

(3) Otherwise, enter “N/A” in both spaces.

e. N-Number, Model, Serial No., and Manufacturer/Builder.

(1) Production Flight Testing.

(a) Non-light-sport category aircraft. For an aircraft that is not a light-sport category aircraft and for the purpose of production flight testing or for multiple purpose of production flight testing and customer demonstration, enter “N/A” in all spaces.

(b) Light-sport category aircraft. For production flight testing of light-sport category aircraft, enter the registration number, aircraft S/N, aircraft model, and builder. For transferable SFPs, enter “N/A” in all spaces.

(2) All Others. For all other categories and purposes, information to complete this section would be contained in section I or VII of the application for an airworthiness certificate. For surplus military aircraft, enter the builder’s S/N and, if applicable, the military S/N in parentheses.

(3) For antique or replica aircraft displaying marks as provided for under § 45.22(b), do not include the additional symbols allowed for marking the aircraft in the aircraft registration number on the airworthiness certificate.

f. Date of Issuance.

(1) Prefix. When a certificate is being amended, exchanged, or replaced, insert the letter “A” or “E” or “R,” respectively, before the date.

(2) Date. For a replacement or exchange certificate, enter the date of the original certificate. Replacing an airworthiness certificate for an N-number change is an administrative function and does not require a finding of airworthiness; therefore, the original certificate date is entered. For amended certificates, enter the date the amended certificate is issued.

g. Expiry. Enter the date of expiry or “unlimited,” as applicable.

(1) For transferable SFPs, a special flight permit is effective for the period of time specified in the permit.

(2) An experimental airworthiness certificate for R&D, showing compliance with regulations, crew training, or market survey is effective for up to three years after the date of issue or renewal. The certificate should not be effective for longer than the applicant’s intended operations. The applicant should have described its intended operations in sufficient detail for you to identify risks related to those operations and mitigate those risks via appropriate operating limitations.

(3) The duration of an experimental airworthiness certificate issued for exhibition, air-racing, operating amateur-built aircraft, operating primary kit-built aircraft, operating

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light-sport aircraft, operating light-sport category kit-built aircraft, operating former light-sport category aircraft, and operating former military aircraft is unlimited unless the FAA establishes a specific period for good cause.

(4) For a provisional certificate, the entry should be per § 21.217.

h. Signature of FAA Representative and Designation or Office No. Complete this block for all categories and purposes per paragraphs A-3.g and A-3.h of this appendix.

i. Conditions and Limitations. Draft the operating limitations per appendix D of this order. When using AWC, printing the completed certificate will automatically include the operating limitations. When using the paper GPO pad version of FAA Form 8130-7, attach the operating limitations to the completed certificate. When using the PDF version of FAA Form 8130-7, add the applicable operating limitations from Table D-1 into the Conditions and Limitations section of FAA Form 8130-7.

j. Printing a Certificate. Except for the size of the printed form, refer to paragraph A-3.i for instructions for printing a certificate using the PDF version of the form. When the printed certificate from AWC spans multiple pages, the registration number, S/N, and date are populated at the top of each page. When the printed certificate from PDF version spans multiple pages, the registration number and date are populated at the bottom of each page. Attach the pages together.

k. Advising the Applicant. Advise the applicant that it is acceptable to duplicate and trim the first page for displaying the certificate per § 91.203.

Figure A-2. Example FAA Form 8130-7, Special Airworthiness Certificate for Restricted Category Aircraft

UNITED STATES OF AMERICA DEPARTMENT OF TRANSPORTATION FEDERAL AVIATION ADMINISTRATION SPECIAL AIRWORTHINESS CERTIFICATE			
CATEGORY/DESIGNATION Restricted			
PURPOSE Agriculture; Patrolling			
MANU-FACTURER	NAME N/A		
	ADDRESS N/A		
FLIGHT	FROM N/A		
	TO N/A		
N 54321		MODEL S2R-H80	SERIAL NO. H80-000
MFG/BUILDER Thrush Aircraft Inc.		DATE OF ISSUANCE 4/28/25	
Unless sooner surrendered, suspended, revoked, or the termination date of <u>unlimited</u> , this airworthiness certificate is effective under the conditions prescribed in 14 CFR, Part 21, Section 21.181 or 21.217.			
SIGNATURE OF FAA REPRESENTATIVE		DESIGNATION OR OFFICE NO. AFS-8XX	
This airworthiness certificate is issued under the authority of Title 49 United States Code 44704 and Title 14 Code of Federal Regulations. Any alteration, misuse or reproduction for a fraudulent purpose of this certificate may be punishable by certificate revocation, fine, and / or imprisonment. THIS PORTION OF THE CERTIFICATE MUST BE DISPLAYED IN THE AIRCRAFT PER THE APPLICABLE REGULATIONS.			

-- Conditions and Limitations --

In addition to the Restricted use(s) listed in the Purpose box, this special airworthiness certificate is issued for the following

Restricted category use(s) and special purpose operation(s):

Agricultural: 1) Spraying, dusting, and seeding; 2) Insect control.

Patrolling: 1) Pipelines; 2) Power lines; 3) Railroads; 4) Harbors.

- END -

UNITED STATES OF AMERICA DEPARTMENT OF TRANSPORTATION FEDERAL AVIATION ADMINISTRATION SPECIAL AIRWORTHINESS CERTIFICATE			
CATEGORY/DESIGNATION Restricted			
PURPOSE Agriculture: Insect Control			
MANU-FACTURER	NAME N/A		
	ADDRESS N/A		
FLIGHT	FROM N/A		
	TO N/A		
N 54321		MODEL S2R-H80	SERIAL NO. H80-000
MFG/BUILDER Thrush Aircraft Inc.		DATE OF ISSUANCE 6/12/25	
Unless sooner surrendered, suspended, revoked, or the termination date of <u>unlimited</u> , this airworthiness certificate is effective under the conditions prescribed in 14 CFR, Part 21, Section 21.181 or 21.217.			
SIGNATURE OF FAA REPRESENTATIVE		DESIGNATION OR OFFICE NO. AFS-8XX	
This airworthiness certificate is issued under the authority of Title 49 United States Code 44704 and Title 14 Code of Federal Regulations. Any alteration, misuse or reproduction for a fraudulent purpose of this certificate may be punishable by certificate revocation, fine, and / or imprisonment. THIS PORTION OF THE CERTIFICATE MUST BE DISPLAYED IN THE AIRCRAFT PER THE APPLICABLE REGULATIONS.			

Figure A-3. Example FAA Form 8130-7, Special Airworthiness Certificate for Research and Development

UNITED STATES OF AMERICA DEPARTMENT OF TRANSPORTATION FEDERAL AVIATION ADMINISTRATION SPECIAL AIRWORTHINESS CERTIFICATE			
CATEGORY/DESIGNATION Experimental			
PURPOSE Research and Development			
MANU-FACTURER	NAME	N/A	
	ADDRESS	N/A	
FLIGHT	FROM	N/A	
	TO	N/A	
N 54321		MODEL F-104A	SERIAL NO. 65-730
MFG/BUILDER Lockheed			DATE OF ISSUANCE 6/12/25
Unless sooner surrendered, suspended, revoked, or the termination date of <u>6/11/2028</u> , this airworthiness certificate is effective under the conditions prescribed in 14 CFR, Part 21, Section 21.181 or 21.217.			
SIGNATURE OF FAA REPRESENTATIVE			DESIGNATION OR OFFICE NO. AIR-8XX
This airworthiness certificate is issued under the authority of Title 49 United States Code 44704 and Title 14 Code of Federal Regulations. Any alteration, misuse or reproduction for a fraudulent purpose of this certificate may be punishable by certificate revocation, fine, and / or imprisonment. THIS PORTION OF THE CERTIFICATE MUST BE DISPLAYED IN THE AIRCRAFT PER THE APPLICABLE REGULATIONS.			

Figure A-4. Example FAA Form 8130-7, Special Flight Permit for Production Flight Test, Customer Demonstration or both

UNITED STATES OF AMERICA DEPARTMENT OF TRANSPORTATION FEDERAL AVIATION ADMINISTRATION SPECIAL AIRWORTHINESS CERTIFICATE			
CATEGORY/DESIGNATION Special Flight Permit			
PURPOSE Production flight test, Customer demonstration			
MANU-FACTURER	NAME	The Boeing Company	
	ADDRESS	1901 Oakesdale Ave SW, Renton, WA 98057-2623	
FLIGHT	FROM	N/A	
	TO	N/A	
N N/A		MODEL N/A	SERIAL NO. N/A
BUILDER N/A			DATE OF ISSUANCE 2/27/25
Unless sooner surrendered, suspended, revoked, or the termination date of <u>27 Feb 2026</u> , this airworthiness certificate is effective under the conditions prescribed in 14 CFR, Part 21, Section 21.181 or 21.217.			
SIGNATURE OF FAA REPRESENTATIVE			DESIGNATION OR OFFICE NO. AIR-8XX
This airworthiness certificate is issued under the authority of Title 49 United States Code 44704 and Title 14 Code of Federal Regulations. Any alteration, misuse or reproduction for a fraudulent purpose of this certificate may be punishable by certificate revocation, fine, and / or imprisonment. THIS PORTION OF THE CERTIFICATE MUST BE DISPLAYED IN THE AIRCRAFT PER THE APPLICABLE REGULATIONS.			

Figure A-5. Example FAA Form 8130-7, Special Airworthiness Certificate for Light-Sport Category Aircraft Certificated Under § 21.190

UNITED STATES OF AMERICA DEPARTMENT OF TRANSPORTATION FEDERAL AVIATION ADMINISTRATION SPECIAL AIRWORTHINESS CERTIFICATE			
CATEGORY/DESIGNATION Light-Sport			
PURPOSE Airplane			
MANU-FACTURER	NAME	N/A	
	ADDRESS	N/A	
FLIGHT	FROM	N/A	
	TO	N/A	
N 54321		MODEL CTSW	SERIAL NO. 9876
MFG/BUILDER Flight Design GmbH			DATE OF ISSUANCE 6/12/25
Unless sooner surrendered, suspended, revoked, or the termination date of <u>unlimited</u> , this airworthiness certificate is effective under the conditions prescribed in 14 CFR, Part 21, Section 21.181 or 21.217.			
SIGNATURE OF FAA REPRESENTATIVE			DESIGNATION OR OFFICE NO. AIR-8XX
This airworthiness certificate is issued under the authority of Title 49 United States Code 44704 and Title 14 Code of Federal Regulations. Any alteration, misuse or reproduction for a fraudulent purpose of this certificate may be punishable by certificate revocation, fine, and / or imprisonment. THIS PORTION OF THE CERTIFICATE MUST BE DISPLAYED IN THE AIRCRAFT PER THE APPLICABLE REGULATIONS.			

Figure A-6. Example FAA Form 8130-7, Special Airworthiness Certificate for Experimental Light-Sport Category Aircraft Certificated Under § 21.191(k) or (l)

UNITED STATES OF AMERICA DEPARTMENT OF TRANSPORTATION FEDERAL AVIATION ADMINISTRATION SPECIAL AIRWORTHINESS CERTIFICATE			
CATEGORY/DESIGNATION Experimental			
PURPOSE Operating Light-Sport Category Kit-Built Aircraft			
MANU-FACTURER	NAME	N/A	
	ADDRESS	N/A	
FLIGHT	FROM	N/A	
	TO	N/A	
N 54321		MODEL Wile E	SERIAL NO. Roxanne-10001
MFG/BUILDER The ACME Corporation			DATE OF ISSUANCE 6/12/27
Unless sooner surrendered, suspended, revoked, or the termination date of <u>unlimited</u> , this airworthiness certificate is effective under the conditions prescribed in 14 CFR, Part 21, Section 21.181 or 21.217.			
SIGNATURE OF FAA REPRESENTATIVE			DESIGNATION OR OFFICE NO. AIR-8XX
This airworthiness certificate is issued under the authority of Title 49 United States Code 44704 and Title 14 Code of Federal Regulations. Any alteration, misuse or reproduction for a fraudulent purpose of this certificate may be punishable by certificate revocation, fine, and / or imprisonment. THIS PORTION OF THE CERTIFICATE MUST BE DISPLAYED IN THE AIRCRAFT PER THE APPLICABLE REGULATIONS.			

Section 5. Light-Sport Category Aircraft/Kit Statement of Compliance.

A-5. Reviewing FAA Form 8130-15, Light-Sport Category Aircraft/Kit Statement of Compliance (SOC). This form is used by the manufacturer of light-sport category aircraft and of kits for kit-built light-sport category aircraft. When reviewing the aircraft manufacturer's SOC, ensure the form is current, accurate, and complete per § 21.190(d), Issuance of a special airworthiness certificate for a light-sport category aircraft.

a. General.

- (1) A SOC that was issued before July 24, 2026, for a kit that was not fully assembled before this date, is acceptable if the SOC was completed properly per paragraph A-5, including stating compliance with applicable FAA-accepted consensus standards as of the date of issuance of that SOC.
- (2) A SOC that was issued on or after July 24, 2026, must state compliance with the applicable consensus standards accepted by the FAA for use on or after this date.
- (3) Verify the information is correct and appropriate for the aircraft identified by the application, registration, the required documentation, and the physical inspection of the aircraft and the aircraft's identification plate.
- (4) Verify the date of manufacture on the aircraft's identification plate matches the date on the registration and FAA Form 8130-15, and that the aircraft serial number is correctly entered on pages 2, 3, and 4.
- (5) Verify FAA-accepted consensus standards listed on the form are appropriate for the aircraft class (airplane, rotorcraft, glider, weight-shift-control, powered parachute, lighter-than-air, gyroplane, helicopter, or powered lift). Manufacturers must use the FAA-accepted consensus standards in effect at the time the aircraft or kit was manufactured.
- (6) If a Compliance Package is used by the manufacturer, (see block 15), a copy of the Compliance Package is attached to the form as part of the aircraft's permanent records.

Note: Refer to the [Light-Sport Category Aircraft public website](#) for the listing of consensus standards which are acceptable for inclusion on the manufacturer's SOC.

Note: FAA employees can access subscriptions to consensus standards through the [FAA Technical Library](#). These subscription services are available exclusively to current FAA employees.

b. Blocks 1 through 18.

- (1) Block 1. Aircraft Manufacturer's Name. Verify the official legal name of the manufacturer matches the name as it appears on the aircraft identification plate per § 45.13.
- (2) Block 2. Aircraft Manufacturer's Street Address. Verify the street address of the aircraft manufacturer is a physical location.

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- (3) Block 3, Country of Aircraft Manufacture. See Chapter 9, Light Sport Category Aircraft, § 21.190 or Chapter 17, Experimental Purposes of Operating Light-Sport Aircraft § 21.191(i), Operating Light-Sport Category Kit-Built Aircraft § 21.191(k), and Operating Former Light-Sport Category Aircraft § 21.191(l)
- (4) Block 4, Date of Manufacture. Verify the date the aircraft was manufactured on the SOC matches the aircraft identification plate. The date of manufacture for kit aircraft is the date the kit was completed by the manufacturer.
- (4) Block 5, Aircraft Serial Number. Verify the aircraft serial number on the SOC matches the aircraft identification plate and the serial number entered on page 3 and 4.
- (5) Block 6, Aircraft Make. Verify the name is the same name assigned by the aircraft manufacturer when it was produced. For example, see the [FAA Make and Model Directory](#) for Light Sport Category Aircraft.
- (6) Block 7, Aircraft Model. Verify the aircraft's model designation matches the aircraft's identification plate. For example, see the [FAA Make and Model Directory](#) for Light Sport Category Aircraft.
- (7) Block 8, Amended Per § 21.190(e). Verify that "yes or no" is selected. If "yes" is selected, verify the applicant lists in block 18 and states compliance to the standards from the original statement of compliance or adds a reference to an attached copy of the original statement of compliance. Blocks 9 through 16 are not used for an amended SOC.
Note: As an option, the manufacturer may attach supplementary documentation. If attachments are included, ensure the comments section indicates that additional documentation is attached. Each page of the attached document should reference the aircraft manufacturer, model, and serial number at the top. To maintain the integrity of the form, the attached documentation becomes part of the SOC.
- (8) Block 9, First Make or Model. Verify the make and model have not been previously issued a light-sport category airworthiness certificate.
- (9) Block 10, Airworthiness Certificate. Verify the aircraft certification basis selected is either a light-sport category aircraft per § 21.190, or a kit of a previously certificated light-sport category model per § 21.191(k).
- (10) Block 11, Aircraft Class. Verify the correct class is selected and applicable FAA-accepted consensus standards and manufactures documentation reflect the correct class (airplane, rotorcraft, glider, weight-shift-control, powered parachute, lighter-than-air, gyroplane, helicopter, or powered lift).
- (11) Block 12, Simplified Flight Controls. If applicable, verify the SOC indicates the aircraft meets the design requirements of § 22.180 and the corresponding statement on page 3, block 16, is selected for simplified flight controls.
Note: Verify that any modifications or additions were authorized by the aircraft manufacturer or person acceptable to the FAA.

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Note: To meet the requirement for submitting an amended SOC to the FAA, advise aircraft owner to submit each amended SOC to the FAA Registry for inclusion in the airworthiness file for the aircraft.

- (12) Block 13, Aerial Work. If applicable, verify towing and any aerial work operations per §21.190(d)(3), § 22.195(d) and operations are specified in block 13. The corresponding statement on page 3, block 16 is selected for aerial work. If the manufacturer has not authorized aerial work operations in the POH, verify this box reads N/A.

Note: An experimental kit-built or former light-sport category aircraft may not be used for aerial work or towing operations, regardless of whether these operations were included by the aircraft manufacturer in the original AOI or POH, or in documentation for an amended SOC per § 21.190(e).

- (13) Block 14, Applicable FAA-Accepted Consensus Standard(s) for Light-Sport Category Aircraft. Each FAA-accepted consensus standard listed should include the source, standard number, and version level.

Note: A consensus standard may address multiple topics and can be listed more than once. The version referenced must be the FAA-accepted version effective on the date of manufacture shown in Block 4. For details on FAA-accepted consensus standards, including their effective dates, refer to the applicable FAA Notice of Availability or [Light-Sport Category Aircraft public website](#).

- (14) Block 15. Manufacturer's Documentation. Verify that the manufacturer's documentation includes the title of each document or company identifier, along with the version and issue date, and that it meets the corresponding consensus standard used for compliance. The manufacturer may prepare a Compliance Package validates compliance with the applicable FAA-accepted consensus standards for all related compliance documents.

Note: A Compliance Package, as defined by F2930, may be used in place of documenting Block 15. If used, the manufacturer includes a set of documents which provides objective evidence, verifiable evidence for compliance to applicable FAA-accepted consensus standards. The manufacturer should indicate a "Compliance Package" is attached in the comment section of block 15. The aircraft make; model and serial number are documented on each page of the "Compliance Package" to maintain the integrity of the documentation.

- (15) Block 16. Statement of Compliance Affidavit. Verify the name, signature, and title representing the manufacturer is documented in the documentation. A second signature may be required in accordance with the manufacturer's procedures.

Note: The FAA Form 8130-15 is signed by the manufacturer's authorized representative who is certified and trained on the requirements associated with the issuance of a SOC by an organization that certifies and trains quality assurance staff in accordance with a consensus standard that has been accepted by the FAA per § 21.190(d)(1). Other signatures by non-manufacturer employees are unacceptable.

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- (16) Block 17. Verify the aircraft's serial number is listed to maintain the integrity of the form.
- (17) Block 18. Amendments to the Statement of Compliance. This block should be completed if block 8, Amended per § 21.190(e), is entered as "yes." The applicant lists and states compliance to an attached copy of the original statement of compliance per § 21.190(e)(4), (e)(5) and (e)(6).

Section 6. Application for Export Certificate of Airworthiness

A-6. Reviewing FAA Form 8130-1, Application for Export C of A. The applicant completes part I of the application. Part II is no longer applicable because an export C of A is no longer issued for aircraft engines, propellers, or articles; refer to FAA Order 8130.21, *Procedures for Completion and Use of the Authorized Release Certificate, FAA Form 8130-3, Airworthiness Approval Tag*, for procedures for the application and issuance of export approvals for aircraft engines, propellers, and articles. Part III is for FAA use only. Instructions for completing part I are intended to help the FAA review submitted applications.

- a.** Part I (For Aircraft). This is part of the form is completed by the applicant. Refer to AC 21-44.
- b.** Part II (For Aircraft Engines, Propellers, and Articles). An export C of A is no longer issued for aircraft engines, propellers, or articles. Refer to FAA Order 8130.21 and AC 21-44 for procedures for issuance of export approvals for aircraft engines, propellers, and articles.
- c.** Part III. Approval (For FAA Use Only).
 - (1)Item No. 21. Check the “Part I” box. Check the “Denial Letter Issued” box if applicable. Digitally sign the form or enter your typed or printed name and sign the form. The number should be the office identifier or designee number. An ODA must use its authorization number as assigned by the FAA. Enter the date as shown on the issued Export Certificate of Airworthiness. Signing an application received with the applicant’s original signature, or as a paper copy, fax, or PDF copy is acceptable.
 - (2)Item No. 22. Not applicable to issuing an export C of A for aircraft.
 - (3)Item No. 23. Review certification file per appendix B, paragraph B-3. Enter typed or printed name of reviewing ASI. A completed file review is indicated by the signature of the reviewing ASI in the “FAA Supervisory Inspector” block along with the office number and date.

Section 7. Export Certificate of Airworthiness.

A-7. Completion of FAA Form 8130-4, Export Certificate of Airworthiness.

a. General.

(1) Completion of FAA Form 8130-4 is accomplished in AWC. The use of the PDF version found on the [FAA Forms](#) website is acceptable if conditions in paragraph 2-3a(4) apply.

(2) See Figure A-9 of this appendix for a completed sample FAA Form 8130-4.

(3) All entries should be typewritten or completed in ink with no erasures, however, following the export of an aircraft, minor strikeovers and corrections are allowed as long as the form remains legible.

b. Certifying Statement. Enter the specification or TCDS for the aircraft, aircraft engine, and propeller, as applicable, in the space provided in the certifying statement.

c. Product, Manufacturer, Model, Serial Number. Enter the manufacturer, model, and S/N of the aircraft and each installed engine and propeller.

d. New/Used Blocks. Mark the applicable block to specify whether the aircraft is new or used.

e. Country Block. Enter the official name of the importing country as indicated in the applicable bilateral agreement, for example, People's Republic of China, Canada, and Federative Republic of Brazil.

f. Exceptions.

(1) Nonconformities and Noncompliance. List each nonconformity to the applicable TC(s) and each noncompliance with the SIR of the importing country. If there are no exceptions, enter "None."

(2) Temporary Installations. The following instructions apply to preparation of the export C of A when temporary installations, such as provisions for extra fuel or navigational equipment, have been made for export delivery:

(a) If the export C of A is issued after the installation has been made, either by the manufacturer or by other persons, the following statement or equivalent should be inserted under Exceptions: "A temporary [*insert type of installation*] has been installed in this aircraft in conformity with [*insert drawing numbers, or other data to which conformity was shown*] to facilitate its delivery flight. This certificate is valid when the temporary installation is removed." Copies of all referenced drawings and data should accompany the original export C of A when it is submitted to the applicant or the applicant's representative.

(b) If the export C of A is issued before making the temporary installation, such as at the manufacturer's plant, and the aircraft is then flown to another location for installation of the temporary equipment, the export C of A should reflect the configuration of the aircraft at the time the certificate was issued. It then becomes the responsibility of the exporter and importer to secure the installation documents

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or data required by the importing CAA. The U.S. export C of A may not be amended, reissued, or revalidated after original issuance.

(3) Third Country Aircraft. When a third country aircraft is exported to a country with which the United States has a bilateral agreement but no third country provision in that agreement, the following statement must be inserted on the export C of A under Exceptions: “This aircraft was not manufactured in the United States and this certificate is not issued pursuant to the bilateral agreement providing for the reciprocal recognition of airworthiness certificates between the United States and [*Country name*], which has stated its willingness to accept this certificate under these conditions, as indicated in their communication, reference _____, dated _____.”

(4) Primary or Restricted Category Aircraft. Include the following comment under Exceptions: “The above is a [*primary/restricted*] category aircraft. This aircraft has not been determined to meet the international standards concerning the airworthiness of aircraft as provided for in Annex 8 to the Convention on International Civil Aviation.”

g. Additional Information. As necessary, enter any additional information or statements required by the applicable TCDS, applicable bilateral agreement, or FAA AC 21-2, appendix 2, *Special Requirements of Importing Countries*. For example, some importing CAAs require the FAA to find conformity and include a statement certifying conformity to the approved TC of the importing CAA. The PDF version of FAA Form 8130-4 provides an “Additional Information” block.

h. Signature of Authorized Representative.

(1) Signature of Authorized Representative. Type the name and FAA authority of the person signing the form adjacent to or under the signature. Authorized representatives may digitally sign the PDF form.

(2) Date. Enter the date the aircraft was inspected by the FAA, found to comply with the applicable requirements, and determined to be airworthy. For a replacement certificate for correction of administrative errors, enter the date of the original certificate with the letter “R” before the date.

(3) District Office or Designee Number.

(a) An ASI must enter the district office designation.

(b) An individual designee must enter his or her designee number.

(c) An ODA must enter the name of the company, “ODA,” and their ODA number.

Figure A-9. Example FAA Form 8130-4, *Export Certificate of Airworthiness*



Export Certificate of Airworthiness

This certifies that the aircraft identified below and particularly described in Specification(s)¹ of the Federal Aviation Administration, Numbered A00009CH, E3SO, and P37EA has been examined as of the date of this certificate, is considered airworthy in accordance with a comprehensive and detailed airworthiness code of the United States Government, and is in compliance with those special requirements of the importing country filed with the United States Government, except as noted below. The certificate in no way attests to compliance with any agreements or contracts between the vendor and purchaser, nor does it constitute authority to operate an aircraft.

Product	Manufacturer	Model	Serial Number(s)
Aircraft	Cirrus Design Corporation	SR22	9999
Engine(s)	Teledyne Continental	IO-550-N	9999
Propeller(s)	Hartzell Propeller Inc.	BHC-J2YF-1BF/F7694	9999

New: ☒ Used: ☐ Country to which exported: The United Kingdom

Exceptions:

Additional Information: The SR22 covered by this certificate conforms to the type design approved under EASA Type Certificate Number EASA.IM.A.007 and is found to be in a condition for safe operation.

Donald Marvel, *Donald Marvel*

Name and Signature of Authorized Representative

April 22, 2024

Date

AIR-842

FAA Office or Designee Number

¹List applicable specification or Type Certificate Data Sheet for the aircraft, engine, and propeller. Applicable specification or Type Certificate Data Sheet, if not attached to this export certificate, will have been forwarded to the appropriate governmental office of the importing country.

Appendix B. Documents for Review and Forwarding.

B-1. Introduction.

This appendix provides procedures for reviewing and forwarding airworthiness certification files. For each certificate or denial issued by a designee, the designee's managing specialist ensures all relevant records are obtained from that designee.

B-2. Airworthiness Certification Files. For the purposes of forwarding documents to the FAA Aircraft Registry, certification files consist of the following:

a. Standard Airworthiness Certificate. Documents supporting the issuance of a standard airworthiness certificate are listed in Table B-1 of this appendix.

b. Special Airworthiness Certificates. Documents supporting the issuance of a special airworthiness certificate are listed in Table B-2 of this appendix.

c. Imports. For imported aircraft, certification files must also include a copy of the export C of A, if provided by the applicant, including any written acceptance of exceptions from the FAA, and, if available, the last foreign airworthiness certificate issued for that aircraft.

d. Export C of A. Documents supporting the issuance of an export C of A include:

(1) The original FAA Form 8130-1.

(2) The original or copy of the statement of acceptance from an importing country listing the specific nonconformities to the approved TC and noncompliance(s) to special requirements of the importing country, as applicable. If the importing CAA provided this acceptance electronically, a copy of that acceptance is acceptable for the certification file.

(3) A copy of the completed FAA Form 8130-4

(4) For a replacement export C of A for correction of administrative errors, the original FAA Form 8130-4, the signed statement requesting the replacement certificate, and a copy of the replacement FAA Form 8130-4.

e. Denials. The application and a copy of any denial letters.

f. FAA Form 8100-1.

B-3. Review Certification Files.

a. If an ASI issued the certificate, another ASI conducts the certification file review. If a designee or ODA unit member issues the certificate, the managing specialist or another ASI from the managing office conducts the certification file review. For special flight permits, the file review required by this paragraph may be performed by the inspector who issued the certificate.

b. Review each certification file to ensure:

(1) Accuracy, completeness, legibility, and compliance with applicable requirements, including all necessary attachments.

(2) Only documents required under this appendix are included. For example, do not include weight and balance documents in the certification file for the FAA Aircraft Registry.

(3) No prohibitive information from the applicant should be included. If the three-view drawings or photographs are marked as proprietary in the program letter, the ASI will review the drawings. Do not send any documents to the Aircraft Registration Branch that are labeled as proprietary (protected under the Uniform Trade Secrets Act), confidential, company-sensitive, subject to International Traffic in Arms Regulation (ITAR) or Export Administration Regulations (EAR) or contain any other label indicating the document is restricted from public or foreign disclosure.

c. The reviewing ASI completes the certification file review block of the application per paragraph A-2a (3) of appendix A to this order.

B-4. Forward Certification Files to FAA Aircraft Registry. The reviewing ASI should complete the review and forward the required documents for applications that have been issued a denial letter no later than 7 calendar days after receipt of the “Certification File Review” AWC email notification. For all other airworthiness certification files required under paragraph B-2 of this appendix, the reviewing ASI should complete the review and forward the required documents no later than 14 business days after receipt of the AWC email notification. The reviewing ASI will use AWC to forward the required documents to the registry.

Note: When AWC cannot be used, the issuer of the airworthiness certificate or denial letter should scan and email the required airworthiness certification files to the reviewing office. The reviewing office should forward files to the FAA Aircraft Registry by email no later than 14 business days after receipt of the files. This alternate method may only be used when AWC does not have an incorporated process or other unforeseen AWC outages. For all other circumstances, use AWC-generated forms or upload forms into AWC and forward documents to the registry using AWC. When emailing documents to the FAA Aircraft Registry, use the following email: Submit.Aircraft.Registry@faa.gov.

**Table B-1. Standard Airworthiness Certificates:
Documents for Review and Forwarding to the Registry**

Amend Certificate	R	R	*	*	A	R	A	*	A	A	A	A	A	*
Exchange Certificate	*	R	*	*	A	*	*	*	*	*	*	*	*	*
Replace Certificate	R	R	*	*	A	R	*	*	*	*	*	*	*	*
Surplus Aircraft of the U.S. Armed Forces	R	R	*	*	A	*	A	*	A	A	A	A	A	R
Used	R	R	*	R	A	*	A	*	A	A	A	*	A	*
Manufactured in a Bilateral Country	R	R	*	R	A	*	A	*	A	A	A	A	*	*
Manufactured Under a TC	R	R	R	*	A	*	A	A	A	A	A	A	*	*
Manufactured Under a PC	R	R	*	*	A	*	A	A	A	A	A	A	*	*
Document/Record Notes: “ R ” means required for forwarding to the Registry, AFB-710 “ A ” means should be available during the process for issuing the certificate “ * ” means not applicable Original FAA Form 8130-6, <i>Application for U.S. Airworthiness Certification</i>. For a replacement, a signed statement is acceptable in lieu of FAA Form 8130-6. A copy of the notarized agent letter, if applicable. Original FAA Form 8130-9, <i>Statement of Conformity</i> For imports under a bilateral agreement, export C of A or other certifying statement of conformity to the U.S. TC and the statement of acceptance of exceptions from the FAA, if applicable. AC Form 8050-3, <i>Certificate of Aircraft Registration</i>, or other acceptable evidence of registration AC Form 8050-64, <i>Assignment of Special Registration Numbers</i> (only required for change to N Number) TCDS, maintenance manuals, technical data, ADs, flight manuals, equipment lists, and other pertinent information necessary to support the certification process TC data (for example, drawings and specifications) and manufacturing records Current weight and balance information Maintenance records showing all maintenance is current FAA-approved flight checkoff form to verify flight test completion, when applicable Evidence that the aircraft conforms to its FAA-approved TC, including any repairs and alterations Evidence that the aircraft has been inspected per the performance rules for 100-hour inspections as set forth in § 43.15 and found to be airworthy FAA Form 8130-10, <i>Surplus Military Aircraft Inspection Record</i>														

**Table B-1. Standard Airworthiness Certificates:
Documents for Review and Forwarding to the Registry (Continued)**

Amend Certificate	*	R	R	R	R
Exchange Certificate	*	*	R	R	R
Replace Certificate	*	*	R	R	R
Surplus Aircraft of the U.S. Armed Forces	R	R	R	R	R
Used	*	R	R	R	R
Manufactured in a Bilateral Country	*	*	R	R	R
Manufactured Under a TC	*	*	R	R	R
Manufactured Under a PC	*	*	R	R	R
Document/Record Notes: “ R ” means required for forwarding to the Registry, AFB-710 “ A ” means should be available during the process for issuing the certificate “ * ” means not applicable	FAA Form 8130-31, <i>Statement of Conformity—Military</i> FAA Form 337 if the approved alteration changes the aircraft category, model, or compliance with Annex 8 of the ICAO Convention. Do not include referenced data forming the basis for approval of the repair or alteration. Copy of FAA Form 8100-2, <i>Standard Airworthiness Certificate</i> Superseded, terminated, or canceled airworthiness certificates must be included and marked accordingly, if a recurrent certificate is issued. Does not apply when replacing a lost certificate FAA Form 8100-1, <i>Conformity Inspection Record</i>				

**Table B-2. Special Airworthiness Certificates:
Documents for Review and Forwarding to the Registry**

Amended Certificate		R	R	*	*	R	R	A	A	*	A	A	R	R	
Exchange Certificate		*	R	*	*	*	*	A	*	*	*	*	*	R	
Replacement Certificate		R	R	*	*	*	*	A	R	*	*	*	*	R	
Provisional		R	R	R	*	*	*	A	*	*	A	A	*	R	
SFP for overweight operations		R	R	*	*	*	*	A	*	*	A	A	*	*	
Experimental	Operating Former Light-Sport Category Aircraft	R	R	*	*	R	*	A	*	*	A	A	*	R	
	Operating Light-Sport Category Kit-Built Aircraft	R	R	*	*	R	*	A	*	*	A	A	*	R	
	Primary Kit-Built	R	R	*	*	R	*	A	*	*	A	A	*	R	
	Amateur-Built	R	R	*	*	R	R	A	*	*	A	A	*	R	
	Market Survey	R	R	*	*	R	R	A	*	*	A	A	*	R	
	Air Racing	R	R	*	*	R	R	A	*	*	A	A	*	R	
	Exhibition	R	R	*	*	R	R	A	*	*	A	A	*	R	
	Crew Training	R	R	*	*	R	R	A	*	*	A	A	*	R	
	Show Compliance	R	R	*	*	R	*	A	*	A	A	A	*	R	
	R&D	R	R	*	*	R	R	A	*	A	A	A	*	R	
Light-Sport		R	R	*	*	*	*	A	*	*	A	A	*	R	
Limited		R	R	*	*	*	*	A	*	*	A	A	R	R	
Multiple Certificates		R	R	R	R	*	*	A	*	A	A	A	R	R	
Restricted		R	R	R	R	*	*	A	*	A	A	A	R	R	
Primary		R	R	R	R	*	*	A	*	A	A	A	R	R	
Document/Record Notes: “ R ” means required for forwarding to the Registry, AFB-710 “ A ” means should be available during the process for issuing the certificate “ * ” means not applicable		Original FAA Form 8130-6. For a replacement, a signed statement is acceptable in lieu of FAA Form 8130-6	A copy of the notarized agent letter, if applicable.	Original FAA Form 8130-9, <i>Statement of Conformity</i> , for new aircraft manufactured under a TC only	For aircraft imported under a bilateral agreement, export C of A and the statement of acceptance of exceptions from the FAA, if applicable	Applicant's program letter	Except for aircraft converted from a previously certificated type without appreciable change in the external configuration, three-view drawings, or three-view dimensioned photographs of the aircraft. See paragraph B-3(b)(3) if proprietary.	AC Form 8050-3, <i>Certificate of Aircraft Registration</i> , or other acceptable evidence of registration	AC Form 8050-64, <i>Assignment of Special Registration Numbers</i> (only for changing N-number)	TC data (for example, TCDS, STCs, drawings and specifications) and manufacturing records for new aircraft or prototype alterations	Maintenance records, historical records, repairs, and alterations	Current weight and balance records	Parts catalogs, maintenance manuals, technical data, CAR/CAM, ADs, flight manuals, equipment lists, and other pertinent information necessary to support the certification process	FAA Form 337 if the approved alteration changes the aircraft category or model. Do not include referenced data forming the basis for approval of the repair or alteration	Copy of FAA Form 8130-7, <i>Special Airworthiness Certificate</i>

**Table B-2. Special Airworthiness Certificates: Documents for Review and Forwarding to the Registry
(Continued)**

Amended Certificate	R	R	R	*	*	*	A	A	*	*
Exchange Certificate	*	R	R	*	*	*	*	*	*	*
Replacement Certificate	*	R	R	*	*	*	*	*	*	*
Provisional	R	R	R	*	*	*	*	*	*	*
SFP for overweight ops	A	*	R	*	*	*	*	*	*	*
Experimental	Operating Former Light-Sport Category Aircraft	R	R	R	*	*	*	*	*	*
	Operating Light-Sport Category Kit-Built Aircraft	R	R	R	*	*	*	*	*	*
	Primary Kit-Built	R	R	R	*	*	*	*	*	*
	Amateur-Built	R	R	R	R	A	A	*	*	*
	Market Survey	R	R	R	*	*	A	A	A	*
	Air Racing	R	R	R	*	*	A	A	A	*
	Exhibition	R	R	R	*	*	A	A	A	*
	Crew Training	R	R	R	*	*	A	A	A	*
	Show Compliance	R	R	R	*	*	A	*	*	*
	R&D	R	R	R	*	*	A	A	A	*
Light-Sport	R	R	R	*	*	*	*	*	*	*
Limited	R	R	R	*	*	*	*	*	*	*
Multiple Certificates	R	R	R	*	*	*	*	*	*	*
Restricted	R	R	R	*	*	*	*	*	*	*
Primary	R	R	R	*	*	*	*	*	*	A
Document/Record										
Notes: “ R ” means required for forwarding to the Registry, AFB-710 “ A ” means should be available during the process for issuing the certificate “ * ” means not applicable										
Copy of operating limitations, including applicable attachments										
Superseded, terminated, or canceled airworthiness certificates must be included and marked accordingly, if a recurrent certificate is issued. N/A to replacing a lost certificate										
FAA Form 8100-1, <i>Conformity Inspection Record</i>										
FAA Form 8130-12, <i>Eligibility Statement, Amateur Built Aircraft</i>										
FAA Amateur-Built Aircraft Fabrication/Assembly Checklist, as appropriate										
Documentation of completed in-process and pre cover inspections										
For turbine powered aircraft, an inspection program selected, established, identified, and used per § 91.409(e) through (h)										
Responsible Flight Standards Office approved inspection program, to include ejection seat maintenance and jettisonable stores (if fitted or required) or required										
For former-military aircraft imported with weapons, ATF Forms 6 and 6A										
Submission of a special inspection and preventive maintenance program										

**Table B-2. Special Airworthiness Certificates:
Documents for Review and Forwarding to the Registry (Continued)**

Amended Certificate	*	*	*	*	*	*
Exchange Certificate	*	*	*	*	*	*
Replacement Certificate	*	*	*	*	*	*
Provisional	*	*	*	*	*	*
SFP for overweight operations	*	*	*	*	*	*
Experimental	Operating Former Light-Sport Category Aircraft	R	A	A	*	*
	Operating Light-Sport Category Kit-Built Aircraft	R	*	A	*	A
	Primary Kit-Built	*	*	*	*	*
	Amateur-Built	*	*	*	*	*
	Market Survey	*	*	*	*	*
	Air Racing	*	*	*	*	*
	Exhibition	*	*	*	*	*
	Crew Training	*	*	*	*	*
	Show Compliance	*	*	*	*	*
	R&D	*	*	*	*	*
	Light-Sport	R	R	A	A	*
	Limited	*	*	*	*	*
	Multiple Certificates	*	*	*	*	*
	Restricted	*	*	*	*	*
	Primary	*	*	*	*	*
Document/Record Notes: “ R ” means required for forwarding to the Registry, AFB-710 “ A ” means should be available during the process for issuing the certificate “ * ” means not applicable		A copy of FAA Form 8130-15, <i>Light-Sport Category Aircraft/Kit SOC</i>				
		A copy of the manufacturer’ s production flight test record(s)				
		AOI, maintenance and inspection procedures, and a flight training supplement written in the English language				
		Statement that the light-sport aircraft was not altered and/or modified without manufacturer approval. When the manufacturer’ s approval is given, it will be in written form and be S/N-specific				
		Evidence that required maintenance and inspections were accomplished and documented in the light-sport aircraft records per part 43 for the configuration. Proof the aircraft was inspected and is in a condition for safe operation				
		Manufacturer’ s assembly instructions, approved flight test procedures, and final inspection acceptance record(s)				

Appendix C. Program Letters

C-1. General. Section 21.193 prescribes information that must be submitted by an applicant for the issuance of an experimental certificate. If the person, as defined in § 1.1, named in the program letter is not the registered owner, refer to chapter 2, paragraph 2-3(b)(2). The letter should be dated to show that the information is current.

C-2. Program Letter Content.

a. Applicants for an experimental airworthiness certificate are required to provide information per § 21.193. The FAA uses this information to understand the purpose and scope of an applicant's experiment, equipment, and operations to prescribe the necessary operating limitations.

Section 21.193(a) through (g) specifies minimum requirements for a program letter for any experimental purpose:

- The experimental purpose for which the aircraft is to be used;
- Enough information to describe the operation, equipment, and test as applicable. Describing this information in sufficient detail enables the FAA to assess safety risks and implement appropriate measures to mitigate them via operating limitations, protecting both the test crew and the general public;
- The estimated time or number of flights required for the operation for experimental airworthiness certificates specified in § 21.191(a) through (f);
- The areas over which flights will be conducted;
- Enough data to identify the aircraft;
- Three-view drawings or three-view dimensional photographs of the aircraft unless the aircraft was converted from a previously type-certificated model without appreciable change in its external configuration, and;

Pertinent information to safeguard the public. Certificate issuers must ensure there is enough information provided by the applicant to correctly apply the certification basis and conditions in the Table D-1 columns titled "For these aircraft..." and "...and these conditions." If the information is insufficient, request the applicant provide greater detail. Advise the applicant that insufficient information, regulatory violations, unsafe and high-risk activities described in the program letter, may result in delay or denial of the issuance of the certificate. The FAA neither approves nor disapproves program letters. If identified risks are not mitigated by the operating limitations in Table D-1, issue additional operating limitations to mitigate those risks. Additional operating limitations require coordination in accordance with appendix D, paragraph D-5.

C-3. Considerations Specific to Each Experimental Purpose.

a. **R&D.** For each project, the program letter—

- (1) Describes the project in sufficient detail to demonstrate it meets the regulatory requirements of § 21.191(a).

- (2) Includes the duration.
- (3) Includes the number of flights and/or flight hours.
- (4) Describes the area and airports in which the aircraft will be operated.
- (5) Provides contact information of the customer if the project will be performed under contract.
- (6) States whether weapons or armaments will be tested or whether stores or equipment will be dropped as part of the testing.

Note: An applicant may be seeking a project with another company, but may not yet have been awarded a specific contract to conduct the R&D. The applicant may still submit an application that includes information about the project they intend to conduct. The operating limitations issued should be specific and only valid for that project.

- (7) Describes any high-risk factors and corresponding mitigations.

b. Showing Compliance with Regulations. The program letter—

- (1) References the FAA project number of the TC/STC project, if applicable.
- (2) Identifies applicable regulations for the issuance of the following: Type Certificates, Supplemental Type Certificates, and Major design changes.
- (3) Identifies applicable functional and reliability requirements.
- (4) Identifies any supporting aircraft needed to support the project.

c. Crew Training. The program letter contains the following:

- (1) The name of the person within the company who will provide the training and that person's qualifications,
- (2) A training syllabus,

Note: Coordinate review of the training syllabus with an operations inspector. Although there is no FAA approval or acceptance of the syllabus, an operations inspector has the expertise to review and validate pilot training procedures.

- (3) The time needed to complete the training (that is, approximate number of hours over a defined period of time),
- (4) When a pilot would need this training (for example, every six months, annually, or after a specified period of inactivity),
- (5) The performance standards to complete the training, and
- (6) The airport(s) and area(s) of operation where the training will be conducted.

Note 1: The Airline Transport Pilot and Aircraft Type Rating Practical Test Standards are used as guides for the tasks taught and flight crew performance standards.

Note 2: For training in a former-military aircraft, the training syllabus for instructor pilots and line pilots follows an appropriate military training standard (for example, Naval Air Training and Operating Procedures Standardization (NATOPS)) or other appropriate training standard deemed acceptable.

Note 3: The occupants of the aircraft must be flight crew employees of the company/applicant.

Note 4: A training syllabus should be provided to the responsible FS office operations inspector for validation.

d. Exhibition. The program letter—

- (1) Provides event names, locations, and estimated dates of exhibitions.
- (2) Describes any major alterations that have been made to the aircraft.

e. Market Surveys. The program letter—

- (1) Describes the market survey in detail.
- (2) Describes the area and airports in which the aircraft will be operated.
- (3) Identifies intended customers.
- (4) Specifies dates for the market survey activity.
- (5) Provides justification for reducing the flight hour requirement in § 21.195(d)(2), if applicable.
- (6) For customer crew training, provides:
 - (a) The name of the person within the company who will provide the training and that person's qualifications,
 - (b) A training syllabus, **Note:** The training syllabus should be coordinated with the responsible FS office operations inspector to review and validate pilot training procedures. Although there is no FAA approval or acceptance of the syllabus, a validation is required when review is completed. See note 1 and 2 for this section.
 - (c) The time needed to complete the training (that is, approximate number of hours over a defined period of time), and

(d) The airport(s) and area(s) of operation where the training will be conducted.

Note 1: Applicable Airman Certification Standards are used as guides for the tasks taught and flight crew performance standards.

Note 2: For training in a former-military aircraft, the training syllabus for instructor pilots and line pilots follows an appropriate military training standard (for example, Naval Air Training and Operating Procedures Standardization (NATOPS)) or other appropriate military training standard deemed acceptable.

f. Air Racing. The program letter —

(1) Provides event names and dates for air races the aircraft will participate in.

(2) Describes any major alterations that have been made to the aircraft.

C-4. Proprietary Information. Proprietary information in program letters containing labels such as classified, company-sensitive, proprietary, or similar, or containing information restricted by International Traffic in Arms Regulations or Export Administration Regulations, must be returned to the applicant. The ASI should advise the applicant that such proprietary and restricted information would be improperly disclosed to the public through the aircraft registry's database. Ask the applicant to provide a program letter with higher level information that excludes proprietary and restricted information. If additional proprietary or restricted information is necessary to fully understand the project or test(s), ask the applicant to submit the additional details in a form and manner that will not be submitted to the registry.

C-5. Additional Considerations for Research & Development (§ 21.191(a)). Because R&D tests can vary considerably, it is especially important that the program letter provide detailed information about the tests and operations to enable the Administrator to safeguard the public. Some considerations include:

a. Safety: Understanding the specifics of the R&D project allows the certificate issuer to assess safety risks and implement appropriate measures to mitigate them via operating limitations, protecting both the test crew and the general public.

b. Regulatory Compliance: Advise the applicant that compliance with applicable regulations is the applicant's responsibility, regardless of the information provided in the program letter. Certificate issuance may be withheld if the applicant's program letter contains unsafe activities or regulatory non-compliances.

c. Identification of Unique Risks: R&D projects may involve new technologies, unconventional designs, or operation of equipment that could serve a military function that may present unique risks. A detailed understanding of the equipment and tests may be needed to assist the certificate issuer with identifying, understanding, and evaluating unique risks.

d. Verification of Operational Limits: Understanding the project helps the certificate issuer apply appropriate operating limitations, such as restricting airspace or specific weather conditions, to safeguard the public.

e. Effective Communication: Contact the applicant for any information needing further clarification or details.

C-6 Additional Considerations for Operating Former Military Aircraft (§ 21.191(j)). Verify the program letter includes sufficient detail to show support of the following operations:

- a.** Identifies the government/military customer.
 - (1) Provide the name of the government/military entity and a point of contact.
 - (2) Include the length of the contract.
- b.** Describes the types of operations/intended use of this aircraft and area of operation.
- c.** Includes aircraft configuration and any other information relevant to how the aircraft will be operated, to include a description of any weapons, special military mission systems, or hydrazine fuel requirements.
- d.** For operations in the United States, determine whether the applicant has obtained or intends to obtain a declaration of PAO from the applicable government contracting entity.

C-7 Space Support Vehicles. A space support vehicle operating in accordance with § 91.331 may hold any experimental airworthiness certificate. If the aircraft will be operated as a space support vehicle, the program letter must describe how the operation will comply with § 91.331. Space support vehicle operations must simulate space flight in support of one of the following requirements:

- a.** Training of potential space flight participants, government astronauts, or crew;
- b.** Testing of hardware to be used in space flight; or
- c.** Research and development tasks, which require the unique capabilities of the aircraft conducting the flight.

Appendix D. Issuing Operating Limitations with Special Airworthiness Certificates

D-1. Introduction. This appendix provides procedures for issuing operating limitations for aircraft issued special airworthiness certificates.

Note: See Chapter 19 for considerations when issuing additional operating limitations for an SFP.

D-2. General.

a. Refer to paragraph A-4 of this order for guidance on completing FAA Form 8130-7. The airworthiness certificate and operating limitations must be generated using the AWC online application tool. If there is a conflict between the AWC-generated operating limitations and Table D-1, the procedures and operating limitations in this appendix (including active memos) take precedence.

D-3. Phase I Operating Limitations.

a. Phase I operating limitations specify various operational requirements during flight testing. Phase I operating limitations allow the operator to flight test the aircraft without applying for a research and development experimental airworthiness certificate. Per Table D-1, Phase I operating limitations are always issued to all experimental purposes *except* for § 21.191(a), (b), (c), and (f), regardless of whether the applicant intends to operate in Phase I; operators of aircraft certificated for any of these four purposes must conduct flight testing under an experimental airworthiness certificate for the purpose of R&D.

D-4. Procedures for Creating Operating Limitations. Title 49 United States Code § 44704(d) provides the legal authority for the issuance of operating limitations. It states, “The Administrator may include in an airworthiness certificate terms required in the interest of safety.”

a. Do not issue an operating limitation that specifies an area of operation outside of the United States, as defined in 49 USC § 40102(a)(46). Advise applicants that no person may operate an aircraft with a special airworthiness certificate over any foreign country without permission from that country’s applicable authority. The applicable authority of that country may impose any operating limitation it deems appropriate, including any limitation concerning geographic areas for flight operations.

b. Review AIR policy statements, memos, and deviations related to this order in DRS for any policy clarifications or corrections.

c. Do not change “aircraft” references in Table D-1 to “vehicle” or “space support vehicle” when issuing operating limitations to space support vehicles.

d. When issuing a recurrent Special Airworthiness Certificate, the operating limitations must be reissued using the current version of this order—specifically Appendix D and Table D-1. Inspectors and designees should not reuse previously issued operating limitations, even if they were attached to an earlier certificate..

e. Begin at the top of Table D-1 and work downward. Issue operating limitations from the table based on the aircraft category or experimental purpose and applicable conditions. You must issue the limitations as worded, except for italicized or bracketed instructions. Where italicized or bracketed instruction exist within an operating limitation, insert the appropriate option or information. Number the limitations sequentially, starting with “1,” and include the Table D-1 limitation number in parentheses at the end of the limitation. For a multipurpose experimental airworthiness certificate, include the applicable purpose(s) in parentheses at the end of each operating limitation, e.g., (R&D), (Showing Compliance), (Crew Training).

f. Aircraft with the following very high-risk characteristics or safety of flight issues must have those factors properly mitigated by issuing the D-4(f) limitations in Table D-1:

- (1) Former military or large aircraft in which a single system failure could render the aircraft uncontrollable;
- (2) Aircraft whose required or recommended airspeed in clean configuration during cruise flight below 10,000’ MSL is greater than 250 kts;
- (3) Rocket-powered aircraft;
- (4) Any other unmitigated very high-risk or safety of flight issue identified by the inspector. In this situation, additional operating limitations may be issued in accordance with paragraph D-5;
- (5) Aircraft that utilize hydrazine; and
- (6) Aircraft with variable sweep wing design.

g. Restrict operations of aircraft having any of the very high-risk characteristics or safety of flight issues described in paragraph D-4.f as follows:

- (1) Prohibit the carriage of passengers.
- (2) Prohibit flight over densely populated areas for all phases of flight.
- (3) Prohibit flight at night.
- (4) Restrict operations under instrument flight rules (IFR) to visual meteorological conditions (VMC) only.

h. After the last limitation, place “-END-” approximately centered on the page.

i. Compare AWC-generated operating limitations with Table D-1 to ensure the operating limitations are complete and correct.

j. Before a designee issues a special airworthiness certificate, it is recommended that the designee forward the operating limitations to their managing specialist for review to ensure that the operating limitations accurately reflect Table D-1.

D-5. Requests to Add, Modify, or Omit an Operating Limitation.

a. The operating limitations in Table D-1 of this appendix may not be sufficient to mitigate every safety risk for every aircraft or operation. Additional limitations may be

necessary to address unmitigated risk. Do not issue an operating limitation that is more relieving than a regulation. An applicant may request to modify or omit an operating limitation by submitting a justification and an equivalent level of safety proposal. If the ASI determines the modification or omission is appropriate, or an additional limitation is necessary, prior to adding, modifying, or omitting an operating limitation, or adding a new limitation you must coordinate with the responsible FAA office by emailing: 9-AVS-AIR-81302OPSLIMITS@faa.gov. The coordination mailbox provides direct access to policy offices, including AIR-632, AFS-300, AFS-800, and AFS-700 (for UAS). No certificate may be issued with additions, modifications, or omissions without this coordination. This includes, additions, modifications, or omissions that were issued on a previous certificate.

b. Certain Aircraft. Operating limitations of the following experimental aircraft must be coordinated with the responsible office by emailing the coordination mailbox:

- (1) Aircraft for which operating limitations were previously issued with modifications or omissions from those required by Table D-1;
- (2) Aircraft that meet any characteristics of paragraph D-4.f;
- (3) All turbine-powered former-military aircraft (e.g., turbojet, turboprop, turbine rotorcraft);
- (4) All aircraft with weapons or special military mission systems; or
- (5) Aircraft of military-like design built independently or by manufacturers and not having military identification (refer to paragraph 23-3).

NOTE: Operating limitations for special flight permits do not require the coordination described in paragraph D-5.a.

c. Before sending your request for coordination, compare your AWC-generated operating limitations to Table D-1 and the AIR-issued policy memos in DRS to verify they are accurate. Include with your email the AWC application number and the following attachments:

- (1) Form 8130-6 application;
- (2) Program letter;
- (3) AWC-generated operating limitations with proposed modifications, additions, and omissions; and

Note: The operating limitations must be generated from AWC on the FAA Form 8130-7 to avoid cut-and-paste errors. Word documents will not be accepted.

- (4) The applicant's justification and proposed equivalent level of safety for any modified, added, or omitted operating limitation.

d. Designees must coordinate through their managing specialist. The managing specialist sends the request to the coordination mailbox. Before issuing the airworthiness

certificate, the designee must receive concurrence from the office of primary responsibility (OPR) listed in appendix D. The concurrence letter or email is uploaded to AWC to document the additions, omissions, or modifications to the operating limitations.

D-6. Review with Applicant. When issuing an airworthiness certificate:

a. Review the operating limitations with the applicant to ensure they have a clear understanding of the operating limitations.

b. Remind the applicant of their obligation to comply with regulations and operating limitations, with particular emphasis on §§ 91.9 and 91.203. Remind the applicant of the following regulatory operating limitations, as applicable:

(1) Restricted category operating limitations in § 91.313.

(2) Light-sport category operating limitations in § 91.327.

(3) Experimental aircraft operating limitations in §§ 91.319 and 91.305.

(4) Limited category operating limitations in § 91.315.

(5) Primary category operating limitations in § 91.325.

(6) Operating limitations for space support vehicle flights in § 91.331.

(7) Provisional TC operating limitations:

(a) Class I, refer to §§ 21.81(e) and 91.317.

(b) Class II, refer to §§ 21.83(h), 91.317, and 121.207.

(8) Operating limitations for provisional amendments to a TC, as in §§ 21.85(g), 91.317, and 121.207.

c. Inform the applicant they may not operate the aircraft over any foreign country without the permission of that country. The CAA of that country may impose any operating limitation it considers necessary.

d. Gliders. Refer to FAA AC 21.17-2 for any applicable guidance.

Table D-1 Operating Limitations

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
1	Primary § 21.184, Restricted § 21.185, Limited § 21.189, Light-sport category § 21.190, All experimental purposes § 21.191 OPR: AIR-600	All conditions.	This aircraft does not meet the airworthiness standards of Annex 8 to the Convention on International Civil Aviation. This aircraft may not be operated in foreign airspace without the permission of the applicable foreign authorities. (1)
2	Primary § 21.184, Restricted § 21.185, Limited § 21.189, Light-sport category § 21.190, All experimental purposes § 21.191 OPR: AFS-300 & AFS-800	All conditions.	This special airworthiness certificate is not in effect during public aircraft operations (PAO). The determination of public aircraft status is made on a flight-by-flight basis. The aircraft cannot be operated as a civil aircraft and as a public aircraft at the same time. PAO declarations are discussed in AC 00-1.1, as amended. Before operating this aircraft under this special airworthiness certificate following a PAO, the aircraft must be returned to the condition and configuration at the time of inspection for the issuance of this airworthiness certificate. The operator must have written procedures for returning the aircraft to the civil configuration. This action must be documented in the maintenance records. (2)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
3	Primary § 21.184, Restricted § 21.185, Limited § 21.189, Light-sport category § 21.190, All experimental purposes § 21.191 OPR: AFS-300 & AFS-800	All conditions.	The addition of weapons or special military mission systems is prohibited. Flights related to providing military services or flights for which no civil regulatory framework exists (for example, air combat maneuvering, air-to-air gunnery, target towing, electronic countermeasures simulation, cruise missile simulation, and aerial refueling) are prohibited under this airworthiness certificate. These activities are inherently military, not civil activities. (3)
4	Experimental amateur-built § 21.191(g) OPR: AFS-800	All conditions.	No person may operate this aircraft for other than recreation and education, except a person may perform demonstrations in accordance with an aviation event waiver. No persons may be carried in this aircraft during such demonstrations. (4)
5	Experimental research and development § 21.191(a) OPR: AIR-600 & AFS-800	All conditions.	Tests and configurations not described in the applicant's program letter that was submitted as the basis for issuance of this airworthiness certificate are prohibited. (5)
6	All experimental purposes § 21.191 OPR: AFS-800	Hot-air airships (e.g., Thunder & Colt AS-56).	The pilot in command must hold a pilot certificate with a lighter-than-air category rating and an airborne heater privilege. The PIC must hold all required ratings or authorization, and endorsements required by 14 CFR Part 61. (6)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
7	Experimental purposes: Exhibition § 21.191(d), Amateur-built § 21.191(g), Primary kit-built § 21.191(h), Light-sport category kit-built § 21.191(k), Former light-sport category § 21.191(l) OPR: AFS-800	All except hot-air airships and powered-lift manufacturers. Note: For an aircraft that is not identified by category or class under § 61.5(b), coordinate with AFS-800 prior to issuance.	The pilot in command must hold a pilot certificate with the category and class rating or privilege appropriate to this aircraft and all authorizations and endorsements identified in Part 61 applicable to this aircraft. Student pilot operation: After completion of Phase I flight testing, holders of a student pilot certificate may operate this aircraft in solo flight without holding the aforementioned category or class rating only for the purpose of meeting aeronautical experience requirements of Part 61 pursuant to a sport, recreational, or private pilot certificate. (7)
8	Experimental purposes: Research and development § 21.191(a), Showing compliance § 21.191(b), Crew training § 21.191(c), Air racing § 21.191(e), Market Survey § 21.191(f), Operating Former Military § 21.191(j) OPR: AFS-800	All except hot-air airships and powered-lift manufacturers. Note: For an aircraft that is not identified by category or class under § 61.5(b), coordinate with AFS-800 prior to issuance	The pilot in command must hold a pilot certificate with the category and class rating or privilege appropriate to this aircraft and all authorizations and endorsements identified in Part 61 applicable to this aircraft. (8)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
9	Experimental purposes: Research and development § 21.191(a), Showing compliance § 21.191(b), Crew training § 21.191(c), Exhibition § 21.191(d), Market surveys § 21.191(f) OPR: AFS-800	Issue only to manufacturers of powered-lift Note: Powered-lift must also be issued operating limitation 10.	The pilot in command (PIC) may hold a pilot certificate with airplane multiengine land, rotorcraft helicopter, or powered-lift, provided no passengers, as defined in § 61.1, are on board the aircraft. The PIC must hold a pilot certificate with powered-lift category if passengers are on board. (9)
10	All experimental purposes § 21.191 OPR: AFS-800	Aircraft with any of the following: -Large aircraft (more than 12,500 pounds maximum take-off weight.) -Turbojet or turbofan, except gliders capable of having only a single seat. -Airplanes with conventional gear having 800 or greater total horsepower and VNE or VMO greater than 250 knots indicated airspeed. -Powered-lift.	The pilot in command (PIC) must hold— (a) An appropriate type rating (if one has been established); (b) An experimental aircraft authorization, by make and model, on their pilot certificate; or (c) A temporary letter of authorization (LOA) issued by a Flight Standards operations inspector. (d) For FAA personnel acting in an official capacity, an authorization issued in accordance with the applicable FAA policy Information to obtain an experimental aircraft authorization or temporary LOA can be found in FAA Order 8900.1, volume 5, chapter 9, section 2 . The PIC must meet § 61.57 recent flight experience. (10)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
11	All experimental purposes § 21.191 OPR: AFS-800	Single seat or single control aircraft that are subject to OL 10.	An authorized instructor, after providing ground and flight training, may make an endorsement to allow the airman to be pilot in command for completing a practical test for the issuance of an experimental aircraft authorization. The endorsement may allow solo operation of the aircraft. The endorsement may be valid for a period up to 30 days and must specify the flight conditions authorized (for example, day, night, IMC) and flying area. The flying area may not exceed 3/8 of the fuel range of the aircraft. (a) Information to obtain an experimental aircraft authorization or temporary LOA can be found in FAA Order 8900.1, volume 5, chapter 9, section 2. (11)
12	All experimental purposes § 21.191 OPR: AFS-800	Aircraft previously type certificated for more than one required flight crew member, required by the military or manufacturer to operate with additional flight crew members (pilots or flight engineers). For manufacturers of powered-lift, use OL 13.	This aircraft must be operated with an appropriately rated second in command (SIC) [<i>add "and flight engineer", if applicable</i>]. The SIC must comply with the SIC qualification and familiarization requirements specified in § 61.55. (12)
13	All experimental purposes § 21.191 OPR: AFS-800	Manufacturers of powered-lift that require two pilots.	This aircraft must be operated with a second in command (SIC). The SIC must comply with the SIC qualification and familiarization requirements specified in § 61.55. The SIC may hold a pilot certificate with airplane multiengine land, rotorcraft helicopter, or powered-lift, provided no passengers, as defined in § 61.1, are carried on board. The SIC must hold a pilot certificate with powered-lift category if any other persons are on board. (13)
14	All experimental purposes § 21.191 OPR: AFS-800	All conditions	This aircraft must not be used to tow or for intentional parachute jumping. (14)
15	All experimental purposes § 21.191 OPR: AFS-300	All conditions	If a manufacturer's operating limit for the aircraft, engine, or propeller is exceeded, unless the aircraft is modified in accordance with these operating limitations and applicable regulation, appropriate inspections to ensure the aircraft is in a condition for safe operation must be completed, and an entry must be made in the maintenance records prior to continued operation. (15)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
16	All experimental purposes § 21.191 OPR: AFS-300	Aircraft with any of the following (except former military): -Large aircraft more than 12,500 pounds gross takeoff weight. -Turbine-powered multi-engine airplanes. -Powered-lift per SFAR 194	No person may operate this aircraft unless it is maintained per one of the following inspection programs. The registered owner or operator must select the applicable program below and specify it in the aircraft maintenance records. (a) For previously type-certificated aircraft: 1) A current inspection program recommended by the manufacturer (§ 91.409(f)(3)); or 2) An FAA-approved inspection program under (§ 91.409(f)(4)). (b) Non-type certificated aircraft: An FAA-approved inspection program under (§ 91.409(f)(4)). This program should not be used for an aircraft model under production; instead, use item (a) above. Note: To adjust an inspection interval, the registered owner or operator must submit a request with supporting documentation and data to the responsible Flight Standards Office and obtain concurrence from that office. Inspections must be recorded in the aircraft maintenance records showing the following, or a similarly worded, statement: "I certify that this aircraft has been inspected on <i>[insert date]</i> per <i>[identify applicable inspection program]</i> and found to be in a condition for safe operation." The entry will include the aircraft's total time-in-service (or total cycles, if appropriate), and the name, signature, certificate number, and type of certificate held by the person who performed the inspection. (16)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
17	All experimental purposes § 21.191 OPR: AFS-300	Aircraft with any of the following (except former military): -Turbine-powered rotorcraft. -Turbine-powered single-engine airplanes	No person may operate this aircraft unless within the preceding 12 calendar months it has had a condition inspection including at least the scope and detail of Part 43, Appendix D and was found to be in a condition for safe operation. The inspections must be recorded in the aircraft maintenance records showing the following, or a similarly worded, statement: "I certify that this aircraft has been inspected on <i>[insert date]</i> per the <i>[insert either: scope and detail of Part 43, Appendix D; or aircraft manufacturer's inspection procedures]</i> and was found to be in a condition for safe operation." The entry will include the aircraft's total time-in-service (or total cycles, if appropriate), and the name, signature, certificate number, and type of certificate held by the person performing the inspection. Alternatively, turbine powered rotorcraft and single-engine turbine-powered airplanes may elect to use: (a) For previously type-certificated aircraft: 1) A current inspection program recommended by the manufacturer (§ 91.409(f)(3)); or 2) An FAA-approved inspection program under (§ 91.409(f)(4)). (b) Non-type certificated aircraft: An FAA-approved inspection program under (§ 91.409(f)(4)). This program should not be used for an aircraft model under production; instead, use item (a) above. These inspections must be recorded in the aircraft maintenance records showing the following, or a similarly worded, statement: "I certify that this aircraft has been inspected on <i>[insert date]</i> per <i>[identify applicable inspection program]</i> and found to be in a condition for safe operation." The entry will include the aircraft's total time-in-service (or total cycles, if appropriate), and the name, signature, certificate number, and type of certificate held by the person who performed the inspection. (17)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
18	All experimental purposes § 21.191 OPR: AFS-300	All other aircraft that are not described in OL 16 or OL 17, (except former military aircraft.)	No person may operate this aircraft unless within the preceding 12 calendar months it has had a condition inspection including at least the scope and detail of Part 43, Appendix D and was found to be in a condition for safe operation. The inspections must be recorded in the aircraft maintenance records showing the following, or a similarly worded, statement: "I certify that this aircraft has been inspected on <i>[insert date]</i> per the <i>[insert either: scope and detail of Part 43, Appendix D; or aircraft manufacturer's inspection procedures]</i> and was found to be in a condition for safe operation." The entry will include the aircraft's total time-in-service (or total cycles, if appropriate), and the name, signature, certificate number, and type of certificate held by the person performing the inspection. (18)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
19	All experimental purposes § 21.191 OPR: AFS-300	All former military aircraft with any of the following: -Large aircraft more than 12,500 pounds gross takeoff weight. -Turbine powered airplanes. -Turbine-powered rotorcraft. -Powered-lift per SFAR 194.	No person may operate this aircraft unless it is maintained per one of the following inspection programs The operator must select and identify in the aircraft maintenance records the applicable inspection programs for the aircraft: (a) For type certificated aircraft (Military Commercial Derivative Aircraft (MCDA)): 1)A current inspection program recommended by the manufacturer § 91.409(f)(3); or 2)An FAA-approved inspection program under § 91.409(f)(4). (b) For non-type certificated aircraft (aircraft has only a military certification basis): 1) An FAA-approved inspection program. § 91.409(f)(4) will be based on the military inspection program the military aircraft was maintained under at the time of surplus. This aircraft must not be operated unless it is operated, inspected, and maintained per appropriate military technical publications and/or manufacturer's recommendations. Inspections must be recorded in the aircraft maintenance records showing the following or a similarly worded statement: "I certify that this aircraft has been inspected on <i>[insert date]</i> per <i>[identify applicable inspection program]</i> and found to be in a condition for safe operation." The entry will include the aircraft's total time-in-service (or total cycles, if appropriate), and the name, signature, certificate number, and type of certificate held by the person who performed the inspection. Note: To extend an inspection interval, the owner/operator must submit a request with supporting documentation and data to the responsible Flight Standards Office and obtain concurrence from that responsible Flight Standards Office. (19)

20	All experimental purposes § 21.191 OPR: AFS-300	All other former military aircraft that are not described in OL 19. Select only the option that applies (i.e., A or B).	No person may operate this aircraft unless it is maintained per one of the following inspection programs. The operator must select and identify in the aircraft maintenance records the applicable inspection programs for the aircraft: (A) For aircraft with 800 or more total horsepower or a V_{NE} greater than 250 knots: (1) An FAA-approved inspection program under § 91.409(f)(4), combined with the military inspection program that was utilized at the time of surplus, shall be used to develop an inspection program for such aircraft. (2) Inspections must be recorded in the aircraft maintenance records showing the following or a similarly worded statement: "I certify that this aircraft has been inspected on <i>[insert date]</i> in accordance with <i>[identify by name and number of the applicable inspection program that was used]</i> and found to be in a condition for safe operation." (3) The entry will include the aircraft's total time-in-service (or cycles, if appropriate), and the name, signature, certificate number, and type of certificate held by the person who performed the inspection. (B) For aircraft with less than 800 total horsepower and a V_{NE} less than 250 knots: (1) No person may operate this aircraft unless, within the preceding 12 calendar months, it has had a condition inspection performed per the scope and detail of Part 43, Appendix D; or (2) A military inspection program that was developed for the specific aircraft and which must at least meet the scope and detail of Part 43, Appendix D, can be used; or (3) An FAA-approved inspection program under § 91.409(f)(4) with the military inspection program that was utilized at the time of surplus shall be used to develop an inspection program for such aircraft. Note: To extend an inspection interval, the owner/operator must submit a request with supporting documentation and data to the responsible Flight Standards Office and obtain concurrence from that responsible Flight Standards Office. The inspection must be recorded in the aircraft maintenance records showing the following or a similarly worded statement: "I certify that this aircraft has been inspected on <i>[insert date]</i> per the <i>[insert either "scope or detail of Part 43" or identify</i>
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No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
			<i>by name and number of the applicable inspection program that was used</i>]. The entry will include the aircraft's total time-in-service (or total cycles, if appropriate), and the name, signature, certificate number, and type of certificate held by the person who performed the inspection. (20)
21	All experimental purposes § 21.191 OPR: AFS-300	Former military turbine powered airplanes with a maximum takeoff weight greater than 100,000lbs.	The owner or operator of a former military airplane must include the military aircraft structural integrity program (ASIP) requirements (or similar military requirements) in the aircraft inspection program (AIP) and comply with those inspection requirements. (21)
22	Experimental purposes: Light-sport kit-built § 21.191(k), Former light-sport category § 21.191(l) OPR: AFS-300	All conditions.	A light-sport repairman certificated under § 65.107 (in accordance with their certificate privileges under § 65.109), an appropriately rated FAA-certificated mechanic, or an appropriately rated FAA-certificated repair station may perform the condition inspection required by these operating limitations. (22)
23	Experimental amateur-built § 21.191(g) OPR: AFS-300	All conditions.	An experimental aircraft builder certificated as a repairman for this aircraft under § 65.104, a light-sport repairman certificated under § 65.107 (in accordance with their certificate privileges under § 65.109), an appropriately rated FAA-certificated mechanic, or an appropriately rated FAA-certificated repair station may perform the condition inspection required by these operating limitations. (23)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
24	Experimental purposes: Research and development § 21.191(a), Showing compliance § 21.191(b), Crew training § 21.191(c), Exhibition § 21.191(d), Air racing § 21.191(e), Market surveys § 21.191(f), Primary kit-built § 21.191(h), Operating former military aircraft § 21.191(j) OPR: AFS-300	All conditions.	Only appropriately rated FAA-certificated repair stations, FAA-certificated mechanics with appropriate ratings, or a manufacturer as authorized by § 43.3 may perform inspections required by these operating limitations. (24)
25	Light-sport category § 21.190, All experimental purposes § 21.191 OPR: AFS-300	All conditions.	The aircraft may not be operated unless the replacement interval for a life-limited article, as defined in § 43.10(a), specified in the applicable technical publications pertaining to the aircraft, and its articles are complied with in one of the following manners: (a) Type-certificated or previously type-certificated products: Replacement of life-limited parts required by § 91.409(e) applies to experimental aircraft when the required replacement times are specified in the aircraft specifications, type certificate data sheets or other documents approved by the Administrator. (b) Non-type-certificated products: Comply with intervals for replacing life-limited parts or articles in accordance with the manufacturer's maintenance procedures or FAA-approved inspection program. (25)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
26	Light-sport category § 21.190, All experimental purposes § 21.191 OPR: AFS-300	All conditions (except amateur-built aircraft).	For aircraft originally incorporating fatigue life recording systems, the owner/operator must maintain and use the system as prescribed by the aircraft manufacturer and comply with the manufacturer's fatigue life limits. (26)
27	Experimental purposes: Research and development § 21.191(a), Showing compliance § 21.191(b) OPR: AFS-300, AFS-800	All conditions.	<u>Major changes or alterations:</u> After making a major change per § 21.93 or major alteration per § 1.1 to the aircraft: (a) Notify the responsible Flight Standards Office or designee prior to operation of the aircraft. This notice enables the FAA to determine if operating limitations require amendment to reduce exposure of the public to risk due to the major change. Note: Multiple notifications are not necessary if changing configurations and the operator has previously notified the FAA of the change and demonstrated and recorded compliance with § 91.319(b)(1) and (2). (b) Demonstrate compliance with § 91.319(b)(1) and (2) prior to conducting any other operations. Compliance with § 91.319(b)(1) and (2) must be completed under day visual meteorological conditions and over an area described by § 91.305. (c) Record compliance with § 91.319(b)(1) and (2) in the maintenance records with the following, or a similarly worded, statement: "I certify that the prescribed flight test has been completed and the aircraft is controllable throughout its normal range of speeds and throughout all maneuvers to be executed, has no hazardous characteristics or design features, and is safe for operation." Notify the FAA certification office responsible or designee to update the FAA Registry of any changes to the engine or propeller model by contacting an FAA certification office or designee. (27)
28	Experimental purposes: Crew training § 21.191(c), Market surveys § 21.191(f) OPR: AFS-300	All conditions.	<u>Major changes or alterations:</u> After a major change per § 21.93 or major alteration per § 1.1, you must obtain a research & development airworthiness certificate and demonstrate compliance with § 91.319(b)(1) and (2). (28)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
29	Experimental purposes: Exhibition § 21.191(d), Air racing § 21.191(e), Amateur-built § 21.191(g), Primary kit-built § 21.191(h), Operating former military aircraft § 21.191(j), Light-sport category kit-built § 21.191(k), Former light-sport category § 21.191(l) OPR: AFS-300	All conditions.	<u>Major changes or alterations:</u> After incorporation of a major change per § 21.93 or major alteration per § 1.1: (a) Notify the responsible Flight Standards office of the change and receive its response in writing. An amended airworthiness certificate and operating limitations may be required (e.g., changing from a piston to a turbine engine). (b) Comply with Phase I operating limitations. Notify the FAA certification office responsible or designee to update the FAA Registry of any changes to the engine or propeller model by contacting an FAA certification office or designee. (29)
30	Restricted category § 21.185, Multiple certifications § 21.187 OPR: AFS-300	All conditions.	Conversion to or from the restricted category must be accomplished per maintenance instructions. Each conversion must be documented via a maintenance record entry. (30)
31	All experimental purposes § 21.191 OPR: AIR-600 & AFS-300	All conditions.	When changing between experimental purposes, the operator must determine that the aircraft is in the appropriate configuration for the purpose intended. A maintenance record entry must be made to document that finding. (31)
32	Light-sport category § 21.190 OPR: AFS-800	All conditions	The pilot may only conduct the flight maneuvers and aerial work operations authorized by the aircraft manufacturer in the AOI or POH. The pilot must comply with any limitations, markings, and placards specified by the aircraft and engine manufacturers. (32)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
33	Experimental purpose: Research & Development § 21.191(a) Showing compliance § 21.191(b) OPR: AFS-800	Issue only when the test requires exceedance of a manufacturer limitation.	Compliance with all the manufacturers operating limitations (e.g., POH, light-sport AOI, aircraft flight manual, military manual), placards, and markings is required unless exceedance of a limitation is required for flight testing that was described in the program letter used as the basis for this airworthiness certificate. Intentional exceedance of a limitation that was not described in the program letter is prohibited. Intentional exceedance of a manufacturer limitation described in the program letter requires compliance with the operating limitation related to major changes appended to this airworthiness certificate, except that notification to the responsible flight standards office is not required. (33)
34	Experimental purpose: Air Racing § 21.191(e) OPR: AFS-800	Issue only when exceedance of a manufacturer limitation is required for air racing.	Compliance with all the manufacturers operating limitations (e.g., POH, light-sport AOI, aircraft flight manual, military manual), placards, and markings is required unless exceedance is required for air racing. Intentional exceedance of a manufacturer limitation requires compliance with the operating limitation related to major changes appended to this airworthiness certificate. (34)
35	All experimental purposes § 21.191 OPR: AFS-800	All others not described in OL 33 or OL 34.	Compliance with all the manufacturers operating limitations (e.g., POH, light-sport AOI, aircraft flight manual, military manual), placards, and markings is required. (35)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
36	All experimental purposes § 21.191 OPR: AFS-800	Former military turbine-powered airplanes.	Preflight planning runway length requirements: Takeoff is prohibited unless takeoff planning determines it is possible to stop the airplane safely on the runway, as shown by the accelerate-stop distance data. For aircraft without accelerate-stop distance data, the airplane must be able to safely stop within the effective length of the runway, from any point during the takeoff, before reaching 105 percent of VMCA or 115 percent of the power-off stalling speed in the takeoff configuration, whichever is greater. In addition, the aircraft must be able to clear all obstacles by at least 50 feet vertically. Landing will not be attempted unless landing planning determines that a full stop landing can be made within 60 percent of the effective length of the runway from a point 50 feet above the runway. When calculating takeoff or landing performance, corrections must be made for any runway gradient. Performance data based on still air may be corrected by taking into account not more than 50 percent of any reported headwind component and not less than 150 percent of any reported tailwind component. Calculations may not include the use of reverse thrust or drag chute. (36)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
37	All experimental purposes § 21.191 OPR: AFS-300 & AFS-800	Ejection seat installed or part of original design.	Ejection seats and explosive devices must be clearly marked on the exterior of the aircraft. Operational ejection seat systems must be maintained and inspected per the manufacturer's procedures or military technical publications. The manufacturer or military service must have approved the ejection seat system, as configured, for installation in the aircraft. Operational ejection seat systems must be secured per the manufacturer's procedures or military technical publication to prevent inadvertent operation of the system when the aircraft is parked or out of service. All crew members aboard an aircraft equipped with operational ejection seat(s), whether armed or not armed, must have successfully completed an ejection seat training program within the previous 24 calendar months prior to operations for this make and model of aircraft. Reference FAA AC 91-87 for further information. Prior to flight, the pilot in command must brief passengers on the passenger safety briefing items listed in FAA AC 91-87. (37)
38	Experimental purposes: Exhibition § 21.191(d), Air racing § 21.191(e) Operating former military aircraft § 21.191(j). OPR: AFS-300	All conditions.	When an aircraft's home base is changed or there is a transfer of ownership, the owner/operator will, within 30 days: (a) Notify the responsible Flight Standards Office. (b) If an accepted or approved inspection program is specified in these operating limitations, submit a copy to the responsible Flight Standards Office. (38)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
39	Experimental purposes: Research and development § 21.191(a), Crew training § 21.191(c), Exhibition § 21.191(d), Air racing § 21.191(e), Operating former military aircraft § 21.191(j) OPR: AFS-300	All Aero Vodochody L-29 & L-39 series airplanes.	If installed, any emergency stores release handle (ESRH) or master armament safety switch (MASS) must be disabled. Both must be disabled and disconnected from all systems. Weapon related buttons and switches must also be disabled and disconnected from all systems. Selection of all simulated failure modes from the rear cockpit (instructor position) must be disabled. (39)
40	Experimental purposes: Research and development § 21.191(a), Crew training § 21.191(c), Exhibition § 21.191(d), Air racing § 21.191(e), Operating former military aircraft § 21.191(j) OPR: AFS-300	All Hawker and BAE Hawk series airplanes.	If installed, all four jettison switches must be disabled and disconnected from all systems. Flight with asymmetric wing mounted equipment is prohibited. (40)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
41	Experimental purposes: Research and development § 21.191(a), Crew training § 21.191(c), Exhibition § 21.191(d), Air racing § 21.191(e), Operating former military aircraft § 21.191(j) OPR: AFS-800	All Douglas A-4 series airplanes.	Any passenger must receive, before flight, adequate training concerning: (a) Any rear-seat responsibilities per the applicable flight manual; (b) "Off-limits" equipment and switches in the rear-seat; and (c) Any other safety-related task not covered under the ejection seat training program. The maximum authorized speed for aircraft operations is 10 percent below the published M _{MO} . (41)
42	All experimental purposes § 21.191 OPR: AFS-300	All aircraft equipped or originally equipped with a drag chute.	The drag chute must be maintained and packed in accordance with applicable maintenance instructions. (42)
43	All experimental purposes § 21.191 OPR: AFS-300 & AFS-800	All aircraft equipped or originally equipped for in-flight refueling.	Operation of aerial refueling equipment is prohibited. Fueling or defueling the aircraft with an engine operating is prohibited. (43)
44	All experimental purposes § 21.191 OPR: AFS-800	Aircraft using hydrazine (e.g., F-16).	Operations are prohibited for aircraft equipped with hydrazine, unless the following are met: (a) For aircraft equipped with a hydrazine-based emergency power unit (EPU), trained ground support personnel must be available to secure the EPU before shutdown. (b) A trained emergency hydrazine response team using U.S. military, Dept. of Defense, or NATO hydrazine response procedures that is capable of responding as specified Coordinate with the airport manager or airport sponsor of the requirements listed in (a) and (b). (44)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
45	Restricted category 21.185 OPR: AFS-800	All conditions	When equipped with operational jettisonable stores, flight operations are restricted to areas described by § 91.305 at all times. (45)
46	Research and development § 21.191(a) OPR: AFS-800	Aircraft capable of carrying jettisonable stores Note: Issue only if jettisoning of stores is part of the R&D test.	When equipped with operational jettisonable stores, explosive bolts, explosive pylon charges (ejectors), military armament installed, or in a configuration not specified by the manufacturer or military, flight operations are limited to: (a) Flight testing described in the program letter that was presented for issuance of this airworthiness certificate; and (b) Areas described in § 91.305 at all times. When configured in accordance with military or manufacturer specifications, compliance with military or manufacturer limitations and performance data is required. Explosive devices must be clearly marked on the exterior of the aircraft and must be maintained in accordance with manufacturer or military instructions. Air Traffic Control must be notified prior to any jettison operations. The airport manager/airport sponsor must be notified of the presence of jettisonable stores, any explosive devices, and planned operations of the aircraft from that airport. When not testing the stores, operation of the aircraft with explosive bolts, explosive pylon charges (ejectors), operational jettisonable stores, military armament installed, or in a configuration not specified by the manufacturer or military is prohibited. (46)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
47	Operating former military aircraft § 21.191(j) OPR: AFS-800	Aircraft capable of carrying jettisonable stores.	Jettisonable stores must be made non-jettisonable (removed or secured in such a manner that in-flight jettison is not possible). When carrying stores, flight operations are limited to: (a) Repositioning the aircraft for operation as a public aircraft; and (b) Areas described in § 91.305 at all times. When configured in accordance with military or manufacturer specifications, compliance with military or manufacturer limitations and performance data is required. Explosive devices must be clearly marked on the exterior of the aircraft and must be maintained in accordance with manufacturer or military instructions. (c) The airport manager/airport sponsor must be notified of the presence of jettisonable stores, any explosive devices, and planned operations of the aircraft from that airport. (47)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
48	Limited category § 21.189, Research and development § 21.191(a), Showing compliance with regulations § 21.191(b), Crew training § 21.191(c), Exhibition § 21.191(d), Air racing § 21.191(e), Market surveys § 21.191(f), Operating amateur-built aircraft § 21.191(g), Operating primary kit-built aircraft § 21.191(h), Operating light-sport category kit-built aircraft § 21.191(k), and Operating former light-sport category aircraft § 21.191(l) OPR: AFS-300, AFS-800, and AJV-P32	Aircraft capable of carrying jettisonable stores that are not described in OL 46.	Jettisonable stores must be made non- operational (removed or secured in such a manner that in-flight jettison is not possible). Installation of explosive bolts, explosive pylon charges (ejectors), operational jettisonable stores, military armament, or operation in a configuration not specified by the manufacturer or military is prohibited. (48)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
49	Experimental purposes: Exhibition § 21.191(d), Air racing § 21.191(e), Amateur-built § 21.191(g), Primary kit-built § 21.191(h), Operating former military aircraft § 21.191(j), Light-sport category kit-built § 21.191(k), Former light-sport category § 21.191(l) OPR: AFS-300 & AFS-800	All conditions.	<u>Phase I:</u> During Phase I flight, all operations must be conducted under day visual meteorological conditions and over an area compliant with § 91.305 until it is shown that the aircraft is controllable throughout its normal range of speeds and throughout all the maneuvers to be executed, and the aircraft has no hazardous operating characteristics or design features. Document compliance with § 91.319(b)(1) and (2) in the maintenance records. (49)
50	Experimental purposes: Exhibition § 21.191(d), Air racing § 21.191(e), Amateur-built § 21.191(g), Primary kit-built § 21.191(h), Light-sport category kit-built § 21.191(k) OPR: AFS-300 & AFS-800	Issue this operating limitation for newly-built or newly-assembled aircraft that have never shown compliance with § 91.319(b)(1) and (2) Note: For amateur-built flight test hours, see paragraph 15-4(f). For kit-built LSA flight test hours, see paragraph 17-3(g).	<u>Phase I:</u> Newly-built aircraft or aircraft that have not demonstrated initial compliance with § 91.319(b)(1) and (2) must complete Phase I flight testing using either of the following: (a) Completion of a task-based flight test program described in FAA Advisory Circular (AC) 90-89 (latest version); or (b) The aircraft is flown for at least _____ hours with at least _____ takeoffs and landings. (50)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
51	Experimental purposes: Exhibition § 21.191(d), Air racing § 21.191(e), Operating former military aircraft § 21.191(j), Former light-sport category § 21.191(l) OPR: AFS-800	All conditions.	<u>Phase I:</u> During Phase I, no person may be carried in this aircraft during flight unless that person is a required flight crew member. (51)
52	Experimental purposes: Amateur-built § 21.191(g), Primary kit-built § 21.191(h), Light-sport category kit-built § 21.191(k) OPR: AFS-800	All conditions.	<u>Phase I:</u> During Phase I flight, no person may be carried in this aircraft during flight unless that person is a required flight crew member or is authorized in accordance with the Additional Pilot Program (see Advisory Circular 90-116.) (52)
53	All experimental purposes § 21.191 OPR: AFS-800	All conditions.	Aerobatic maneuvers are prohibited unless, in the current aircraft configuration: (a) Those maneuvers were tested and recorded in the maintenance record under Phase I; (b) Those maneuvers were tested and recorded under a research & development airworthiness certificate; (c) The aircraft was previously type-certificated for aerobatic flight; or (d) For former military aircraft, the aircraft is operated in accordance with manufacturer or military maneuvering limitations. Testing of aerobatic maneuvers is limited to day operations and must be conducted in areas described by § 91.305. No person may be carried in this aircraft during testing unless that person is a required flight crew member or is authorized in accordance with Phase I limitations. Each tested maneuver must be documented in the maintenance record and include the entry speed for each maneuver. (53)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
54	Experimental purposes: Exhibition § 21.191(d), Air racing § 21.191(e), Amateur-built § 21.191(g), Primary kit-built § 21.191(h), Operating former military aircraft § 21.191(j), Light-sport category kit-built § 21.191(k), Former light-sport category § 21.191(l) OPR: AFS-800	All conditions.	Unless operated in accordance with manufacturer or military performance data and limitations, removable, externally-mounted equipment must be tested to demonstrate that the aircraft is controllable throughout its normal range of speeds and throughout all the maneuvers to be executed and has no hazardous operating characteristics or design features. Testing must be documented in the maintenance record during Phase I. All removable externally-mounted equipment must be maintained to manufacturer or military instructions, if applicable. (54)
55	All experimental purposes: § 21.191 OPR: AFS-800	Aircraft that are not described in paragraph D-4f.	Night flight operations are prohibited unless the instruments specified in § 91.205 (b) and (c) are installed, operational, compliant with the performance requirements of, and maintained per the applicable regulations. (55)
56	All experimental purposes: § 21.191 OPR: AFS-800	Aircraft described in paragraph D-4f.	Night flight is prohibited. (56)
57	All experimental purposes: § 21.191 OPR: AFS-800	Aircraft described in paragraph D-4f.	Instrument flight rules (IFR) operations are prohibited unless the instruments specified in § 91.205(b) and (d) are installed, operational, compliant with the performance requirements of, and maintained per the applicable regulations. If the aircraft's instruments are in compliance with these requirements and the aircraft or engine manufacturer does not prohibit IFR operations, then IFR operations are permitted, but limited to visual meteorological conditions. (57)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
58	All experimental purposes: § 21.191 OPR: AFS-800	Aircraft that are not described in paragraph D-4f.	Instrument flight rules (IFR) operations are prohibited unless the instruments specified in § 91.205(b) and (d) are installed, operational, compliant with the performance requirements of, and maintained per the applicable regulations. IFR flight is limited to visual meteorological conditions if there is an aircraft or engine manufacturer limitation prohibiting flight in instrument meteorological conditions. (58)
59	Light-sport category § 21.190 OPR: AFS-800	All conditions.	Night flight and instrument flight rules (IFR) operations are prohibited unless the following conditions are met: (a) There is no limitation prohibiting these operations in the AOI or POH; The instruments specified in § 91.205 (b), (c), and (d) (as applicable to night or IFR operations) are installed, operational, compliant with the performance requirements of, and maintained in accordance with applicable regulations. (59)
60	Experimental purposes: Research & Development § 21.191(a) Showing Compliance § 21.191(b) OPR: AFS-800	All conditions.	Prior to compliance with § 91.319(b)(1) and (2), the aircraft must be operated in areas described by § 91.305 at all times unless otherwise authorized for takeoff and landing in these operating limitations. The pilot in command must not perform any maneuvers that have not been flight tested or operate the aircraft outside the weight, airspeeds, and center of gravity limits tested except in areas compliant with § 91.305. (60)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
61	Experimental purposes: Crew training § 21.191(c), Exhibition § 21.191(d), Air Racing § 21.191(e), Market survey § 21.191(f), Amateur-built § 21.191(g), Primary kit-built § 21.191(h), Operating former military aircraft § 21.191(j), Light-sport category kit-built § 21.191(k), Former light-sport category § 21.191(l) OPR: AFS-800	All Conditions.	The pilot in command must not perform any maneuvers that have not been flight tested or operate the aircraft outside the weight, airspeeds, and center of gravity limits tested except in accordance with Phase I operating limitations. If you do not have Phase I operating limitations, you must obtain a research & development airworthiness certificate and demonstrate compliance with § 91.319(b)(1) and (2). (61)
62	All experimental purposes: § 21.191 OPR: AFS-800	Aircraft described in paragraph D-4f.	The carriage of passengers, as defined in § 61.1, is prohibited. (62)
63	All experimental purposes: § 21.191 OPR: AFS-800	Aircraft described in paragraph D-4f.	Flight over a densely populated area or in a congested airway is prohibited. The aircraft must be in a position to make a safe emergency landing without hazard to persons or property on the surface at all times. (63)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
64	All experimental purposes § 21.191 OPR: AFS-800	Production Certificate Holders, provided the aircraft was manufactured by the PC holder and is not described in D-4f.	Prior to compliance with § 91.319(b)(1) and (2), flight over a densely populated area is authorized only for the purpose of takeoff and landing, provided the aircraft is in a position to make a safe emergency landing without undue hazard to persons or property on the surface. Multiple traffic patterns over densely populated areas are prohibited. Except during flight testing, which must be conducted in accordance with § 91.305, once compliance with § 91.319(b)(1) and (2) has been established, this aircraft may be operated over densely populated areas, provided the aircraft- (a) For the purpose of takeoff and landing, is in a position to make a safe emergency landing without undue hazard to persons or property on the surface; (b) At all other times is in a position to make a safe emergency landing without hazard to persons or property on the surface. (64)
65	All experimental purposes § 21.191 OPR: AFS-800 All experimental purposes § 21.191 OPR: AFS-800	All other aircraft that are not described in OL 63 or 64.	Except during flight testing, which must be conducted in accordance with § 91.305, this aircraft may be operated over densely populated areas, provided the aircraft- (a) For the purpose of takeoff and landing, is in a position to make a safe emergency landing without undue hazard to persons or property on the surface; (b) At all other times is in a position to make a safe emergency landing without hazard to persons or property on the surface; (c) Has demonstrated compliance with 91.319(b)(1) and (2) and, if applicable, Phase I operating limitations both initially and after any major change as defined in § 21.93 or major alteration as defined in § 1.1; and (d) Is not equipped with operational external jettisonable stores, explosive bolts, explosive pylon charges (ejectors), military armament installed, and, if non-operational stores are installed, is in a configuration specified by the manufacturer or military. (65)

No.	For these aircraft:	...and these conditions:	...issue this operating limitation:
66	Experimental Exhibition § 21.191(d) OPR: AFS-800	Aircraft that are not described in paragraph D-4f.	No person may be carried in this aircraft during the exhibition of the aircraft's flight capabilities, performance, or unusual characteristics at air shows, or for motion picture, television, or similar productions, unless essential for the purpose of the flight. This limitation does not enable the carriage of persons for compensation or hire. (66)
67	All experimental purposes § 21.191 OPR: AFS-300	Gliders.	A placard must be displayed in the cockpit in full view of the pilot that contains the following: maximum gross weight, CG limits, maximum tow speed, maximum airspeed in smooth air (V_{NE}), and maximum airspeed in rough air (V_{NO}). (67)
68	Experimental purposes: Research and development § 21.191(a), Showing compliance § 21.191(b), Crew training § 21.191(c), Operating former military aircraft § 21.191(j) OPR: AFS-800	All conditions	Carriage of passengers, as defined in § 61.1, is prohibited. (68)
69	All experimental purposes: § 21.191 OPR: AFS-800	Space Support Vehicle operating in accordance with § 91.331.	When conducting a space support vehicle flight in accordance with the requirements of § 91.331, any rule or term of a certificate, including operating limitations, issued by the Administrator that would prohibit carrying persons or property for compensation or hire are not in effect during the flight. (69)

Appendix E. CAA Assistance with U.S. Airworthiness Certificates for New Aircraft Manufactured Outside the United States.

E-1. Introduction. The FAA may obtain assistance from a CAA of the SOM in the final processing, dating, and delivery of a U.S. airworthiness certificate for newly manufactured, type-certificated aircraft destined for export to the United States. The FAA issues the form, and the CAA helps with specific process steps. The use of this procedure is only allowed if no conflict exists with the applicable bilateral agreement. A certificate management office may apply the procedures in this appendix only when approved by AIR-630. Upon request from the U.S. aircraft owner or the CAA of the SOM, the FAA may, at its discretion, authorize the CAA to act on its behalf.

E-2. Applicability. The provisions of this appendix apply only to an aircraft that--

- a.** Is newly manufactured in a country/jurisdiction that has a bilateral agreement concerning airworthiness with the United States;
- b.** Has been exported via an export C of A from the exporting CAA;
- c.** Is registered in the United States;
- d.** Conforms to its FAA TC; and
- e.** Is in a condition for safe operation.

E-3. Procedure.

- a. Notify the CAA.** The aircraft manufacturer notifies the CAA of the SOM that an aircraft has been sold to a U.S. owner and is to be placed on the U.S. registry. As a result, the U.S. aircraft owner will be requesting issuance of a U.S. airworthiness certificate at the point of manufacture.
- b. Reserve a U.S. Registration Number.** The U.S. owner submits a request to reserve a U.S. registration number [hereafter, N-number] for the aircraft and, upon receipt, provides it to the aircraft manufacturer for permanent marking of the aircraft. The U.S. owner also provides this information to the CAA.
- c. Provide FAA Form 8130-6.** To avoid delays, at least 20 days before the expected delivery date of the aircraft, the U.S. owner provides FAA Form 8130-6 to the FAA office that issues airworthiness certificates for import aircraft, including its request that the airworthiness certificate be issued at the point of manufacture. Sections I, II, and III are to be completed, as applicable; however, the following items in section III are to be left blank because the information for these items is unknown until final delivery/export of the aircraft:
 - (1) The status of compliance with the current versions of applicable ADs as indicated in the airworthiness directives block of subsection B, Aircraft Certification Basis.
 - (2) Total airframe hours (including production flight test time) in the applicable block of subsection C, Total Airframe Hours.

(3) The date of the application as indicated in subsection D, Certification.

d. CAA Notifies FAA Office. The CAA notifies the manager of the certificate-issuing, FAA office of its desire to act on behalf of the FAA in the delivery of an airworthiness certificate for the aircraft. The following information will be provided by the CAA:

- (1) N-number, make, model, and S/N of the aircraft.
- (2) Name of the U.S. aircraft owner.
- (3) Scheduled aircraft delivery date.

e. Airworthiness Certificate Prepared. The FAA certificate-issuing office prepares the airworthiness certificate, including two copies.

(1) Block 5 of a standard airworthiness certificate includes pre-printed statements of airworthiness and of compliance with ICAO Annex 8. It also includes introductory text for recording exceptions to the statement of compliance with ICAO Annex 8. Although unusual, such an exception occurs when an applicable FAA exemption does not comply with ICAO Annex 8. Review the applicable aircraft TCDS to identify applicable exemptions by make, model, and S/N. Review each exemption to determine if it contains language indicating a noncompliance with Annex 8 of the ICAO Convention or limiting the aircraft to domestic operations only. You may find FAA exemptions granted to date in [DRS](#). If an exemption indicates noncompliance with Annex 8 or limits the aircraft to domestic operations only, add the following statement to block 5 of the standard airworthiness certificate: “This aircraft is limited to domestic operations only because exemption [number/numbers] [insert exemption number(s)] [does/do] not meet Annex 8 of the ICAO Convention.” Otherwise, enter “None.”

(2) Leave the Date of Issuance block blank.

(3) Sign the original and two copies.

(4) Forward the certificates to the CAA. Do not deliver the airworthiness certificate more than 45 days before the expected aircraft delivery date unless specific circumstances warrant an earlier delivery. Any costs of overnight courier services for the delivery of forms needing expedited delivery to the CAA will be paid for by the applicant or the manufacturer. The following sentence must be included in the transmittal letter: “Do not deliver this airworthiness certificate, issued for the subject aircraft above, until this office notifies you that the aircraft has been properly registered.”

f. Data Plate Preparation and Installation. Upon completion of manufacture and required inspections and tests, including production flight tests, the aircraft manufacturer installs an ID plate on the aircraft that meets the requirements of § 45.11. The aircraft manufacturer also applies the N-number to the aircraft per § 45.21.

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g. Export C of A Issued. The CAA issues an export C of A for the aircraft after completing all tasks and inspections necessary to determine that the aircraft conforms to the FAA-approved type design and is in a condition for safe operation. This certificate must contain the certification statement prescribed in the applicable FAA TCDS under the Import Requirements heading.

Note: If any nonconformities, deviations, or exceptions exist, the CAA must obtain written acceptance of these conditions from the certificate-issuing, FAA office before issuing the export C of A. Such conditions may disqualify the aircraft from receiving the intended airworthiness certificate due to its inability to fully meet the requirements of the U.S. TC.

h. Additional Work Performed After Issuance of the CAA's Export C of A.

(1) If any additional work (for example, alterations or repairs) is performed on the aircraft by the manufacturer after issuance of the export C of A, and before delivery of the U.S. airworthiness certificate, the following is required:

(a) The work must be controlled, documented, and completed by the manufacturer under its CAA-approved production quality system and associated procedures.

(b) The exporting CAA will review the manufacturer's additional completed work to ensure the aircraft remains in compliance with its FAA-approved type design and is in a condition for safe operation. If the CAA is satisfied that these requirements are met, the U.S. airworthiness certificate may be released to the registered owner/operator per this appendix.

(2) If any additional work (for example, alterations or repairs) is performed on the aircraft by someone other than the manufacturer (which also may include any other persons or organizations under the direct control of the manufacturer) after issuance of the export C of A, the aircraft may be disqualified from receipt of the U.S. airworthiness certificate issued under this special procedure.

Note: The CAA is not responsible for the review and acceptance of any additional work performed outside of its direct control and oversight. This includes any additional work performed on the aircraft directly by the new U.S. owner before receiving the U.S. airworthiness certificate. The certificate-issuing, FAA office could not, therefore, be assured of the continued validity of the export C of A upon which this special procedure and the issuance of the U.S. airworthiness certificate are based.

(3) The new U.S. owner/operator (or their authorized agent) may at times perform the various functions and activities which may be necessary to prepare the newly acquired aircraft for their departure flight from the manufacturer and placement into operation.

(a) These functions and activities (for example, preoperational servicing/maintenance, pre-flight inspections, aircraft systems functional checks, navigation/communication equipment, and operational software installation) must

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be properly documented and may be undertaken after issuance of the export C of A and before receipt of the U.S. airworthiness certificate.

(b) After the completion of these tasks, the new U.S. owner/operator must ensure the aircraft remains in conformity to the FAA-approved type design and in a condition for safe operation. The CAA may, at its discretion, monitor these functions and activities in advance of the release of the U.S. airworthiness certificate.

(4) The CAA should contact the FAA certificate-issuing office when any problems arise pertaining to these requirements which would preclude the release and delivery of the U.S. airworthiness certificate to the new U.S. owner/operator.

i. Part 47. The aircraft manufacturer or the U.S. owner must ask the CAA in the SOM to confirm to the FAA Aircraft Registry that they have not issued registration for the aircraft or that registration in that country has ended, is canceled, or is invalid. Refer to Part 47, Aircraft Registration, §§ 47.15(a)(1) and 47.37(b).

j. Submit Information. The U.S. owner submits to the FAA Aircraft Registry all information required to register the aircraft. Permanent aircraft registration is sent by email or mail as Form 8050-3 (Certificate of Aircraft Registration), which serves as official proof of registration.. The FAA may discontinue issuing the temporary certificates (Form REGAR-FAX-4) since the 8050-3 can be electronically sent the same day by email.

k. Provide Complete FAA Form 8130-6. The U.S. owner provides the CAA with a complete FAA Form 8130-6 after receipt of the registration. Sections IV and V, and the entire reverse side of FAA Form 8130-6 are to be left blank. However, the items left blank from the initial form, and one additional item, should now be completed with the following:

- (1) The status of compliance up to the most current and applicable ADs as indicated in the Airworthiness Directives block of subsection B.
- (2) The total airframe hours (including production flight test time) in the applicable block of subsection C.
- (3) The final date entered by the applicant must be the same as or later than the date of the export C of A issued by the exporting CAA.
 - (a) The date cannot be later than the date entered on the airworthiness certificate for the aircraft.
 - (b) The signature of the person in subsection D must be the registered owner's (or an authorized employee of the corporation or company signified as the registered owner) identified under subsection A of section III. If the signature is other than one of these persons, the application must be accompanied with a notarized agent letter or a notarized and current power of attorney delegating the authority to act as

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an agent on the owner's behalf to apply for the airworthiness certificate. A copy of the notarized agent letter or power of attorney is acceptable.

- (4) The specific 14 CFR reference listed in section III, subsection C, should be crossed out and changed from § 91.173 to § 91.417.

l. CAA Verifies Information. The CAA verifies the following before delivery of the airworthiness certificate:

- (1) The registered owner identified on the application for the airworthiness certificate is the same person, company, or corporation shown on Aeronautical Center Form 8050-3 or Form REGAR-FAX-4, *Temporary Certificate of Registration*, from the FAA Aircraft Registry.
- (2) The aircraft N-number on the registration, application, airworthiness certificate, and aircraft match. In addition, the N-number marks on the aircraft must meet the requirements of Part 45.
- (3) The aircraft's ID plate has all of the required data and proper information and is mounted in the proper location on the aircraft.
- (4) The CAA has issued the export C of A.

m. CAA Completes Airworthiness Certificate. The CAA finalizes the airworthiness certificate as follows. The date on which the certificate was issued is entered in the Date of Issuance block on the original and on the two copies. Dates on certificates should be in DD MMM YYYY or DD Mmm YYYY format; for example, "25 DEC 2016" or "25 Dec 2016." Date of issuance entries must be typewritten or made with another appropriate instrument, for example, a mechanical date stamping device for the date entry. Handwritten entries are not permitted.

n. CAA Provides Documents. The CAA provides (electronic transmission is acceptable) the documents listed below to the FAA certificate-issuing office before the first flight of the aircraft under the U.S. airworthiness certificate. This action is necessary because the FAA certificate-issuing office must be in possession of legal documentation in the form of on-hand file records of proper airworthiness certification before the aircraft is operated.

- (1) A copy of the export C of A.
- (2) A copy of the registration.
- (3) A copy of the completed and dated airworthiness certificate.

o. CAA Displays Airworthiness Certificate. The CAA places the airworthiness certificate in the aircraft for display at the cabin or cockpit entrance in such a way that the certificate is legible to passengers or crew per § 91.203(b). The following statement is entered into the aircraft maintenance record: "U.S. [*Standard or Special*] Airworthiness Certificate, issued [*date*], has been placed in the aircraft on behalf of [*FAA certificate-issuing office*] on [*date*]." The person from the CAA who issued and placed the airworthiness certificate in the aircraft must sign the aircraft maintenance record and include a functional title or other evidence of authorization to act on behalf of the CAA.

p. CAA Disposition of Documents. The CAA retains one copy of the airworthiness certificate and forwards the following documents to the manager of the FAA certificate-issuing office:

- (1) The remaining copy of the dated airworthiness certificate.
- (2) The original and one copy of the completed application.
- (3) The original and one copy of the export C of A issued by the CAA of the SOM.
- (4) A copy of the registration.

q. Application Endorsed. The FAA certificate-issuing office endorses the application by entering a statement on the form in a location that can be easily read. At a minimum, this statement must:

- (1) Indicate that the airworthiness certificate was issued on the basis of the export C of A for Export No. [*insert number where applicable*], and
- (2) Include the issuing CAA's name and the date of issuance of the export C of A.

r. FAA Disposition of Documents. The FAA certificate-issuing office then forwards the application, the copy of the airworthiness certificate, and the original export C of A to the FAA Aircraft Registry for permanent filing. A copy of the registration is only forwarded if it is part of documentation of a falsified registration being provided to the CAA.

s. Delivery of the Aircraft. The U.S. owner takes delivery of the aircraft and installs the U.S. registration in the aircraft.

Appendix F. Sample Special Flight Authorizations.

Figure F-1. Example SFA for the Flight of an Aircraft to a Place Where Repairs or Alterations Are to Be Made



U.S. Department
of Transportation

**Federal Aviation
Administration****FOREIGN CIVIL AIRCRAFT SPECIAL FLIGHT AUTHORIZATION (SFA)**

Authorization No.:	<u>NE-03-09</u>
Aircraft Make:	<u>Cessna</u>
Aircraft Model:	<u>180H</u>
Serial No.:	<u>18051515</u>
Nationality and Registration Marks:	<u>CF-ABC</u>
Name and address of Registered Owner:	<u>Mr. Richard A. Roe</u> <u>777 Quebec Street</u> <u>Smithton, Ontario, Canada</u>

Pursuant to Title 14 of the Code of Federal Regulations (14 CFR) 91.715, Mr. Richard A. Roe is hereby authorized to operate the aircraft identified above for the purpose of flying it from Hartford, Connecticut, to Ontario, Canada, for permanent repair of damage incurred during a landing accident at Hartford. A representative of Transport Canada Civil Aviation has inspected the aircraft and found it safe for the intended flight provided that the airspeed does not exceed 130 knots, and no passengers are carried aboard the aircraft. All operations must be per the following restrictions and limitations.

1. A copy of this authorization must be available to the pilot in command when operating under the terms of this SFA.
2. You must comply with all limitations imposed by the State of Registry and this authorization.
3. Persons or property must not be carried for compensation or hire.
4. This authorization is valid in the United States only.
5. Upon request, this authorization must be made available to an FAA inspector.
6. All flights must be conducted under visual flight rules, day only.

Figure F-1. Example SFA for the Flight of an Aircraft to a Place Where Repairs or Alterations Are to Be Made (Continued)

7. All flights must be conducted at airspeeds not to exceed 130 knots.
8. No person may be carried in the aircraft during flight unless that person is required crew or maintenance personnel essential to the purpose of the flight and has been advised of the contents of this SFA and of the airworthiness status of the aircraft.
9. This SFA is valid until July XX, 2023, unless superseded or rescinded.

J. A. Smith, Manager
Bradley Flight Standards District Office
Office of General Aviation Safety Assurance

Issued in Windsor Locks, Connecticut, July XX, 2023.

Figure F-2. Example SFA for the Flight of an Aircraft to a New SOR
(This Format Is Generally Applicable to a Single Aircraft Authorization for
Ferry Flights)



U.S. Department
of Transportation

**Federal Aviation
Administration**

CIVIL AIRCRAFT SPECIAL FLIGHT AUTHORIZATION (SFA)

Authorization No.:	<u>SO-11-01</u>
Aircraft Make:	<u>Beech</u>
Aircraft Model:	<u>D185</u>
Serial No.:	<u>A-23456</u>
Nationality and Registration Marks:	<u>HK-ABC</u>
Name and address of Registered Owner:	<u>Mr. Hernando Restrepo</u> <u>22 Calle de Presidente</u> <u>Fusagasuga, Colombia</u>

Pursuant to Title 14 of the Code of Federal Regulations (14 CFR) 91.715, Mr. Hernando Restrepo is hereby authorized to operate the aircraft identified above for the purpose of flying from Quebec, QC, Canada, to Fusagasuga, Colombia. The aircraft identified above was under Canadian registry and held a current and valid Canadian airworthiness certificate before its sale to Mr. Restrepo. A current and valid Colombian airworthiness certificate will not be issued until after its entry into Colombia. All operations of the aircraft must be per the following restrictions and limitations.

1. A copy of this authorization must be available to the pilot in command when operating under the terms of this SFA.
2. You must comply with all limitations imposed by the State of Registry and this authorization.
3. Persons or property must not be carried for compensation or hire.
4. This authorization is valid in the United States only.
5. Upon request, this authorization must be made available to an FAA inspector.
6. The identification marks assigned to the aircraft by the State of Columbia must be displayed on the aircraft according to that country's applicable requirements.

**Figure F-2. Example SFA for the Flight of an Aircraft to a New SOR
(This Format Is Generally Applicable to a Single Aircraft Authorization for Ferry Flights)
(Continued)**

7. No person may be carried in the aircraft during flight unless that person is required crew or maintenance personnel essential to the purpose of the flight and has been advised of the contents of this SFA and of the airworthiness status of the aircraft.
8. This SFA is valid until August XX, 2023, or unless superseded or rescinded.

J. A. Smith, Manager
Portland ME, Flight Standards District Office
Office of General Aviation Safety Assurance

Issued in Portland, Maine, on July XX, 2023.

Figure F-3. Example SFA for the Purpose of Flight Testing

U.S. Department
of Transportation**Federal Aviation
Administration**

FOREIGN CIVIL AIRCRAFT SPECIAL FLIGHT AUTHORIZATION (SFA)

Authorization No.:	<u>ASW-1</u>
Aircraft Make:	<u>McDonnell Douglas</u>
Aircraft Model:	<u>DC-9-11</u>
Serial No.:	<u>12345</u>
Nationality and Registration Marks:	<u>CF-POH</u>
Name and address of Registered Owner:	<u>Canada Air, Montreal, Canada</u>
Name and Address of Agent:	<u>John Doe Company,</u> <u>21 Blackfoot Drive</u> <u>San Antonio, Texas 78216</u>

Pursuant to Title 14 of the Code of Federal Regulations (14 CFR) 91.715, the John Doe Company is hereby authorized to operate the aircraft identified above for the purpose of conducting flight test(s) required to obtain a supplemental type certificate (STC) covering the installation in the aircraft of General Electric CGY2 turbofan engines. All operations of the aircraft must be per the following restrictions and limitations.

1. A copy of this authorization must be available to the pilot in command when operating under the terms of this SFA.
2. You must comply with all limitations imposed by the State of Registry and this authorization.
3. Persons or property must not be carried for compensation or hire.
4. This authorization is valid in the United States only.
5. Upon request, this authorization must be made available to an FAA inspector.
6. No person may be carried in the aircraft during flight unless that person is required crew or maintenance personnel essential to the purpose of the flight and has been advised of the contents of this SFA and of the airworthiness status of the aircraft.

Figure F-3. Example SFA for the Purpose of Flight Testing (Continued)

7. All flight tests must be conducted in compliance with § 91.305.
8. All flights must be conducted under visual flight rules, day only.
9. All maintenance and inspection of the aircraft must be conducted under the direct supervision of qualified personnel holding appropriate licenses issued or rendered valid by the Canadian Department of Transportation and according to Canadian aircraft maintenance requirements.
10. This SFA is valid until July XX, 2023, or unless superseded or rescinded.

J. A. Smith, Manager
Central Certificate Management Branch
System Oversight Division

Issued in Fort Worth, Texas, on July XX, 2023.

Figure F-4. Example Blanket SFA for Customer Crew Training

U.S. Department
of Transportation**Federal Aviation
Administration****CIVIL AIRCRAFT SPECIAL FLIGHT AUTHORIZATION (SFA)**

Authorization No.: NE-02-43

Pursuant to Title 14 of the Code of Federal Regulations (14 CFR) 91.715 to the Yankee Airplane Company, John Hancock Airport, Boston, Massachusetts 02111, this constitutes authority instead of an airworthiness certificate. For the purpose of giving customer crew training to the buyer, its employees, or designees in any aircraft manufactured by the Yankee Airplane Company when the aircraft has been placed under foreign registry, each aircraft operated under this SFA must be operated according to the following restrictions and limitations.

1. Each aircraft operated for customer crew training flights must carry this SFA attached with a statement including the name and address of the aircraft owner, the aircraft's assigned nationality and registration marks, and the dates on which the customer crew training flights are scheduled to begin and end.
2. All customer crew training and aircraft maintenance must be conducted under the direct supervision of qualified Yankee Airplane Company personnel.
3. Customer crew training flights on any one aircraft must be conducted during an interval not to exceed 30 days.
4. Before beginning customer crew training flights with any one aircraft, the Yankee Airplane Company must submit to the local FAA Manufacturing Inspector the information specified in paragraph 1 of this authorization pertaining to that aircraft.
5. The identification marks assigned to the aircraft by the State of Registry must be displayed on the aircraft according to that country's applicable requirements.
6. Persons or property must not be carried for compensation or hire.
7. No person may be carried in the aircraft during flight unless that person is required crew or maintenance personnel essential to the purpose of the flight and has been advised of the contents of this SFA and of the airworthiness status of the aircraft.

Figure F-4. Example Blanket SFA for Customer Crew Training (Continued)

8. This authorization is valid in the United States only.
9. This SFA is valid until superseded or rescinded.

J. A. Smith,
Division Manager
Office of General Aviation Safety Assurance

Issued in Burlington, Massachusetts, on July XX, 2023.

Figure F-5. Example SFA for Export Delivery



CIVIL AIRCRAFT SPECIAL FLIGHT AUTHORIZATION (SFA)

Authorization No.: NE-03-59
Aircraft Make: Piper
Model: PA 84
Serial No.: 1334
Nationality and Registration Marks: I-JAB
Registered Owner: Joseph A. Banco, Via Banco, Rome, Italy

Pursuant to Title 14 of the Code of Federal Regulations (14 CFR) 91.715, Mr. Joseph A. Banco is hereby authorized to operate the aircraft identified above for the purpose of export and delivery from Westfield, Massachusetts, to Rome, Italy. This aircraft is on Italian registry and an airworthiness certificate has not yet been issued.

1. A copy of this authorization must be available to the pilot in command when operating under the terms of this SFA.
2. You must comply with all limitations imposed by the State of Registry and this authorization.
3. Persons or property must not be carried for compensation or hire.
4. This authorization is valid in the United States only.
5. Upon request, this authorization must be made available to an FAA inspector.
6. No person may be carried in this aircraft during flight unless that person is required crew or maintenance personnel essential to the purpose of the flight and has been advised of the content of this SFA and of the airworthiness status of the aircraft.
7. The aircraft must not be operated with temporary fuel system(s) or temporary navigation equipment installed or at a weight in excess of its maximum certificated takeoff weight, unless approved, by the civil aviation authority (CAA) State of Registry in writing.

Figure F-5. Example SFA for Export Delivery (Continued)

8. The kinds of operations authorized are visual flight rules, instrument flight rules, day, and night.
9. This SFA is valid until August XX, 2023, unless superseded or rescinded.

J. A. Smith,
Division Manager
Office of General Aviation Safety Assurance

Issued in Burlington, Massachusetts, on July XX, 2023.

Figure F-6. Example Blanket SFA for Delivering Aircraft
for the Purpose of Export Delivery



U.S. Department
of Transportation

**Federal Aviation
Administration**

FOREIGN CIVIL AIRCRAFT SPECIAL FLIGHT AUTHORIZATION (SFA)

Authorization No.: WP-26-22

This SFA is issued to the John Smith Airplane Company, 711 Water Boulevard, San Diego, California 82101, pursuant to Title 14 of the Code of Federal Regulations (14 CFR) 91.715. A copy of this SFA furnished by the above constitutes authority instead of an airworthiness certificate for the purpose of export delivery of aircraft manufactured by that Company. This SFA is applicable to aircraft that are on a foreign registry and have no airworthiness certificate.

1. A copy of this authorization must be available to the pilot in command when operating under the terms of this SFA.
2. No person may be carried in this aircraft during flight unless that person is required crew or maintenance personnel essential to the purpose of the flight and has been advised of the content of this SFA and of the airworthiness status of the aircraft.
3. The aircraft must not be operated with temporary fuel system(s) or temporary equipment installed, or at a weight in excess of its maximum certificated takeoff weight, unless approved by the civil aviation authority (CAA) of the State of Registry in writing.
4. Persons or property may not be carried in the aircraft for compensation or hire.
5. This authorization is valid in the United States only.
6. The kinds of operations authorized are visual flight rules, instrument flight rules, day, and night.
7. This SFA is valid until December XX, 2023.

Monico Ramirez
Aviation Safety Inspector
Los Angeles International Field Office

Issued in Los Angeles, California, on July XX, 2023.

Figure F-7. Example SFA for the Purpose of Demonstration

U.S. Department
of Transportation**Federal Aviation
Administration****CIVIL AIRCRAFT SPECIAL FLIGHT AUTHORIZATION (SFA)**

Authorization No.: NE-01-31
Aircraft Make: Hansa
Model: HFB-320
Serial No.: 1024
Nationality and Registration Marks: D-CARO
Name and Address of Registered Owner: Hamburger Flugzeugbau G.M.B.H.
2103 Hamburg
Finkenwerder Postfact 109, Germany

Pursuant to Title 14 of the Code of Federal Regulations (14 CFR) 91.715, Hamburger Flugzeugbau G.M.B.H. is hereby authorized to operate the aircraft identified above for the purpose of conducting demonstration flights in the United States. The aircraft has been issued a provisional certificate of airworthiness by the Luftfahrt-Bundesamt and has been shown to meet standards equivalent to those required for provisional certification of a U.S.-registered civil aircraft. All operations of the aircraft must be per the following restrictions and limitations.

1. A copy of this authorization must be available to the pilot in command when operating under the terms of this SFA.
2. You must comply with all limitations imposed by the State of Registry and this authorization.
3. Persons or property must not be carried for compensation or hire.
4. This authorization is valid in the United States only.
5. Upon request, this authorization must be made available to an FAA inspector.
6. The identification marks assigned to the aircraft by the State of Registry must be displayed on the aircraft according to that country's applicable requirements.

Figure F-7. Example SFA for the Purpose of Demonstration (Continued)

7. No person may be carried in this aircraft during flight unless that person is required crew or maintenance personnel essential to the purpose of the flight and has been advised of the content of this SFA and of the airworthiness status of the aircraft.
8. All flights must be conducted under visual flight rules, day only.
9. This SFA is valid until September XX, 2023, unless superseded or rescinded.

J. A. Smith,
Division Manager
Office of General Aviation Safety Assurance

Issued in Burlington, Massachusetts, on July 21, 2023.

Figure F-8. Example SFA for Participating in Aviation Events



U.S. Department
of Transportation

**Federal Aviation
Administration**

FOREIGN CIVIL AIRCRAFT SPECIAL FLIGHT AUTHORIZATION (SFA)

Authorization No.:	<u>NE-01-31</u>
Aircraft Make:	<u>Aero Vodochody</u>
Model:	<u>L-39C</u>
Serial No.:	<u>1024</u>
Nationality and Registration Marks:	<u>ES-XXX</u>
Name and Address of Registered Owner:	<u>Hamburger Flugzeugbau G.M.B.H.</u> <u>2103 Hamburg</u> <u>Finkenwerder Postfact 109, Germany</u>

Pursuant to Title 14 of the Code of Federal Regulations (14 CFR) 91.715, Hamburger Flugzeugbau G.M.B.H. is hereby authorized to operate the aircraft identified above for the purpose of participating in aviation events. The aircraft does not meet the airworthiness requirements specified in Annex 8 to the Convention on International Civil Aviation and has been issued an Estonian permit to fly. The aircraft will enter the United States on or about August 1, 2017, from Canada and will exit to Mexico approximately December 1, 2017. The intended aviation events include—

- (1) Rhode Island National Guard Open House & Air Show, North Kingstown, RI
- (2) Bethpage Air Show at Jones Beach, Wantagh, NY
- (3) Virginia Beach Patriotic Festival, Virginia Beach, VA
- (4) Westmoreland County Air Show, Latrobe, PA
- (5) Denton Air Show, Denton, TX
- (6) Spectacle Aerien International Bagotville, Bagotville, QC, Canada
- (7) Battle Creek Field of Flight Air Show & Balloon Festival, Battle Creek, MI
- (8) EAA AirVenture, Oshkosh, WI
- (9) National Championship Air Races, Reno, NV

All operations of the aircraft must abide by the following restrictions and limitations.

Figure F-8. Example SFA for Participating in Aviation Events (Continued)

1. A copy of this authorization must be available to the pilot in command when operating under the terms of this SFA.
2. You must comply with all limitations imposed by the State of Registry and this authorization.
3. Persons or property must not be carried for compensation or hire.
4. This authorization is valid in the United States only.
5. Upon request, this authorization must be made available to an FAA inspector.
6. Operation is restricted to airports that are within airspace class C, D, E, or G, except in the case of a declared emergency or authorized operations under an airshow waiver.
7. Kinds of operations authorized by the State of Registry are authorized, with the following restriction. If instrument flight operations are authorized, the pilot in command must have a method to avoid operating over densely populated areas or in congested airways.
8. Flight over densely populated areas is authorized only for the purpose of takeoff or landing.
9. Flight in Reduced Vertical Separation Minimum (RVSM) designated airspace is prohibited.
10. Flight with any externally mounted equipment is prohibited.
11. Preflight planning runway length requirements:
 - a. Takeoff is prohibited unless takeoff planning determines that it is possible to stop the airplane safely on the runway, as shown by the accelerate-stop distance data. In addition, the aircraft must be able to clear all obstacles by at least 50 feet vertically.
 - b. Landing will not be attempted unless landing planning determines that a full stop landing can be made within 60 percent of the effective length of the runway from a point 50 feet above the runway.
 - c. When calculating takeoff or landing performance, corrections must be made for any runway gradient. Performance data based on still air may be corrected by taking into account not more than 50 percent of any reported headwind component and not less than 150 percent of any reported tailwind component. Calculations may not include the use of reverse thrust or drag chute.

Figure F-8. Example SFA for Participating in Aviation Events (Continued)

12. No person may be carried in the aircraft during flight unless that person is required crew or maintenance personnel essential to the purpose of the flight and has been advised of the contents of this SFA and of the airworthiness status of the aircraft.
13. This authorization is valid until October 31, 2023, unless superseded or rescinded, or the Estonian permit to fly becomes invalid.

A.A. Smith, Manager
New York International
Field Office

Issued in Jamaica, New York, on July XX, 2023.

Appendix G. Limitations for Operation of an Aircraft with a Door Removed

Figure G-1. Example Limitations for Operation of an Aircraft with a Door Removed



Make _____
Model _____ Serial No. _____
Registration No. _____

AIRCRAFT OPERATING LIMITATIONS

The aircraft described above may be flown with not more than one cabin door removed for the purpose of (see note below), provided the aircraft is operated per the applicable sections of Title 14 of the Code of Federal Regulations and the following limitations:

Note: Show specific operations; for example, intentional parachute jumping, skydiving, etc.

1. Maximum speed must not exceed any of the following:
 - The approved maneuvering speed.
 - 70 percent maximum level flight speed.
 - 70 percent maximum structural cruising speed.
2. Aerobatic maneuvers are not permitted.
3. Maximum yaw angle 10 degrees; maximum bank angle 15 degrees.
4. A Federal Aviation Administration (FAA)-approved safety belt must be provided and worn by each occupant during takeoff and landing and at all other times when required by the pilot-in-command.
5. All occupants must wear parachutes when intentional parachute jumping and skydiving operations are conducted.
6. Smoking is not permitted.
7. When operations other than intentional parachute jumping and skydiving are conducted, a suitable guardrail or equivalent safety device must be provided for the doorway.
8. All loose articles must be tied down or stowed.
9. No baggage may be carried.

**Figure G-1. Example Limitations for Operation of an Aircraft
With a Door Removed (Continued)**

10. Parachutists' static lines must be kept free of pilot's controls and control surfaces.
11. Operations are limited to visual flight rules conditions.
12. Cabin door hold-open clips installed on wing brace struts and/or under surface of wing must be removed before conducting intentional parachute jumping or skydiving operations.
13. When intentional parachute jumping, skydiving, or other specified operations are being conducted, the pilot at the controls must hold at least a private pilot certificate and appropriate rating.
14. This aircraft must not be operated in solo flight by the holder of a student pilot certificate.
15. Operation of this aircraft with a door removed for any purpose other than that for which it is certificated is prohibited.
16. The following placard must be placed on the instrument panel in full view of the pilot:
 "For flight with door removed, see aircraft operating limitations dated
 _____."
17. A copy of these limitations must be carried in the aircraft when flight operations are conducted with the door removed.
18. These operating limitations are a part of the airworthiness certificate.

FAA Inspector _____ Date _____

Office No. _____

Appendix H. Acronyms

14 CFR	Title 14 of the Code of Federal Regulations
49 USC	Title 49 of the United States Code
AC	Advisory Circular or Aeronautical Center
AD	Airworthiness Directive
AEE	Office of Environment and Energy
AED	Aircraft Evaluation Division
AFS	Flight Standards Service
AFS-300	Aircraft Maintenance Division
AFB-710	Civil Aviation Registry
AFS-800	General Aviation and Commercial Division
AIR	Aircraft Certification Service
AIR-40	International Office
AIR-600	Policy and Standards Division
AIR-630	Organization and Systems Policy Branch
AIR-631	Design and Maintenance Systems Section
AIR-632	Production and Airworthiness Section
AIR-634	System Oversight and Delegation Section
AIR-500	Integrated Certificate Management Division
AIR-700	Compliance and Airworthiness Division
AIR-800	System Oversight Division
AOI	Aircraft Operating Instructions
ASI	Aviation Safety Inspector
ASTM	ASTM, International (formerly American Society for Testing and Materials)
ATC	Air Traffic Control
ATF	Department of Justice; Bureau of Alcohol, Tobacco, Firearms, and Explosives
AWC	Airworthiness Certification Application (electronic database)
BAA	Bilateral Airworthiness Agreement
BASA	Bilateral Aviation Safety Agreement
CAA	Civil Aviation Authority
CAM	Civil Aeronautics Manual
CAMP	Continuous Airworthiness Maintenance Program
CAR	Civil Air Regulations
CFR	Code of Federal Regulations
CG	Center of Gravity
CM Section	AIR Certificate Management Section
DAR	Designated Airworthiness Representative
DER	Designated Engineering Representative
DMIR	Designated Manufacturing Inspection Representative
DMO	Directives Management Officer
DOD	Department of Defense
DOT	Department of Transportation
DRS	Dynamic Regulatory System
EAA	Experimental Aircraft Association

EASA	European Union Aviation Safety Agency
EDRS	Electronic Document Retrieval System
EPU	Emergency Power Unit
Export C of A	Export Certificate of Airworthiness
FAA	Federal Aviation Administration
FAARA 2018	FAA Reauthorization Act of 2018
FSIMS	Flight Standards Information Management System
FSCAP	Flight Safety-Critical Aircraft Part
GPO	Government Publishing Office
ICA	Instructions for Continued Airworthiness
ICAO	International Civil Aviation Organization
ICMD	Integrated Certificate Management Division (AIR-500)
ID	Identification
IFO	International Field Office
IFR	Instrument Flight Rules
IP	Implementation Procedures
IPA	Implementation Procedures for Airworthiness
IPC	Illustrated Parts Catalog
LOA	Letter of Authorization
MCAI	Mandatory Continuing Airworthiness Information
NATO	North Atlantic Treaty Organization
N/A	Not applicable
NOA	Notice of Availability
NTSB	National Transportation Safety Board
ODA	Organization Designation Authorization
PAH	Production Approval Holder
PAO	Public Aircraft Operations
PC	Production Certificate
PDF	Portable Document Format
PIC	Pilot in Command
PIV	Personal Identification Verification
PLR	Production Limitation Record
PMA	Parts Manufacturer Approval
POH	Pilot's Operating Handbook
PPC	Powered Parachute
R&D	Research and Development
RVSM	Reduced Vertical Separation Minimum
SFA	Special Flight Authorization
SFP	Special Flight Permit
SIF	Special Interest Flight
SIP	Schedule of Implementation Procedures
SIR	Special Import Requirements
S/N	Serial Number
SOC	Statement of Compliance
SOD	State of Design
SOM	State of Manufacture

SOR	State of Registry
STC	Supplemental Type Certificate
TC	Type Certificate
TCDS	Type Certificate Data Sheet
TIP	Technical Implementation Procedures
TSO	Technical Standard Order
UAS	Unmanned Aircraft System
VFR	Visual Flight Rules
VLA	Very Light Aircraft
WebOPSS	Web-based Operations Safety System
WSC	Weight-Shift-Control

Appendix I. Definitions

List of terms and definitions used in this order:

- **Aerial Work Operations.** Aerial work operations are undefined; however, FAA has consistently interpreted the term “aerial work” to mean work done from the air with the same departure and destination points, while no property of another is carried on the aircraft, and only persons essential to the operation are carried on board the aircraft. As used with respect to light-sport category aircraft, aerial work operations are specific aerial work tasks listed in the aircraft’s pilot operating handbook or operating limitations, as applicable, and specified in the aircraft’s statement of compliance, in accordance with § 21.190. Examples of aerial work operations include crop dusting, seeding, spraying, bird chasing, banner towing, aerial photography or survey, firefighting, and powerline or pipeline patrol. These limited examples are for information and should not be used to prevent the use of additional aerial work operations terms.
- **Airworthy.** An aircraft with a type certificate (TC) is airworthy when it conforms to its U.S. TC and is in a condition for safe operation. For the purpose of this order, a non-type-certificated aircraft is airworthy when it is in a condition for safe operation.
- **Airworthiness Certificate.** A standard airworthiness certificate, FAA Form 8100-2, and special airworthiness certificate, FAA Form 8130-7.
- **Bilateral Agreement.** A document signed by the United States and a foreign jurisdiction that allows reciprocal acceptance of specified aeronautical products and of procedures for approving different types of aeronautical products. For the purposes of this order, [bilateral agreements](#) include Bilateral Aviation Safety Agreements (BASA), Bilateral Airworthiness Agreements (BAA), and the Aviation Safety Agreement with the European Union.
- **Category.** Refer to § 1.1 for this definition.
- **Classification.** Refers to the two types of airworthiness certificates, standard and special.
- **Condition for Safe Operation.** Refers to the condition of the aircraft relative to wear and deterioration, for example, skin corrosion, window delamination/crazing, fluid leaks, and tire wear.
- **Conformity to the TC.** Means the aircraft configuration and the engine, propeller, and articles installed are consistent with the drawings, specifications, and other data that are part of the TC. This includes any supplemental type certificate (STC), repairs, and alterations incorporated into the aircraft.
- **Consensus Standard.** For the purpose of certifying light-sport category aircraft, an industry-developed consensus standard that applies to aircraft design, production, and airworthiness. It includes, but is not limited to, standards for aircraft design and performance, required equipment, manufacturer quality assurance systems, production acceptance test procedures, operating instructions, maintenance and inspection procedures, identification and recording of major repairs and major alterations, and continued airworthiness instructions.
- **Continued Airworthiness System.** For the purpose of eligibility in airworthiness certifying light-sport category aircraft, the manufacturer’s closed-loop system consisting of controls, procedures, and requirements for monitoring and correcting safety-of-flight issues through the issuance of safety directives meeting the identified consensus standard.

- **Critical Characteristic.** Any feature throughout the life cycle of a Flight-Safety Critical Aircraft Part (FSCAP) that, if nonconforming, missing, or degraded, could cause a catastrophic failure resulting in loss or serious damage to the aircraft or an uncommanded engine shutdown resulting in an unsafe condition. A characteristic can be critical in terms of dimension, tolerance, finish, or material; an assembly, manufacturing, or inspection process; or an operation, field maintenance, or depot overhaul requirement. A manufacturing-critical characteristic is produced during the manufacturing process. An installation-critical characteristic, such as torque, is critical in terms of assembly or installation.
- **Designee.** A person delegated to act as a representative of the Administrator under Part 183.
- **Exemption.** Relief from requirements of a current regulation for an individual or entity.
- **Export.** Refers to the transfer of an aviation product or article from the regulatory jurisdiction of one CAA to another.
- **Flight Safety-Critical Aircraft Part (FSCAP).** Any article containing a critical characteristic whose failure, malfunction, or absence could cause (1) a catastrophic failure resulting in loss or serious damage to the aircraft, or (2) an uncommanded engine shutdown resulting in an unsafe condition.
- **Implementation Procedures (IP).** A document under a bilateral agreement that specifies detailed procedures on cooperation between the FAA and another CAA in the discipline of airworthiness. Depending on the agreement, this document may be called Implementation Procedures for Airworthiness (IPA), Technical Implementation Procedures (TIP), or Schedule of Implementation Procedures (SIP).
- **Import.** Refers to the receipt of an aviation product or article from the regulatory jurisdiction of one CAA by another.
- **Maintenance Records.** The records of maintenance for an aircraft, aircraft engine, or propeller. Commonly referred to as a “logbook.”
- **Make.** As used in this order concerning the “make” of a product, make is the name assigned to the aircraft by the manufacturer when it was produced.
- **Manufacturer’s Statement of Compliance (SOC).** A signed statement made by the aircraft manufacturer stating that the aircraft (by make, model, serial number, and date of manufacture) meets applicable requirements per § 21.190(d).
- **Military Surplus Product or Article.** A product or article that originally was released as surplus by the U.S. military, even if subsequently resold by a manufacturer, owner/operator, repair facility, or any other parts supplier.
- **Military-Unique Flight Safety-Critical Aircraft Part (FSCAP).** Any FSCAP specifically and uniquely designed and manufactured for the U.S. military, for which there is no corresponding FAA-approved TC or PAH engine, propeller, or article produced for civilian application. Breakout products or articles produced specifically for military use by a manufacturer other than an FAA PAH using military-provided designs, drawings, and specifications are also considered military-unique
- **N-number.** The national and registration number for an aircraft registered in the United States.

Appendix I

• **New Aircraft.** An aircraft may be considered new if ownership is retained by the manufacturer, distributor, dealer, or their trustee; if there is no intervening private owner, lease, or time-sharing arrangements; and if the aircraft has not been used in any pilot school and/or air taxi operation. An aircraft is still considered new regardless of the operating time logged by the manufacturer, distributor, or dealer if the aircraft has been maintained per Part 43, as applicable, and the aircraft has remained under the operational control of the manufacturer, distributor, or dealer.

• **Original Certification.** This term applies to the issuance of standard or special airworthiness certificates and export certificates of airworthiness for (meeting one of the following):

- An aircraft that has never left the control of the production certificate (PC) or type certificate (TC) holder that manufactured it;
- An aircraft that has never held a U.S. or foreign export certificate of airworthiness (C of A), airworthiness certificate, or equivalent document except for an aircraft that has only held a special flight permit (SFP);
- An aircraft involved in a certification project such as developing a TC or STC, including issuing a standard airworthiness certificate after the certification project is completed; or
- The issuance of a provisional airworthiness certificate under § 21.225 for a provisional amendment to a TC.

• **Production Approval Holder (PAH).** A holder of a PC, Parts Manufacturer Approval (PMA), or Technical Standard Order (TSO) authorization who controls the design and quality of a product or article thereof.

• **Purpose.** The definition varies depending on use:

- As used with respect to issuance of a special airworthiness certificate for an experimental purpose, the purpose refers to the various intended flight operations under § 21.191. Examples include research and development, showing compliance with regulations, crew training, exhibition, etc.
- As used with respect to issuance of a special flight permit (SFP), the purpose refers to the various intended flight operations under § 21.197. Examples include flying an aircraft to a base where repairs are to be performed, delivering an aircraft, production flight testing new aircraft, etc.
- As used with respect to restricted category aircraft, the purpose refers to the various special purpose operations eligible for receipt of a restricted category TC under § 21.25.

• **Recurrent Certification.** This term applies to the issuance of standard or special airworthiness certificates or export C of A that do not meet the definition of original certification. Recurrent certification verifies an aircraft has been properly maintained and altered while in service.

• **Registration.** Means any of the following:

- Aeronautical Center Form 8050-3, *Certificate of Aircraft Registration*.
- Aeronautical Center Form 8050-6, *Dealer's Aircraft Registration Certificate*.
- Form REGAR-FAX-4, *Temporary Certificate of Registration*.

Appendix I

- Aircraft Registration Application, AC Form 8050-1, under the provisions of § 47.31(c).
- A copy of the Assignment of Special Registration Number, AC Form 8050-64, along with the current AC Form 8050-3, Certificate of Aircraft Registration.
- **Should.** Used to express what is expected. If the “should” expectation cannot be met, contact your managing specialist (designees) or manager (ASIs) for further direction. Document the resulting mitigating actions.
- **Simplified Flight Controls.** Light-sport category aircraft may be designed with simplified flight controls per the design requirements in § 22.180.
- **Space Support Vehicle.** An aircraft that is a launch vehicle, reentry vehicle, or a component of a launch or reentry vehicle.
- **Space Support Vehicle Flight.** A flight in the air that is not a launch or reentry but is conducted by a space support vehicle.
- **State of Design (SOD).** Per § 21.1(b)(8), the country or jurisdiction having regulatory authority over the organization responsible for the design and continued airworthiness of a civil aeronautical product or article.
- **State of Manufacture (SOM).** Per § 21.1(b)(9), the country or jurisdiction having regulatory authority over the organization responsible for the production and airworthiness of a civil aeronautical product or article.
- **State of Registry (SOR).** The country or jurisdiction on whose register an aircraft is recorded.
- **Third-Country Aircraft.** As used concerning exporting/importing an aircraft, an aircraft whose SOM is not the country of the exporting or importing CAA.
- **Type Certificate (TC).** A design approval issued by the FAA when the applicant demonstrates that a product complies with the applicable regulations. As defined by § 21.41, the TC includes the type design, the operating limitations, the TCDS, the applicable regulations of 14 CFR parts 21 through 49, and other conditions or limitations prescribed by the Administrator. The TC is the foundation for other FAA approvals, including production and airworthiness approvals.
- **Type Design.** The engineering definition of a particular product. The type design consists of the following (see § 21.31):
 - Drawings and specifications.
 - Dimensions, materials, and processes.
 - Airworthiness limitations.
 - For primary category aircraft, if desired, a special inspection and preventive maintenance program designed to be accomplished by an appropriately rated and trained pilot/owner.
 - Other data to describe the product design, and to determine the airworthiness, noise characteristics, fuel venting, and exhaust emissions (where applicable).

Appendix J. Related Publications

J-1. Related Publications. This order references several publications; visit the FAA website at FAA Regulations & Policies or the Dynamic Regulatory System at DRS for the current version.

a. FAA Orders.

- (1) Order 0000.1, *FAA Standard Subject Classification System*.
- (2) Order 1350.14, *Records Management*.
- (3) Order 8000.95, *Designee Management Policy*.
- (4) Order 8110.4, *Type Certification*.
- (5) Order 8110.56, *Restricted Category Type Certification*.
- (6) Order 8110.101, *Type Certification Procedures For Military Commercial Derivative Aircraft*.
- (7) Order 8130.34, *Airworthiness Certification of Unmanned Aircraft Systems and Optionally Piloted Aircraft*.
- (8) Order 8300.16, *Major Repair and Alteration Data Approval*.
- (9) Order 8900.1, *Flight Standards Information Management System*.
- (10) Order 8130.21, *Procedures for Completion and Use of the Authorized Release Certificate, FAA Form 8130-3, Airworthiness Approval Tag*.
- (11) Order 8130.35, *Amateur-Built Aircraft National Kit Evaluation Team (NKET)*.
- (12) Order 8130.36, *Special Light-Sport Aircraft Audit Program*.

b. FAA Advisory Circulars.

- (1) AC 00-1.1, *Public Aircraft Operations—Manned and Unmanned*.
- (2) AC 20-27, *Certification and Operation of Amateur-Built Aircraft*.
- (3) AC 21-2, *Complying with the Requirements of Importing Countries or Jurisdictions When Exporting U.S. Products, Articles, or Parts*.
- (4) AC 21-4, *Special Flight Permits of Operation of Overweight Aircraft*.
- (5) AC 21-12, *Application for U.S. Airworthiness Certificate*. AC 21-13, *Standard Airworthiness Certification of Surplus Military Aircraft and Aircraft Built from Spare and Surplus Parts*.
- (6) AC 21-28, *Airworthiness Certification of U.S. Produced Aircraft and Engine Kits Assembled Outside the United States*.
- (7) AC 21-44, *Issuance of Export Airworthiness Approvals Under 14 CFR Part 21 Subpart L*.

- (8) AC 21-54, *Experimental Airworthiness Certification of Certain Former-Military Aircraft*.
- (9) AC 43-209, *Recommended Inspection Procedures for Former Military Aircraft*.
- (10) AC 45-3, *Installation, Removal, or Change of Identification Data and Identification Plates on Aircraft Engines*.
- (11) AC 90-89, *Amateur-Built Aircraft and Ultralight Flight Testing Handbook*.
- (12) AC 90-116, *Additional Pilot Program for Phase I Flight Test*.
- (13) AC 91-85, *Authorization of Aircraft and Operators for Flight in Reduced Vertical Separation Minimum (RVSM) Airspace*.
- (14) AC 91-87, *Ejection Seat Training Programs*.
- (15) AC 21.17-2, *Type Certification—Fixed Wing Gliders (Sailplanes), Including Powered Gliders*.

c. Other FAA Documents.

- (1) *Revised Listing of Amateur-Built Aircraft Kits (“kit list”)* at FAA Amateur-Built Kits Listing.
- (2) *Amateur-Built Fabrication and Assembly Checklists, Amateur-Built Kits Checklists and Approval Letters Evaluated by National Kit Evaluation Team (NKET)* at FAA Amateur-Built Assembly Checklists.
- (3) *Amateur-Built Fabrication and Assembly Checklist (2009) Job Aid* at FAA Job Aid.
- (4) Federal Aviation Agency. *Civil Aeronautics Manual (CAM)*. See DRS.

Directive Feedback Information

Please submit any written comments or recommendations for improving this directive or suggest new items or subjects to be added to it. Also, if you find an error, please tell us about it.

Subject: Order 8130.2M, Airworthiness Certification of Aircraft

To: Directive Management Officer, 9-AVS-AIR-Directives-Management-Officer@faa.gov

(Please mark all appropriate line items)

☐ An error (procedural or typographical) has been noted in paragraph _____ on page _____. .

☐ Recommend paragraph _____ on page _____ be changed as follows:
(attached separate sheet if necessary)

☐ In a future change to this order, please include coverage on the following subject *(briefly describe what you want added)*:

☐ Other comments:

☐ I would like to discuss the above. Please contact me.

Submitted by : _____ Date: _____

Telephone Number _____ Routing Symbol: _____